

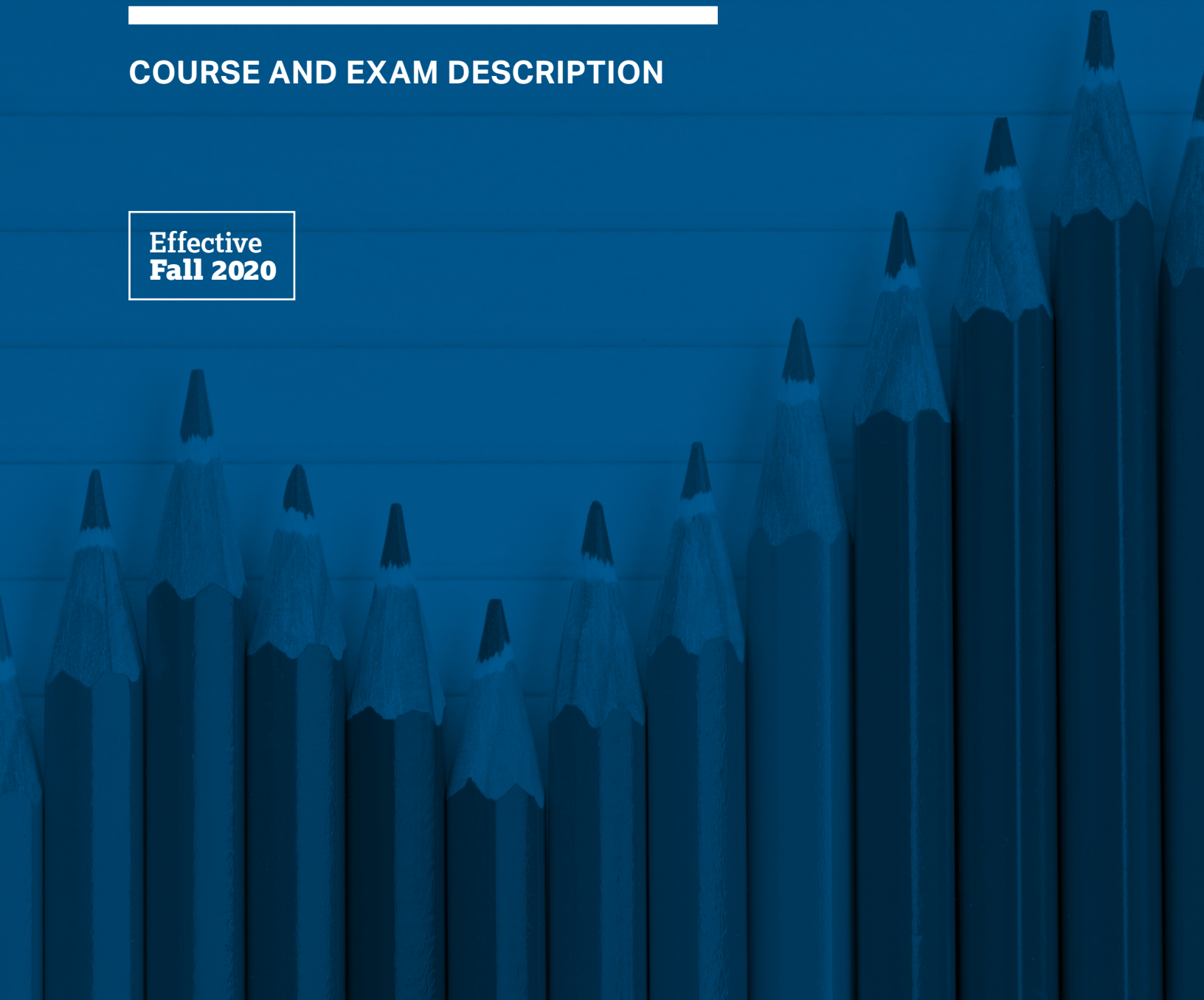
INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions

AP[®] Statistics

COURSE AND EXAM DESCRIPTION

Effective
Fall 2020



AP[®] Statistics

COURSE AND EXAM DESCRIPTION

Effective
Fall 2020

AP COURSE AND EXAM DESCRIPTIONS ARE UPDATED PERIODICALLY

Please visit AP Central (apcentral.collegeboard.org) to determine whether a more recent course and exam description is available.

What AP® Stands For

Thousands of Advanced Placement teachers have contributed to the principles articulated here. These principles are not new; they are, rather, a reminder of how AP already works in classrooms nationwide. The following principles are designed to ensure that teachers' expertise is respected, required course content is understood, and that students are academically challenged and free to make up their own minds.

1. AP stands for clarity and transparency. Teachers and students deserve clear expectations. The Advanced Placement Program makes public its course frameworks and sample assessments. Confusion about what is permitted in the classroom disrupts teachers and students as they navigate demanding work.
2. AP is an unflinching encounter with evidence. AP courses enable students to develop as independent thinkers and to draw their own conclusions. Evidence and the scientific method are the starting place for conversations in AP courses.
3. AP opposes censorship. AP is animated by a deep respect for the intellectual freedom of teachers and students alike. If a school bans required topics from their AP courses, the AP Program removes the AP designation from that course and its inclusion in the AP Course Ledger provided to colleges and universities. For example, the concepts of evolution are at the heart of college biology, and a course that neglects such concepts does not pass muster as AP Biology.
4. AP opposes indoctrination. AP students are expected to analyze different perspectives from their own, and no points on an AP Exam are awarded for agreement with any specific viewpoint. AP students are not required to feel certain ways about themselves or the course content. AP courses instead develop students' abilities to assess the credibility of sources, draw conclusions, and make up their own minds.

As the AP English Literature course description states: "AP students are not expected or asked to subscribe to any one specific set of cultural or political values, but are expected to have the maturity to analyze perspectives different from their own and to question the meaning, purpose, or effect of such content within the literary work as a whole."

5. AP courses foster an open-minded approach to the histories and cultures of different peoples. The study of different nationalities, cultures, religions, races, and ethnicities is essential within a variety of academic disciplines. AP courses ground such studies in primary sources so that students can evaluate experiences and evidence for themselves.
6. Every AP student who engages with evidence is listened to and respected. Students are encouraged to evaluate arguments but not one another. AP classrooms respect diversity in backgrounds, experiences, and viewpoints. The perspectives and contributions of the full range of AP students are sought and considered. Respectful debate of ideas is cultivated and protected; personal attacks have no place in AP.
7. AP is a choice for parents and students. Parents and students freely choose to enroll in AP courses. Course descriptions are available online for parents and students to inform their choice. Parents do not define which college-level topics are suitable within AP courses; AP course and exam materials are crafted by committees of professors and other expert educators in each field. AP courses and exams are then further validated by the American Council on Education and studies that confirm the use of AP scores for college credits by thousands of colleges and universities nationwide.

The AP Program encourages educators to review these principles with parents and students so they know what to expect in an AP course. Advanced Placement is always a choice, and it should be an informed one. AP teachers should be given the confidence and clarity that once parents have enrolled their child in an AP course, they have agreed to a classroom experience that embodies these principles.

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Ellen Breazel, *Clemson University, Clemson, SC*

Christy Brown, *Clemson University, Clemson, SC*

Paul Buckley, *Gonzaga College High School, Washington, DC*

Mine Cetinkaya-Rundel, *Duke University, Durham, NC*

Jeff Eicher, Jr., *Classical Academy High School, Escondido, CA*

Kerri Swails Freeland, *University High School, Morgantown, WV*

Kenneth Koehler, *Iowa State University, Ames, IA*

Michael Lacey, *Peters Township High School, McMurray, PA*

Laura Marshall, *Phillips Exeter Academy, Exeter, NH*

S. Leigh Nataro, *Kent Place School, Summit, NJ*

Kathleen Petko, *Palatine High School, Palatine, IL*

Paul Rodriguez, *Troy High School, Fullerton, CA*

Penny Smeltzer, *Austin Peace Academy, Austin, TX*

David Spohn, *Hudson High School, Hudson, OH*

Daren Starnes, *The Lawrenceville School, Lawrenceville, NJ*

Robert Stephenson, *Iowa State University, Ames, IA*

Jessica Utts, *University of California, Irvine, Irvine, CA*

Adam Yankay, *Western Reserve Academy, Hudson, OH*

College Board Staff

Sara Hunter, *Associate Director, AP Curricular Publications*

Tiffany Judkins, *Director, AP Instructional Design and
PD Resource Development*

Claire Lorenz, *Senior Director, AP Instructional Design and
PD Resource Development*

Daniel McDonough, *Senior Director, AP Content Integration*

Stephanie Ogden, *Director, AP Mathematics Content Development*

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About AP

College Board’s Advanced Placement® Program (AP®) enables willing and academically prepared students to pursue college-level studies—with the opportunity to earn college credit, advanced placement, or both—while still in high school. Through AP courses in 38 subjects, each culminating in a challenging exam, students learn to think critically, construct solid arguments, and see many sides of an issue—skills that prepare them for college and beyond. Taking AP courses demonstrates to college admission officers that students have sought the most challenging curriculum available to them, and research indicates that students who score a 3 or higher on an AP Exam typically experience greater academic success in college and are more likely to earn a college degree than non-AP students. Each AP teacher’s syllabus is evaluated and approved by faculty from some of the nation’s leading colleges and universities, and AP Exams are developed and scored by college faculty and experienced AP teachers. Most four-year colleges and universities in the United States grant credit, advanced placement, or both on the basis of successful AP Exam scores; more than 3,300 institutions worldwide annually receive AP scores.

AP Course Development

In an ongoing effort to maintain alignment with best practices in college-level learning, AP courses and exams emphasize challenging, research-based curricula aligned with higher education expectations.

Individual teachers are responsible for designing their own curriculum for AP courses, selecting appropriate college-level readings, assignments, and resources. This course and exam description presents the content and skills that are the focus of the corresponding college course and that appear on the AP Exam. It also organizes the content and skills into a series of units that represent a sequence found in widely adopted college textbooks and that many AP teachers have told us they follow in order to focus their instruction. The intention of this publication is to respect teachers’ time and expertise by providing a roadmap that they can modify and adapt to their local priorities and preferences. Moreover, by organizing the AP course content and skills into units, the AP Program is able

to provide teachers and students with free formative assessments—Personal Progress Checks—that teachers can assign throughout the year to measure student progress as they acquire content knowledge and develop skills.

Enrolling Students: Access, Opportunity, and Readiness

The AP Program welcomes all students willing to challenge themselves with college-level coursework and career preparation. We strongly encourage educators to invite students into AP classes, including students from ethnic, racial, socioeconomic, geographic, or other groups not broadly participating in a school’s AP program. We believe that readiness for AP is attainable, and that educators can expand readiness by opening access to Pre-AP course work. We commit to support educators and communities in their efforts to make AP courses widely available, advancing students in their plans for college and careers.

Offering AP Courses: The AP Course Audit

The AP Program unequivocally supports the principle that each school implements its own curriculum that will enable students to develop the content understandings and skills described in the course framework.

While the unit sequence represented in this publication is optional, the AP Program does have a short list of curricular and resource requirements that must be fulfilled before a school can label a course “Advanced Placement” or “AP.” Schools wishing to offer AP courses must participate in the AP Course Audit, a process through which AP teachers’ course materials are reviewed by college faculty. The AP Course Audit was created to provide teachers and administrators with clear guidelines on curricular and resource requirements for AP courses and to help colleges and universities validate courses marked “AP” on students’ transcripts. This process ensures that AP teachers’ courses meet or exceed the curricular and resource expectations that college and secondary school faculty have established for college-level courses.

The AP Course Audit form is submitted by the AP teacher and the school principal (or designated administrator) to confirm awareness and understanding of the curricular and resource requirements. A syllabus or course outline, detailing how course requirements are met, is submitted by the AP teacher for review by college faculty.

Please visit the [AP Course Audit](#) website for more information to support the preparation and submission of materials for the AP Course Audit.

How the AP Program Is Developed

The scope of content for an AP course and exam is derived from an analysis of hundreds of syllabi and course offerings of colleges and universities. Using this research and data, a committee of college faculty and expert AP teachers work within the scope of the corresponding college course to articulate what students should know and be able to do upon the completion of the AP course. The resulting course framework is the heart of this course and exam description and serves as a blueprint of the content and skills that can appear on an AP Exam.

The AP Test Development Committees are responsible for developing each AP Exam, ensuring the exam questions are aligned to the course framework. The AP Exam development process is a multiyear endeavor; all AP Exams undergo extensive review, revision, piloting, and analysis to ensure that questions are accurate, fair, and valid, and that there is an appropriate spread of difficulty across the questions.

Committee members are selected to represent a variety of perspectives and institutions (public and private, small and large schools and colleges), and a range of gender, racial/ethnic, and regional groups. A list of each subject’s current AP Test Development Committee members is available on [AP Central](#)®.

Throughout AP course and exam development, College Board gathers feedback from various stakeholders in both secondary schools and higher education institutions. This feedback is carefully considered to ensure that AP courses and exams are able to provide students with a college-level learning experience and the opportunity to demonstrate their qualifications for advanced placement or college credit.

How AP Exams Are Scored

The exam scoring process, like the course and exam development process, relies on the expertise of both AP teachers and college faculty. While multiple-choice questions are scored by machine, the free-response

questions and through-course performance assessments, as applicable, are scored by thousands of college faculty and expert AP teachers. Most are scored at the annual AP Reading, while a small portion is scored online. All AP Readers are thoroughly trained, and their work is monitored throughout the Reading for fairness and consistency. In each subject, a highly respected college faculty member serves as Chief Faculty Consultant and, with the help of AP Readers in leadership positions, maintains the accuracy of the scoring standards. Scores on the free-response questions and performance assessments are weighted and combined with the results of the computer-scored multiple-choice questions, and this raw score is converted into a composite AP score on a 1–5 scale.

AP Exams are **not** norm-referenced or graded on a curve. Instead, they are criterion-referenced, which means that every student who meets the criteria for an AP score of 2, 3, 4, or 5 will receive that score, no matter how many students that is. The criteria for the number of points students must earn on the AP Exam to receive scores of 3, 4, or 5—the scores that research consistently validates for credit and placement purposes—include:

- The number of points successful college students earn when their professors administer AP Exam questions to them.
- The number of points researchers have found to be predictive that an AP student will succeed when placed into a subsequent higher-level college course.
- Achievement-level descriptions formulated by college faculty who review each AP Exam question.

Using and Interpreting AP Scores

The extensive work done by college faculty and AP teachers in the development of the course and exam and throughout the scoring process ensures that AP Exam scores accurately represent students’ achievement in the equivalent college course. Frequent and regular research studies establish the validity of AP scores as follows:

AP Score	Credit Recommendation	College Grade Equivalent
5	Extremely well qualified	A
4	Well qualified	A-, B+, B
3	Qualified	B-, C+, C
2	Possibly qualified	n/a
1	No recommendation	n/a

While colleges and universities are responsible for setting their own credit and placement policies, most private colleges and universities award credit and/or advanced placement for AP scores of 3 or higher. Additionally, most states in the U.S. have adopted statewide credit policies that ensure college credit for scores of 3 or higher at public colleges and universities. To confirm a specific college's AP credit/placement policy, use the search engine available on the [AP Credit Policy Search](#) page.

BECOMING AN AP READER

Each June, thousands of AP teachers and college faculty members from around the world gather for seven days in multiple locations to evaluate and score the free-response sections of the AP Exams. Ninety-eight percent of surveyed educators who took part in the AP Reading say it was a positive experience.

There are many reasons to consider becoming an AP Reader, including opportunities to:

- **Bring positive changes to the classroom:**
Surveys show that the vast majority of returning AP Readers—both high school and college educators—make improvements to the way they

teach or score because of their experience at the AP Reading.

- **Gain in-depth understanding of AP Exam and AP scoring standards:** AP Readers gain exposure to the quality and depth of the responses from the entire pool of AP Exam takers, and thus are better able to assess their students' work in the classroom.
- **Receive compensation:** AP Readers are compensated for their work during the Reading. Expenses, lodging, and meals are covered for Readers who travel.
- **Score from home:** AP Readers have online distributed scoring opportunities for certain subjects. Check the [AP Reader](#) site for details.
- **Earn Continuing Education Units (CEUs):**
AP Readers earn professional development hours and CEUs that can be applied to PD requirements by states, districts, and schools.

How to Apply

Visit the [Become an AP Reader](#) site for eligibility requirements and to start the application process.

AP Resources and Supports

By completing a simple activation process at the start of the school year, teachers and students receive access to a robust set of classroom resources.

AP Classroom

AP Classroom is a dedicated online platform designed to support teachers and students throughout their AP experience. The platform provides a variety of powerful resources and tools to provide yearlong support to teachers and enable students to receive meaningful feedback on their progress.



UNIT GUIDES

Appearing in this publication and on AP Classroom, these planning guides outline all required course content and skills, organized into commonly taught units. Each unit guide suggests a sequence and pacing of content, scaffolds skill instruction across units, organizes content into topics, and provides tips on taking the AP Exam.



PERSONAL PROGRESS CHECKS

Formative AP questions for every unit provide feedback to students on the areas where they need to focus. Available online, Personal Progress Checks measure knowledge and skills through multiple-choice questions with rationales to explain correct and incorrect answers, and free-response questions with scoring information. Because the Personal Progress Checks are formative, the results of these assessments cannot be used to evaluate teacher effectiveness or assign letter grades to students, and any such misuses are grounds for losing school authorization to offer AP courses.*



PROGRESS DASHBOARD

This dashboard allows teachers to review class and individual student progress throughout the year. Teachers can view class trends and see where students struggle with content and skills that will be assessed on the AP Exam. Students can view their own progress over time to improve their performance before the AP Exam.



AP QUESTION BANK

This online library of real AP Exam questions provides teachers with secure questions to use in their classrooms. Teachers can find questions indexed by course topics and skills, create customized tests, and assign them online or on paper. These tests enable students to practice and get feedback on each question.

*To report misuses, please call, 877-274-6474 (International: +1-212-632-1781).

Digital Activation

In order to teach an AP class and make sure students are registered to take the AP Exam, teachers must first complete the digital activation process. Digital activation gives students and teachers access to resources and gathers students' exam registration information online, eliminating most of the answer sheet bubbling that has added to testing time and fatigue.

AP teachers and students begin by signing in to **My AP** and completing a simple activation process at the start of the school year, which provides access to all AP resources, including AP Classroom.

To complete digital activation:

- Teachers and students sign in to or create their College Board accounts.
- Teachers confirm that they have added the course they teach to their AP Course Audit account and have had it approved by their school's administrator.
- Teachers or AP Coordinators, depending on who the school has decided is responsible, set up class sections so students can access AP resources and have exams ordered on their behalf.
- Students join class sections with a join code provided by their teacher or AP Coordinator.
- Students will be asked for additional registration information upon joining their first class section, which eliminates the need for extensive answer sheet bubbling on exam day.

While the digital activation process takes a short time for teachers, students, and AP Coordinators to complete, overall it helps save time and provides the following additional benefits:

- **Access to AP resources and supports:** Teachers have access to resources specifically designed to support instruction and provide feedback to students throughout the school year as soon as activation is complete.
- **Streamlined exam ordering:** AP Coordinators can create exam orders from the same online class rosters that enable students to access resources. The coordinator reviews, updates, and submits this information as the school's exam order in the fall.
- **Student registration labels:** For each student included in an exam order, schools will receive a set of personalized AP ID registration labels, which replaces the AP student pack. The AP ID connects a student's exam materials with the registration information they provided during digital activation, eliminating the need for pre-administration sessions and reducing time spent bubbling on exam day.
- **Targeted Instructional Planning Reports:** AP teachers will get Instructional Planning Reports (IPRs) that include data on each of their class sections automatically rather than relying on special codes optionally bubbled in on exam day.

Instructional Model

Integrating AP resources throughout the course can help students develop skills and conceptual understandings. The instructional model outlined below shows possible ways to incorporate AP resources into the classroom.



Plan

Teachers may consider the following approaches as they plan their instruction before teaching each unit.

- Review the overview at the start of each **unit guide** to identify essential questions, conceptual understandings, and skills for each unit.
- Use the **Unit at a Glance** table to identify related topics that build toward a common understanding, and then plan appropriate pacing for students.
- Identify useful strategies in the **Instructional Approaches** section to help teach the concepts and skills.



Teach

When teaching, supporting resources could be used to build students' conceptual understanding and their mastery of skills.

- Use the topic pages in the **unit guides** to identify the required content.
- Integrate the content with a skill, considering any appropriate scaffolding.
- Employ any of the instructional strategies previously identified.
- Use the available resources on the topic pages to bring a variety of assets into the classroom.



Assess

Teachers can measure student understanding of the content and skills covered in the unit and provide actionable feedback to students.

- At the end of each unit, use **AP Classroom** to assign students the online **Personal Progress Checks**, as homework or as an in-class task.
- Provide question-level feedback to students through answer rationales; provide unit- and skill-level feedback using the progress dashboard.
- Create additional practice opportunities using the **AP Question Bank** and assign them through **AP Classroom**.

About the AP Statistics Course

The AP Statistics course introduces students to the major concepts and tools for collecting, analyzing, and drawing conclusions from data. There are four themes evident in the content, skills, and assessment in the AP Statistics course: exploring data, sampling and experimentation, probability and simulation, and statistical inference. Students use technology, investigations, problem solving, and writing as they build conceptual understanding.

College Course Equivalent

The AP Statistics course is equivalent to a one-semester, introductory, non-calculus-based college course in statistics.

Prerequisites

The AP Statistics course is an excellent option for any secondary school student who has successfully completed a second-year course in algebra and who possesses sufficient mathematical maturity and quantitative reasoning ability. Because second-year algebra is the prerequisite course, AP Statistics is usually taken in either the junior or senior year. Decisions about whether to take AP Statistics and when to take it depend on a student's plans:

- Students planning to take a science course in their senior year will benefit greatly from taking AP Statistics in their junior year.
- For students who would otherwise take no mathematics in their senior year, AP Statistics allows them to continue to develop their quantitative skills.
- Students who wish to leave open the option of taking calculus in college should include precalculus in their high school program and perhaps take AP Statistics concurrently with precalculus.
- Students with the appropriate mathematical background are encouraged to take both AP Statistics and AP Calculus in high school.

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AP STATISTICS

Course Framework



Introduction

The AP Statistics course outlined in this framework reflects a commitment to what college faculty value and mirrors the core curricula found in corresponding college courses. This framework defines content students must know and skills students must master in order to learn and retain an understanding of statistics they can apply in academic and everyday endeavors. Teachers may adjust the framework to meet state and local requirements.

The framework is organized in a logical sequence, based on teacher input and commonly used textbooks. These sequences represent one reasonable learning pathway for the course, among many. Teachers may adjust the suggested sequencing of units or topics, although they will want to carefully consider how to account for such changes as they access course resources for planning, instruction, and assessment.

Balancing guidance and flexibility, this approach helps to prepare students for college credit and placement.

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Course Framework Components

Overview

This course framework provides a clear and detailed description of the course requirements necessary for student success.

The course framework includes two essential components:

1 COURSE SKILLS

The course skills are central to the study and practice of statistics. Students should develop and apply the described skills on a regular basis over the span of the course.

2 COURSE CONTENT

The course content is organized into commonly taught units of study that provide a suggested sequence for the course. These units comprise the content and conceptual understandings that colleges and universities typically expect students to master to qualify for college credit and/or placement. This content is grounded in big ideas, which are cross-cutting concepts that build conceptual understanding and spiral throughout the course.

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Course Skills

The AP Statistics course skills describe what a student should be able to do while exploring course concepts. The table that follows presents these skills, which students should develop during the AP Statistics course. These skills form the basis of the tasks on the AP Exam.

The unit guides later in this publication embed and spiral these skills throughout the course, providing teachers with one way to integrate them in the course content with sufficient repetition to prepare students to transfer those skills when taking the AP Exam. Because the course skills are aligned to specific learning objectives, AP Exam questions will also reflect this pairing.

More detailed information about teaching the course skills can be found in the Instructional Approaches section of this publication.

**Skill Category 1****Selecting Statistical Methods** 1

Select methods for collecting and/or analyzing data for statistical inference.

Skill Category 2**Data Analysis** 2

Describe patterns, trends, associations, and relationships in data.

Skill Category 3**Using Probability and Simulation** 3

Explore random phenomena.

Skill Category 4**Statistical Argumentation** 4

Develop an explanation or justify a conclusion using evidence from data, definitions, or statistical inference.

SKILLS

1.A Identify the question to be answered or problem to be solved (*not assessed*).

1.B Identify key and relevant information to answer a question or solve a problem.

1.C Describe an appropriate method for gathering and representing data.

2.A Describe data presented numerically or graphically.

2.B Construct numerical or graphical representations of distributions.

2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

2.D Compare distributions or relative positions of points within a distribution.

3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.

3.B Determine parameters for probability distributions.

3.C Describe probability distributions.

4.A Make an appropriate claim or draw an appropriate conclusion.

4.B Interpret statistical calculations and findings to assign meaning or assess a claim.

INFERENCE

1.D Identify an appropriate inference method for confidence intervals.

1.E Identify an appropriate inference method for significance tests.

1.F Identify null and alternative hypotheses.

3.D Construct a confidence interval, provided conditions for inference are met.

3.E Calculate a test statistic and find a p -value, provided conditions for inference are met.

4.C Verify that inference procedures apply in a given situation.

4.D Justify a claim based on a confidence interval.

4.E Justify a claim using a decision based on significance tests.

Course Content

Based on the Understanding by Design® (Wiggins and McTighe) model, this course framework provides a clear and detailed description of the course requirements necessary for student success. The framework specifies what students must know, be able to do, and understand, with a focus on three big ideas that encompass the principles and processes in the discipline of statistics. The framework also encourages instruction that prepares students for advanced coursework in statistics or other fields using statistical reasoning and for active, informed engagement with a world of data to be interpreted appropriately and applied wisely to make informed decisions.

Big Ideas

The big ideas serve as the foundation of the course and allow students to create meaningful connections among concepts. They are often overarching concepts or themes that become threads that run throughout the course. Revisiting the big ideas and applying them in a variety of contexts allows students to develop deeper conceptual understanding. Below are the big ideas of the course and a brief description of each.

BIG IDEA 1: VARIATION AND DISTRIBUTION (VAR)

The distribution of measures for individuals within a sample or population describes variation. The value of a statistic varies from sample to sample. How can we determine whether differences between measures represent random variation or meaningful distinctions? Statistical methods based on probabilistic reasoning provide the basis for shared understandings about variation and about the likelihood that variation between and among measures, samples, and populations is random or meaningful.

BIG IDEA 2: PATTERNS AND UNCERTAINTY (UNC)

Statistical tools allow us to represent and describe patterns in data and to classify departures from patterns. Simulation and probabilistic reasoning allow us to anticipate patterns in data and to determine the likelihood of errors in inference.

continued on next page

BIG IDEA 3: DATA-BASED PREDICTIONS, DECISIONS, AND CONCLUSIONS (DAT)

Data-based regression models describe relationships between variables and are a tool for making predictions for values of a response variable. Collecting data using random sampling or randomized experimental design means that findings may be generalized to the part of the population from which the selection was made. Statistical inference allows us to make data-based decisions.

UNITS

The course content is organized into commonly taught units. The units have been arranged in a logical sequence frequently found in many college courses and textbooks.

The nine units in AP Statistics, and their weighting on the multiple-choice section of the AP Exam, are listed below.

Pacing recommendations at the unit level and on the Course at a Glance provide suggestions for how to teach the required course content and administer the Personal Progress Checks. The suggested class

periods are based on a schedule in which the class meets five days a week for 45 minutes each day.

While these recommendations have been made to aid planning, teachers should of course adjust the pacing based on the needs of their students, alternate schedules (e.g., block scheduling), or their school's academic calendar.

TOPICS

Each unit is broken down into teachable segments called topics. The topic pages (starting on p. 36) contain all required content for each topic.

Exam Weighting for the Multiple-Choice Section of the AP Exam

Units	Exam Weighting
Unit 1: Exploring One-Variable Data	15–23%
Unit 2: Exploring Two-Variable Data	5–7%
Unit 3: Collecting Data	12–15%
Unit 4: Probability, Random Variables, and Probability Distributions	10–20%
Unit 5: Sampling Distributions	7–12%
Unit 6: Inference for Categorical Data: Proportions	12–15%
Unit 7: Inference for Quantitative Data: Means	10–18%
Unit 8: Inference for Categorical Data: Chi-Square	2–5%
Unit 9: Inference for Quantitative Data: Slopes	2–5%

Spiraling the Big Ideas

The following table shows how the big ideas spiral across units.

Big Ideas	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8	Unit 9
	Exploring One-Variable Data	Exploring Two-Variable Data	Collecting Data	Probability, Random Variables, and Probability Distributions	Sampling Distributions	Inference for Categorical Data: Proportions	Inference for Quantitative Data: Means	Inference for Categorical Data: Chi-Square	Inference for Quantitative Data: Slopes
Variation and Distribution VAR	✓	✓	✓	✓	✓	✓	✓	✓	✓
Patterns and Uncertainty UNC	✓	✓	✓	✓	✓	✓	✓	✓	✓
Data-Based Predictions, Decisions, and Conclusions DAT		✓	✓			✓	✓	✓	✓

Course at a Glance

Plan

The Course at a Glance provides a useful visual organization of the AP Statistics curricular components, including:

- Sequence of units, along with approximate weighting and suggested pacing. Please note, pacing is based on 45-minute class periods, meeting five days each week for a full academic year.
- Progression of topics within each unit
- Spiraling of the big ideas and course skills across units

Teach

SKILL CATEGORIES

Skill categories spiral throughout the course.

- | | |
|--|---|
| 1 Selecting Statistical Methods | 3 Using Probability and Simulation |
| 2 Data Analysis | 4 Statistical Argumentation |

+ Indicates 3 or more skills for a given topic. See the individual topic for all the relevant skills.

BIG IDEAS

Big ideas spiral across topics and units.

- | | |
|---------------------------------------|---|
| VAR Variation and Distribution | DAT Data-Based Predictions, Decisions, and Conclusions |
| UNC Patterns and Uncertainty | |

Assess

Assign the Personal Progress Checks—either as homework or in class—for each unit. Each Personal Progress Check contains formative multiple-choice and free-response questions. The feedback from the Personal Progress Checks shows students the areas where they need to focus.

UNIT 1		Exploring One-Variable Data
~14–16 Class Periods		15–23% AP Exam Weighting
VAR 1	1.1	Introducing Statistics: What Can We Learn from Data?
VAR 2	1.2	The Language of Variation: Variables
UNC 2	1.3	Representing a Categorical Variable with Tables
UNC 2	1.4	Representing a Categorical Variable with Graphs
UNC 2	1.5	Representing a Quantitative Variable with Graphs
UNC 2	1.6	Describing the Distribution of a Quantitative Variable
UNC 2 4	1.7	Summary Statistics for a Quantitative Variable
UNC 2	1.8	Graphical Representations of Summary Statistics
UNC 2	1.9	Comparing Distributions of a Quantitative Variable
VAR 2 3	1.10	The Normal Distribution

Personal Progress Check 1	
Multiple-choice: ~35 questions	
Free-response: 2 questions	
▪ Exploring Data	
▪ Exploring Data	

UNIT 2		Exploring Two-Variable Data
~10–11 Class Periods		5–7% AP Exam Weighting
VAR 1	2.1	Introducing Statistics: Are Variables Related?
UNC 2	2.2	Representing Two Categorical Variables
UNC 2	2.3	Statistics for Two Categorical Variables
UNC DAT 2	2.4	Representing the Relationship Between Two Quantitative Variables
DAT 2 4	2.5	Correlation
DAT 2	2.6	Linear Regression Models
DAT 2	2.7	Residuals
DAT 2 4	2.8	Least Squares Regression
DAT 2	2.9	Analyzing Departures from Linearity

Personal Progress Check 2	
Multiple-choice: ~35 questions	
Free-response: 2 questions	
▪ Exploring Data	
▪ Investigative Task	

UNIT 3

Collecting Data

~9–10 Class Periods | 12–15% AP Exam Weighting

VAR 1	3.1 Introducing Statistics: Do the Data We Collected Tell the Truth?
DAT 1 4	3.2 Introduction to Planning a Study
DAT 1	3.3 Random Sampling and Data Collection
DAT 1	3.4 Potential Problems with Sampling
VAR 1	3.5 Introduction to Experimental Design
VAR 1	3.6 Selecting an Experimental Design
VAR 4	3.7 Inference and Experiments

Personal Progress Check 3

Multiple-choice: ~20 questions

Free-response: 2 questions

- Exploring Data and Collecting Data
- Collecting Data

UNIT 4

Probability, Random Variables, and Probability Distributions

~18–20 Class Periods | 10–20% AP Exam Weighting

VAR 1	4.1 Introducing Statistics: Random and Non-Random Patterns?
UNC 3	4.2 Estimating Probabilities Using Simulation
VAR 3 4	4.3 Introduction to Probability
VAR 4	4.4 Mutually Exclusive Events
VAR 3	4.5 Conditional Probability
VAR 3	4.6 Independent Events and Unions of Events
VAR 2 4	4.7 Introduction to Random Variables and Probability Distributions
VAR 3 4	4.8 Mean and Standard Deviation of Random Variables
VAR 3	4.9 Combining Random Variables
UNC 3	4.10 Introduction to the Binomial Distribution
UNC 3 4	4.11 Parameters for a Binomial Distribution
UNC 3 4	4.12 The Geometric Distribution

Personal Progress Check 4

Multiple-choice: ~45 questions

Free-response: 2 questions

- Probability
- Investigative Task

UNIT 5

Sampling Distributions

~10–12 Class Periods | 7–12% AP Exam Weighting

VAR 1	5.1 Introducing Statistics: Why Is My Sample Not Like Yours?
VAR 3	5.2 The Normal Distribution, Revisited
UNC 3	5.3 The Central Limit Theorem
UNC 4 3	5.4 Biased and Unbiased Point Estimates
VAR 3 4	5.5 Sampling Distributions for Sample Proportions
UNC 3 4	5.6 Sampling Distributions for Differences in Sample Proportions
UNC 3 4	5.7 Sampling Distributions for Sample Means
UNC 3 4	5.8 Sampling Distributions for Differences in Sample Means

Personal Progress Check 5

Multiple-choice: ~35 questions

Free-response: 2 questions

- Probability and Sampling Distributions
- Investigative Task

UNIT 6

Inference for Categorical Data: Proportions

~16–18

Class Periods

12–15%

AP Exam Weighting

VAR 1	6.1 Introducing Statistics: Why Be Normal?
UNC +	6.2 Constructing a Confidence Interval for a Population Proportion
UNC 4	6.3 Justifying a Claim Based on a Confidence Interval for a Population Proportion
VAR 1 4	6.4 Setting Up a Test for a Population Proportion
VAR DAT 3 4	6.5 Interpreting <i>p</i> -Values
DAT 4	6.6 Concluding a Test for a Population Proportion
UNC +	6.7 Potential Errors When Performing Tests
UNC +	6.8 Confidence Intervals for the Difference of Two Proportions
UNC 4	6.9 Justifying a Claim Based on a Confidence Interval for a Difference of Population Proportions
VAR 1 4	6.10 Setting Up a Test for the Difference of Two Population Proportions
VAR DAT 3 4	6.11 Carrying Out a Test for the Difference of Two Population Proportions

Personal Progress Check 6

Multiple-choice: ~55 questions

Free-response: 2 questions

- Inference
- Investigative Task

UNIT 7

Inference for Quantitative Data: Means

~14–16

Class Periods

10–18%

AP Exam Weighting

VAR 1	7.1 Introducing Statistics: Should I Worry About Error?
VAR UNC +	7.2 Constructing a Confidence Interval for a Population Mean
UNC 4	7.3 Justifying a Claim About a Population Mean Based on a Confidence Interval
VAR 1 4	7.4 Setting Up a Test for a Population Mean
VAR DAT 3 4	7.5 Carrying Out a Test for a Population Mean
UNC +	7.6 Confidence Intervals for the Difference of Two Means
UNC 4	7.7 Justifying a Claim About the Difference of Two Means Based on a Confidence Interval
VAR 1 4	7.8 Setting Up a Test for the Difference of Two Population Means
VAR DAT 3 4	7.9 Carrying Out a Test for the Difference of Two Population Means
	7.10 Skills Focus: Selecting, Implementing, and Communicating Inference Procedures

Personal Progress Check 7

Multiple-choice: ~50 questions

Free-response: 2 questions

- Inference and Collecting Data
- Investigative Task

UNIT 8

Inference for Categorical Data: Chi-Square

~10–11

Class Periods

2–5%

AP Exam Weighting

VAR 1	8.1 Introducing Statistics: Are My Results Unexpected?
VAR +	8.2 Setting Up a Chi-Square Goodness of Fit Test
VAR DAT 3 4	8.3 Carrying Out a Chi-Square Test for Goodness of Fit
VAR 3	8.4 Expected Counts in Two-Way Tables
VAR 1 4	8.5 Setting Up a Chi-Square Test for Homogeneity or Independence
VAR DAT 3 4	8.6 Carrying Out a Chi-Square Test for Homogeneity or Independence
	8.7 Skills Focus: Selecting an Appropriate Inference Procedure for Categorical Data

Personal Progress Check 8

Multiple-choice: ~30 questions

Free-response: 2 questions

- Inference
- Inference and Exploring Data/Collecting Data

UNIT 9

Inference for Quantitative Data: Slopes

~7–8

Class
Periods

2–5%

AP Exam
Weighting

VAR
1
9.1 **Introducing Statistics:
Do Those Points Align?**

UNC
+
9.2 **Confidence Intervals
for the Slope of a
Regression Model**

UNC
4
9.3 **Justifying a Claim
About the Slope
of a Regression
Model Based on a
Confidence Interval**

VAR
1
4
9.4 **Setting Up a Test
for the Slope of a
Regression Model**

VAR
DAT
3
4
9.5 **Carrying Out a Test
for the Slope of a
Regression Model**

9.6 **Skills Focus: Selecting
an Appropriate
Inference Procedure**

Personal Progress Check 9

Multiple-choice: ~25 questions

Free-response: 1 question

- Inference and Exploring Data

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AP STATISTICS

Unit Guides

Introduction

Designed with extensive input from the community of AP Statistics educators, the unit guides offer teachers helpful guidance in building students' skills and knowledge. The suggested sequence was identified through a thorough analysis of the syllabi of highly effective AP teachers and the organization of typical college textbooks.

This unit structure respects new AP teachers' time by providing one possible sequence they can adopt and modify rather than having to build from scratch. An additional benefit is that these units enable the AP Program to provide interested teachers with formative assessments—the Personal Progress Checks—that they can assign their students at the end of each unit to gauge progress toward success on the AP exam. However, experienced AP teachers who are satisfied with their current course organization and exam results should feel no pressure to adopt these units, which comprise an optional sequence for this course.

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Using the Unit Guides

UNIT
1

15–23% AP EXAM WEIGHTING

~14–16 CLASS PERIODS

Exploring One-Variable Data

BIG IDEA 1
Variation and Distribution

Is my cat old compared to other cats?

BIG IDEA 2
Patterns and Uncertainty

How certain are we that what seems to be a pattern is not just a coincidence?

Developing Understanding

Unit 1 introduces students to data and the vocabulary of statistics. Students also learn to talk about data in real-world contexts. Variability in data may seem to suggest certain conclusions about the data distribution, but not all variation is meaningful. Statistics allows us to develop shared understandings of uncertainty and variation. In this unit, students will define and represent categorical and quantitative variables, describe and compare distributions of one-variable data, and interpret statistical calculations to assess claims about individual data points or samples. Students will also begin to apply the normal distribution model as an introduction to how theoretical models for populations can be used to describe some distributions of sample data. Later units will more fully develop probabilistic modeling and inference.

Building Course Skills

Having access to a world of data is meaningless without the ability to organize and analyze that information. To develop these skills, students will need multiple opportunities to interact with data presented in different formats, i.e., as a table, a graph, or even just a list of values. Students should be asked to verbally describe the patterns and characteristics they see in the data (including shape, center, variability, and unusual features for a quantitative variable) and then compare the characteristics of two different sets of data. Students should also create displays that appropriately represent the data (e.g., using a bar graph for categorical data).

Teachers can provide explicit feedback on students' verbal responses so they understand the level of detail needed. For example, when students are asked to describe a distribution of quantitative data, they often provide an acronym associated with that type of distribution (e.g., SOCS or CUSIS) but then struggle to discuss

all the elements the acronym stands for. In particular, students often neglect to discuss unusual features such as gaps or outliers. Teachers can reinforce that these elements must be addressed in their descriptions and that all data has context (e.g., the variable of interest, including any units of measurement).

Preparing for the AP Exam

In preparation for the AP Exam, teachers can encourage students to carefully read each question and completely answer the question asked. When interpreting representations of quantitative data, for example, students should describe shape, center, and variability, as well as unusual features, such as outliers. A response focused only on the center, for example, would be considered incomplete. Students should also provide complete explanations in context for all conclusions made from data. If asked to justify the selection of a particular conclusion over other options, students should include both a reasoning for their choice and rationales for not choosing the others.

AP Statistics Course and Exam Description

UNIT OPENERS

Developing Understanding provides an overview that contextualizes and situates the key content of the unit within the scope of the course.

Big ideas serve as the foundation of the course and help develop understanding as they spiral throughout the course. The **essential questions** are thought-provoking questions that motivate students and inspire inquiry.

Building Course Skills describes specific skills that are appropriate to focus on in that unit.

Preparing for the AP Exam provides helpful tips and common student misunderstandings identified from prior exam data.

UNIT 1 Exploring One-Variable Data			
UNIT AT A GLANCE		Skills	Class Periods
Topic			~14–16 CLASS PERIODS
VAR-1	1.1 Introducing Statistics: What Can We Learn from Data?	1.A Identify the question to be answered or problem to be solved (not assessed).	
	1.2 The Language of Variation: Variables	1.B Describe data presented numerically or graphically.	
	1.3 Representing a Categorical Variable with Tables	1.C Construct numerical or graphical representations of distributions. 1.D Describe data presented numerically or graphically.	
	1.4 Representing a Categorical Variable with Graphs	1.E Construct numerical or graphical representations of distributions. 1.F Describe data presented numerically or graphically. 1.G Compare distributions or relative positions of points within a distribution.	
UNC-1	1.5 Representing a Quantitative Variable with Graphs	1.H Describe data presented numerically or graphically. 1.I Construct numerical or graphical representations of distributions.	
	1.6 Describing the Distribution of a Quantitative Variable	1.J Describe data presented numerically or graphically.	
	1.7 Summary Statistics for a Quantitative Variable	1.K Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response. 1.L Interpret statistical calculations and findings to assign meaning or assess a claim.	
	1.8 Graphical Representations of Summary Statistics	1.M Construct numerical or graphical representations of distributions. 1.N Describe data presented numerically or graphically.	
VAR-2	1.9 Comparing Distributions of a Quantitative Variable	1.O Compare distributions or relative positions of points within a distribution.	
	1.10 The Normal Distribution	1.P Compare distributions or relative positions of points within a distribution. 1.Q Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
Go to AP Classroom to assign the Personal Progress Check for Unit 1. Review the results in class to identify and address any student misunderstandings.			

34 | Course Framework V.1

The **Unit at a Glance** table shows the topics, related enduring understandings, and skills. The class periods column has been left blank so that teachers can customize the time they spend on each topic.

The table includes **skills** for each topic to show how to link the content in that topic to a specific AP Statistics skill or skills. The questions on the Personal Progress Checks are based on this pairing. Because the course skills are aligned to specific learning objectives, AP Exam questions also reflect this pairing.

Using the Unit Guides

Exploring One-Variable Data

UNIT 1

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	1.5	Gallery Walk Have students work in groups of four to construct a dotplot, a stem-and-leaf plot, a histogram, or a boxplot for a set of student-generated data (e.g., time in minutes to get to school). After the gallery walk, discuss what information can be seen more easily in each graph (e.g., boxplots can easily show the IQR).
2	1.6 1.8	FRQ Partner Quiz Have students work in pairs to answer 2017 FRQ 4. Have one student write and the other perform the calculations. (Although the first part of the question does not require any calculations, the second part requires calculations to justify the solution.) Discussing and crafting a solution with a partner may require more time than if students completed the FRQ individually.
3	1.9	Notice and Wonder Display just the graphs from 2018 FRQ 6. Have students think individually for one minute about how the graphs compare. Then ask them, "What do you notice? What do you wonder?" What questions could be answered with these graphs? Have students share their ideas with a partner then debrief the ideas as a class.
4	1.10	Reversing Interpretations Give pairs of students four pictures of normal distributions with various parts shaded. Have students create the question that could have resulted in the picture shown (e.g., if a value of 15 is labeled and the distribution is shaded to the right of 15, students could write "What is the probability that a value is more than 15?").

AP Statistics Course and Exam Description
Course Framework V.1 | 35

The **Sample Instructional Activities** page includes optional activities that can help teachers tie together the content and skills of a particular topic.

Sampling Distributions

UNIT 5

TOPIC 5.5 Sampling Distributions for Sample Proportions

Required Course Content

ENDURING UNDERSTANDING
UNC-3
 Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE
UNC-3.K
 Determine parameters of a sampling distribution for sample proportions. **[Skill 3.B]**

ESSENTIAL KNOWLEDGE
UNC-3.K.1
 For independent samples (sampling with replacement) of a categorical variable from a population with population proportion, p , the sampling distribution of the sample proportion, $\hat{p}$, has a mean, $\mu_{\hat{p}} = p$ and a standard deviation, $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.
UNC-3.K.2
 If sampling without replacement, the standard deviation of the sample proportion is smaller than what is given by the formula above. If the sample size is less than 10% of the population size, the difference is negligible.

UNC-3.L
 Determine whether a sampling distribution for a sample proportion can be described as approximately normal. **[Skill 3.C]**

UNC-3.M
 Interpret probabilities and parameters for a sampling distribution for a sample proportion. **[Skill 4.B]**

UNC-3.L.1
 For a categorical variable, the sampling distribution of the sample proportion, $\hat{p}$, will have an approximate normal distribution, provided the sample size is large enough: $np \geq 10$ and $n(1-p) \geq 10$.

UNC-3.M.1
 Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

SKILLS
3.B Using Probability and Simulation
 Determine parameters for probability distributions.
3.C Describing Probability Distributions
4.B Statistical Argumentation
 Describe probabilistic arguments and findings to assign meaning or assess a claim.

AVAILABLE RESOURCES
 + Classroom Resources +
 + Sampling Distributions
 + Calculators
 + Aren't Enough?
 + The Importance of Communication in AP Statistics

AP Statistics Course and Exam Description
Course Framework V.1 | 115

TOPIC PAGES

The **skills** note the course skills that are paired with the learning objectives for that topic.

Enduring understandings are the long-term takeaways related to the big ideas that leave a lasting impression on students.

Learning objectives define what a student needs to be able to do with content knowledge in order to progress toward the enduring understandings.

Where possible, **available resources** are provided that might help teachers address a particular topic.

Essential knowledge statements describe the knowledge required to perform the learning objective.

Clarifying statements provide guidance about using the formula sheet for certain topics in the course.

REQUIRED COURSE CONTENT LABELING SYSTEM

BIG IDEA

VAR

Variation and Distribution

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.A

Identify questions to be answered, based on variation in one-variable data.

ESSENTIAL KNOWLEDGE

VAR-1.A.1

Numbers may convey meaningful information, when placed in context.

NOTE: Labels are used to distinguish each unique element of the required course content and are used throughout this course and exam description. Additionally, they are used in the AP Question Bank and other resources found in AP Classroom. Enduring understandings are labeled sequentially according to the big idea that they are related to. Learning objectives are labeled to correspond with the enduring understanding they relate to. Finally, essential knowledge statements are labeled to correspond with the learning objective they relate to.

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AP STATISTICS

UNIT 1

Exploring One-Variable Data



15–23%

AP EXAM WEIGHTING



~14–16

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 1

Multiple-choice: ~35 questions

Free-response: 2 questions

- Exploring Data
- Exploring Data

Exploring One-Variable Data



Developing Understanding

BIG IDEA 1 *Variation and Distribution* **VAR**

- Is my cat old, compared to other cats?

BIG IDEA 2 *Patterns and Uncertainty* **UNC**

- How certain are we that what seems to be a pattern is not just a coincidence?

Unit 1 introduces students to data and the vocabulary of statistics. Students also learn to talk about data in real-world contexts. Variability in data may seem to suggest certain conclusions about the data distribution, but not all variation is meaningful. Statistics allows us to develop shared understandings of uncertainty and variation. In this unit, students will define and represent categorical and quantitative variables, describe and compare distributions of one-variable data, and interpret statistical calculations to assess claims about individual data points or samples. Students will also begin to apply the normal distribution model as an introduction to how theoretical models for populations can be used to describe some distributions of sample data. Later units will more fully develop probabilistic modeling and inference.

Building Course Skills

2.A 2.B 2.D

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all the elements the acronym stands for. In particular, students often neglect to discuss unusual features such as gaps or outliers. Teachers can reinforce that these elements must be addressed in their descriptions and that all data has context (e.g., the variable of interest, including any units of measurement).

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UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~14–16 CLASS PERIODS
VAR-1	1.1 Introducing Statistics: What Can We Learn from Data?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
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UNC-1	1.3 Representing a Categorical Variable with Tables	2.B Construct numerical or graphical representations of distributions. 2.A Describe data presented numerically or graphically.	
	1.4 Representing a Categorical Variable with Graphs	2.B Construct numerical or graphical representations of distributions. 2.A Describe data presented numerically or graphically. 2.D Compare distributions or relative positions of points within a distribution.	
	1.5 Representing a Quantitative Variable with Graphs	2.A Describe data presented numerically or graphically. 2.B Construct numerical or graphical representations of distributions.	
	1.6 Describing the Distribution of a Quantitative Variable	2.A Describe data presented numerically or graphically.	
	1.7 Summary Statistics for a Quantitative Variable	2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	1.8 Graphical Representations of Summary Statistics	2.B Construct numerical or graphical representations of distributions. 2.A Describe data presented numerically or graphically.	
	1.9 Comparing Distributions of a Quantitative Variable	2.D Compare distributions or relative positions of points within a distribution.	
VAR-2	1.10 The Normal Distribution	2.D Compare distributions or relative positions of points within a distribution. 3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
 Go to AP Classroom to assign the Personal Progress Check for Unit 1. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

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2	1.6 1.8	FRQ Partner Quiz Have students work in pairs to answer 2017 FRQ 4 . Have one student write and the other perform the calculations. (Although the first part of the question does not require any calculations, the second part requires calculations to justify the solution.) Discussing and crafting a solution with a partner may require more time than if students completed the FRQ individually.
3	1.9	Notice and Wonder Display just the graphs from 2018 FRQ 5 . Have students think individually for one minute about how the graphs compare. Then ask them, "What do you notice? What do you wonder? What questions could be answered with these graphs?" Have students share their ideas with a partner then debrief the ideas as a class.
4	1.10	Reversing Interpretations Give pairs of students four pictures of normal distributions with various parts shaded. Have students create the question that could have resulted in the picture shown (e.g., if a value of 15 is labeled and the distribution is shaded to the right of 15, students could write "What is the probability that a value is more than 15?").

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.



AVAILABLE RESOURCE

- Classroom Resource > [Coke® Versus Pepsi®: An Introductory Activity for Test of Significance](#) (may be used in Topic 1.1 to introduce the course or in Topic 6.4 to introduce inference tests)

TOPIC 1.1

Introducing Statistics: What Can We Learn from Data?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.A

Identify questions to be answered, based on variation in one-variable data.

[Skill 1.A]

ESSENTIAL KNOWLEDGE

VAR-1.A.1

Numbers may convey meaningful information, when placed in context.

TOPIC 1.2

The Language of Variation: Variables

SKILL

 Data Analysis

2.A

Describe data presented numerically or graphically.



ILLUSTRATIVE EXAMPLES

Categorical variables:

- Dominant hand
- Age group (young or old)
- Highest degree earned

Quantitative variables:

- Age of a structure
- Height of a child
- Concentration of a sample

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.B

Identify variables in a set of data. [Skill 2.A]

VAR-1.C

Classify types of variables. [Skill 2.A]

ESSENTIAL KNOWLEDGE

VAR-1.B.1

A variable is a characteristic that changes from one individual to another.

VAR-1.C.1

A categorical variable takes on values that are category names or group labels.

VAR-1.C.2

A quantitative variable is one that takes on numerical values for a measured or counted quantity.

SKILLS

 Data Analysis

2.B

Construct numerical or graphical representations of distributions.

2.A

Describe data presented numerically or graphically.

TOPIC 1.3

Representing a Categorical Variable with Tables

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.A

Represent categorical data using frequency or relative frequency tables. **[Skill 2.B]**

UNC-1.B

Describe categorical data represented in frequency or relative tables. **[Skill 2.A]**

ESSENTIAL KNOWLEDGE

UNC-1.A.1

A frequency table gives the number of cases falling into each category. A relative frequency table gives the proportion of cases falling into each category.

UNC-1.B.1

Percentages, relative frequencies, and rates all provide the same information as proportions.

UNC-1.B.2

Counts and relative frequencies of categorical data reveal information that can be used to justify claims about the data in context.

TOPIC 1.4

Representing a Categorical Variable with Graphs

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.C

Represent categorical data graphically. [Skill 2.B]

UNC-1.D

Describe categorical data represented graphically. [Skill 2.A]

UNC-1.E

Compare multiple sets of categorical data. [Skill 2.D]

ESSENTIAL KNOWLEDGE

UNC-1.C.1

Bar charts (or bar graphs) are used to display frequencies (counts) or relative frequencies (proportions) for categorical data.

UNC-1.C.2

The height or length of each bar in a bar graph corresponds to either the number or proportion of observations falling within each category.

UNC-1.C.3

There are many additional ways to represent frequencies (counts) or relative frequencies (proportions) for categorical data.

UNC-1.D.1

Graphical representations of a categorical variable reveal information that can be used to justify claims about the data in context.

UNC-1.E.1

Frequency tables, bar graphs, or other representations can be used to compare two or more data sets in terms of the same categorical variable.

SKILLS



Data Analysis

2.B

Construct numerical or graphical representations of distributions.

2.A

Describe data presented numerically or graphically.

2.D

Compare distributions or relative positions of points within a distribution.

SKILLS

 Data Analysis

2.A

Describe data presented numerically or graphically.

2.B

Construct numerical or graphical representations of distributions.



ILLUSTRATIVE EXAMPLES

A discrete variable:

- Number of students in a class

A continuous variable:

- Height of a child

TOPIC 1.5

Representing a Quantitative Variable with Graphs

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.F

Classify types of quantitative variables. [Skill 2.A]

ESSENTIAL KNOWLEDGE

UNC-1.F.1

A discrete variable can take on a countable number of values. The number of values may be finite or countably infinite, as with the counting numbers.

UNC-1.F.2

A continuous variable can take on infinitely many values, but those values cannot be counted. No matter how small the interval between two values of a continuous variable, it is always possible to determine another value between them.

UNC-1.G

Represent quantitative data graphically. [Skill 2.B]

UNC-1.G.1

In a histogram, the height of each bar shows the number or proportion of observations that fall within the interval corresponding to that bar. Altering the interval widths can change the appearance of the histogram.

UNC-1.G.2

In a stem and leaf plot, each data value is split into a "stem" (the first digit or digits) and a "leaf" (usually the last digit).

continued on next page

LEARNING OBJECTIVE

UNC-1.G

Represent quantitative data graphically. [Skill 2.B]

ESSENTIAL KNOWLEDGE

UNC-1.G.3

A dotplot represents each observation by a dot, with the position on the horizontal axis corresponding to the data value of that observation, with nearly identical values stacked on top of each other.

UNC-1.G.4

A cumulative graph represents the number or proportion of a data set less than or equal to a given number.

UNC-1.G.5

There are many additional ways to graphically represent distributions of quantitative data.

SKILL

 Data Analysis

2.A

Describe data presented numerically or graphically.

TOPIC 1.6

Describing the Distribution of a Quantitative Variable

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.H

Describe the characteristics of quantitative data distributions.

[Skill 2.A]

ESSENTIAL KNOWLEDGE

UNC-1.H.1

Descriptions of the distribution of quantitative data include shape, center, and variability (spread), as well as any unusual features such as outliers, gaps, clusters, or multiple peaks.

UNC-1.H.2

Outliers for one-variable data are data points that are unusually small or large relative to the rest of the data.

UNC-1.H.3

A distribution is skewed to the right (positive skew) if the right tail is longer than the left. A distribution is skewed to the left (negative skew) if the left tail is longer than the right. A distribution is symmetric if the left half is the mirror image of the right half.

UNC-1.H.4

Univariate graphs with one main peak are known as unimodal. Graphs with two prominent peaks are bimodal. A graph where each bar height is approximately the same (no prominent peaks) is approximately uniform.

UNC-1.H.5

A gap is a region of a distribution between two data values where there are no observed data.

continued on next page

LEARNING OBJECTIVE

UNC-1.H

Describe the characteristics of quantitative data distributions.
[Skill 2.A]

ESSENTIAL KNOWLEDGE

UNC-1.H.6

Clusters are concentrations of data usually separated by gaps.

UNC-1.H.7

Descriptive statistics does not attribute properties of a data set to a larger population, but may provide the basis for conjectures for subsequent testing.

SKILLS

 Data Analysis

2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

 Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

TOPIC 1.7

Summary Statistics for a Quantitative Variable

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.1

Calculate measures of center and position for quantitative data. [Skill 2.C]

ESSENTIAL KNOWLEDGE

UNC-1.1.1

A statistic is a numerical summary of sample data.

UNC-1.1.2

The mean is the sum of all the data values divided by the number of values. For a sample, the mean is denoted by $\bar{x}$: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$,

where x_i represents the i^{th} data point in the sample and n represents the number of data values in the sample.

UNC-1.1.3

The median of a data set is the middle value when data are ordered. When the number of data points is even, the median can take on any value between the two middle values. In AP Statistics, the most commonly used value for the median of a data set with an even number of values is the average of the two middle values.

UNC-1.1.4

The first quartile, Q_1 , is the median of the half of the ordered data set from the minimum to the position of the median. The third quartile, Q_3 , is the median of the half of the ordered data set from the position of the median to the maximum. Q_1 and Q_3 form the boundaries for the middle 50% of values in an ordered data set.

continued on next page

LEARNING OBJECTIVE

UNC-1.I

Calculate measures of center and position for quantitative data. [Skill 2.C]

UNC-1.J

Calculate measures of variability for quantitative data. [Skill 2.C]

UNC-1.K

Explain the selection of a particular measure of center and/or variability for describing a set of quantitative data. [Skill 4.B]

ESSENTIAL KNOWLEDGE

UNC-1.I.5

The p^{th} percentile is interpreted as the value that has $p\%$ of the data less than or equal to it.

UNC-1.J.1

Three commonly used measures of variability (or spread) in a distribution are the range, interquartile range, and standard deviation.

UNC-1.J.2

The range is defined as the difference between the maximum data value and the minimum data value. The interquartile range (IQR) is defined as the difference between the third and first quartiles: $Q3 - Q1$. Both the range and the interquartile range are possible ways of measuring variability of the distribution of a quantitative variable.

UNC-1.J.3

Standard deviation is a way to measure variability of the distribution of a quantitative variable. For a sample, the standard deviation is denoted by s : $s_x = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$. The square of the sample standard deviation, s^2 , is called the sample variance.

UNC-1.J.4

Changing units of measurement affects the values of the calculated statistics.

UNC-1.K.1

There are many methods for determining outliers. Two methods frequently used in this course are:

- An outlier is a value greater than $1.5 \times \text{IQR}$ above the third quartile or more than $1.5 \times \text{IQR}$ below the first quartile.
- An outlier is a value located 2 or more standard deviations above, or below, the mean.

UNC-1.K.2

The mean, standard deviation, and range are considered nonresistant (or non-robust) because they are influenced by outliers. The median and IQR are considered resistant (or robust), because outliers do not greatly (if at all) affect their value.

SKILLS

 Data Analysis

2.B

Construct numerical or graphical representations of distributions.

2.A

Describe data presented numerically or graphically.

TOPIC 1.8

Graphical Representations of Summary Statistics

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.L

Represent summary statistics for quantitative data graphically. [Skill 2.B]

UNC-1.M

Describe summary statistics of quantitative data represented graphically. [Skill 2.A]

ESSENTIAL KNOWLEDGE

UNC-1.L.1

Taken together, the minimum data value, the first quartile (Q1), the median, the third quartile (Q3), and the maximum data value make up the five-number summary.

UNC-1.L.2

A boxplot is a graphical representation of the five-number summary (minimum, first quartile, median, third quartile, maximum). The box represents the middle 50% of data, with a line at the median and the ends of the box corresponding to the quartiles. Lines ("whiskers") extend from the quartiles to the most extreme point that is not an outlier, and outliers are indicated by their own symbol beyond this.

UNC-1.M.1

Summary statistics of quantitative data, or of sets of quantitative data, can be used to justify claims about the data in context.

UNC-1.M.2

If a distribution is relatively symmetric, then the mean and median are relatively close to one another. If a distribution is skewed right, then the mean is usually to the right of the median. If the distribution is skewed left, then the mean is usually to the left of the median.

TOPIC 1.9

Comparing Distributions of a Quantitative Variable

SKILL



Data Analysis

2.D

Compare distributions or relative positions of points within a distribution.

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.N

Compare graphical representations for multiple sets of quantitative data.

[Skill 2.D]**UNC-1.O**

Compare summary statistics for multiple sets of quantitative data. **[Skill 2.D]**

ESSENTIAL KNOWLEDGE

UNC-1.N.1

Any of the graphical representations, e.g., histograms, side-by-side boxplots, etc., can be used to compare two or more independent samples on center, variability, clusters, gaps, outliers, and other features.

UNC-1.O.1

Any of the numerical summaries (e.g., mean, standard deviation, relative frequency, etc.) can be used to compare two or more independent samples.

SKILLS

 Data Analysis

2.D

Compare distributions or relative positions of points within a distribution.

 Using Probability and Simulation

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.



ILLUSTRATIVE EXAMPLES

Variables that can be modeled by a normal distribution:

- Body temperature
- Weight of a loaf of bread

TOPIC 1.10

The Normal Distribution

Required Course Content

ENDURING UNDERSTANDING

VAR-2

The normal distribution can be used to represent some population distributions.

LEARNING OBJECTIVE

VAR-2.A

Compare a data distribution to the normal distribution model.
[Skill 2.D]

ESSENTIAL KNOWLEDGE

VAR-2.A.1

A parameter is a numerical summary of a population.

VAR-2.A.2

Some sets of data may be described as approximately normally distributed. A normal curve is mound-shaped and symmetric. The parameters of a normal distribution are the population mean, μ , and the population standard deviation, σ .

VAR-2.A.3

For a normal distribution, approximately 68% of the observations are within 1 standard deviation of the mean, approximately 95% of observations are within 2 standard deviations of the mean, and approximately 99.7% of observations are within 3 standard deviations of the mean. This is called the empirical rule.

VAR-2.A.4

Many variables can be modeled by a normal distribution.

continued on next page

LEARNING OBJECTIVE**VAR-2.B**

Determine proportions and percentiles from a normal distribution. **[Skill 3.A]**

VAR-2.C

Compare measures of relative position in data sets. **[Skill 2.D]**

ESSENTIAL KNOWLEDGE**VAR-2.B.1**

A standardized score for a particular data value is calculated as $(\text{data value} - \text{mean})/(\text{standard deviation})$, and measures the number of standard deviations a data value falls above or below the mean.

VAR-2.B.2

One example of a standardized score is a z-score, which is calculated as

$$z\text{-score} = \left(\frac{x_i - \mu}{\sigma} \right).$$

A z-score measures how many standard deviations a data value is from the mean.

VAR-2.B.3

Technology, such as a calculator, a standard normal table, or computer-generated output, can be used to find the proportion of data values located on a given interval of a normally distributed random variable.

VAR-2.B.4

Given the area of a region under the graph of the normal distribution curve, it is possible to use technology, such as a calculator, a standard normal table, or computer-generated output, to estimate parameters for some populations.

VAR-2.C.1

Percentiles and z-scores may be used to compare relative positions of points within a data set or between data sets.

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AP STATISTICS

UNIT 2

Exploring Two-Variable Data



5–7%

AP EXAM WEIGHTING



~10–11

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 2

Multiple-choice: ~35 questions

Free-response: 2 questions

- Exploring Data
- Investigative Task

Exploring Two-Variable Data



Developing Understanding

BIG IDEA 1 *Variation and Distribution* **VAR**

- Does the fact that the number of shark attacks increases with ice cream sales necessarily mean that ice cream sales cause shark attacks?

BIG IDEA 2 *Patterns and Uncertainty* **UNC**

- How might you represent incomes of individuals with and without a college degree to help describe similarities and/or differences between the two groups?

BIG IDEA 3 *Data-Based Predictions, Decisions, and Conclusions* **DAT**

- How can you determine the effectiveness of a linear model that uses the number of cricket chirps per minute to predict temperature?

Building on Unit 1, students will explore relationships in two-variable categorical or quantitative data sets. They will use graphical and numerical methods to investigate an association between two categorical variables. Skills learned while working with two-way tables will transfer to calculating probabilities in Unit 4.

Students will describe form, direction, strength, and unusual features for an association between two quantitative variables. They will assess correlation and, if appropriate, use a linear model to predict values of the response variable from values of the explanatory variable. Students will interpret the least-squares regression line in context, analyze prediction errors (residuals), and explore departures from a linear pattern.

Building Course Skills

2.C 2.D 4.B

In Unit 2, students are looking at the relationship between variables. The ability to calculate and describe statistical values, such as a conditional relative frequency or the slope of a regression line, is critical for data analysis because students must be able to analyze patterns before drawing conclusions about the data. Students should be allowed to perform their calculations using technology to help them become more aware of procedural errors. Students will also need practice translating output from technology (“calculator speak”) into appropriate statistical language.

As any statistician will assert, a numerical calculation is only as good as one’s ability to interpret what it means in the real world. Rather than just reporting values from their calculations, students must be able to connect their numerical results to the scenario’s context and formulate a verbal response that makes that connection clear. Teachers can model good communication and provide high-quality feedback to help students use accurate statistical language

when comparing side-by-side bar graphs, for example, and to avoid common errors in reasoning, such as using the word “line” to explain why a relationship is linear.

Preparing for the AP Exam

Students need ongoing practice with interpretation of vocabulary and calculated values in context. It is typically not sufficient to speak generally about the direction of a relationship, for example. If the question is about a linear model for predicting the weight of a wolf based on its length, students should write that a positive relationship means that longer wolves tend to have higher weights (see **2017 FRQ 1**). Students can communicate statistical uncertainty by using words such as “tend to have” and “on average,” being careful to be precise with language. For example, when explaining evidence of a linear relationship, the difference between discussing a *rate* of change, as opposed to a change, is the difference between right and wrong. For the sake of clarity, the word “correlation” should be reserved for discussions about relationships between two quantitative variables.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~10–11 CLASS PERIODS
VAR-1	2.1 Introducing Statistics: Are Variables Related?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
	2.2 Representing Two Categorical Variables	2.D Compare distributions or relative positions of points within a distribution.	
UNC-1	2.3 Statistics for Two Categorical Variables	2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response. 2.D Compare distributions or relative positions of points within a distribution.	
	2.4 Representing the Relationship Between Two Quantitative Variables	2.B Construct numerical or graphical representations of distributions. 2.A Describe data presented numerically or graphically.	
UNC-1, DAT-1	2.5 Correlation	2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	2.6 Linear Regression Models	2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.	
DAT-1	2.7 Residuals	2.B Construct numerical or graphical representations of distributions. 2.A Describe data presented numerically or graphically.	
	2.8 Least Squares Regression	2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	2.9 Analyzing Departures from Linearity	2.A Describe data presented numerically or graphically. 2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.	
 Go to AP Classroom to assign the Personal Progress Check for Unit 2. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	2.5 2.8	Quickwrite Give students a scatterplot and its associated computer output. Have them identify and describe the meaning of the following values in the context of the problem: slope, y-intercept, coefficient of determination, and standard error of the residuals. Also have them calculate the correlation and explain how they found it. Have students compare their write-ups in groups of three to four.
2	2.7	Reversing Interpretations Instead of asking students to interpret a residual, give them the residual and the equation of the least-squares regression line and ask them to make a prediction for a particular observation (e.g., "One wolf in the pack had a length of 1.4 m and a residual of -9.87 . What does that -9.87 tell us about that particular wolf?")
3	2.8	Build the Model Solution Provide students with strips of paper containing portions of the model solution for 2018 FRQ 1 and have them work to assemble the phrases into a solution for the FRQ. Words can be grouped for part a, as follows: [The estimate of the intercept is] [72.95]. [It is] [estimated that] [the average time to] [finish checkout] [if there are no other customers in line] [is 72.95 seconds]. Additional numbers or phrases for part a could include [174.40], [is 174.50 seconds], and [the time to].
4	2.9	Predict and Confirm Have students toss a handful of M&Ms and record how many land M side up. This is trial 1. Then have them remove the ones that were M side up. For trial 2, have students toss the remaining candies (the ones left over after removing the ones that landed M side up) and record how many land M side up on the second toss. Ask students to think about the trend and make a prediction: Will it be linear? A scatterplot of many trials should show a nonlinear relationship.

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

TOPIC 2.1

Introducing Statistics: Are Variables Related?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.D

Identify questions to be answered about possible relationships in data.

[Skill 1.A]

ESSENTIAL KNOWLEDGE

VAR-1.D.1

Apparent patterns and associations in data may be random or not.

TOPIC 2.2

Representing Two Categorical Variables

SKILL



Data Analysis

2.D

Compare distributions or relative positions of points within a distribution.

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.P

Compare numerical and graphical representations for two categorical variables.
[Skill 2.D]

ESSENTIAL KNOWLEDGE

UNC-1.P.1

Side-by-side bar graphs, segmented bar graphs, and mosaic plots are examples of bar graphs for one categorical variable, broken down by categories of another categorical variable.

UNC-1.P.2

Graphical representations of two categorical variables can be used to compare distributions and/or determine if variables are associated.

UNC-1.P.3

A two-way table, also called a contingency table, is used to summarize two categorical variables. The entries in the cells can be frequency counts or relative frequencies.

UNC-1.P.4

A joint relative frequency is a cell frequency divided by the total for the entire table.

SKILLS

 Data Analysis

2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

2.D

Compare distributions or relative positions of points within a distribution.

TOPIC 2.3

Statistics for Two Categorical Variables

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.Q

Calculate statistics for two categorical variables.
[Skill 2.C]

UNC-1.R

Compare statistics for two categorical variables.
[Skill 2.D]

ESSENTIAL KNOWLEDGE

UNC-1.Q.1

The marginal relative frequencies are the row and column totals in a two-way table divided by the total for the entire table.

UNC-1.Q.2

A conditional relative frequency is a relative frequency for a specific part of the contingency table (e.g., cell frequencies in a row divided by the total for that row).

UNC-1.R.1

Summary statistics for two categorical variables can be used to compare distributions and/or determine if variables are associated.

TOPIC 2.4

Representing the Relationship Between Two Quantitative Variables

Required Course Content

ENDURING UNDERSTANDING

UNC-1

Graphical representations and statistics allow us to identify and represent key features of data.

LEARNING OBJECTIVE

UNC-1.S

Represent bivariate quantitative data using scatterplots. **[Skill 2.B]**

ESSENTIAL KNOWLEDGE

UNC-1.S.1

A bivariate quantitative data set consists of observations of two different quantitative variables made on individuals in a sample or population.

UNC-1.S.2

A scatterplot shows two numeric values for each observation, one corresponding to the value on the x -axis and one corresponding to the value on the y -axis.

UNC-1.S.3

An explanatory variable is a variable whose values are used to explain or predict corresponding values for the response variable.

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SKILLS

Data Analysis

2.B

Construct numerical or graphical representations of distributions.

2.A

Describe data presented numerically or graphically.

ENDURING UNDERSTANDING**DAT-1**

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE**DAT-1.A**

Describe the characteristics of a scatter plot. **[Skill 2.A]**

ESSENTIAL KNOWLEDGE**DAT-1.A.1**

A description of a scatter plot includes form, direction, strength, and unusual features.

DAT-1.A.2

The direction of the association shown in a scatterplot, if any, can be described as positive or negative.

DAT-1.A.3

A positive association means that as values of one variable increase, the values of the other variable tend to increase. A negative association means that as values of one variable increase, values of the other variable tend to decrease.

DAT-1.A.4

The form of the association shown in a scatterplot, if any, can be described as linear or non-linear to varying degrees.

DAT-1.A.5

The strength of the association is how closely the individual points follow a specific pattern, e.g., linear, and can be shown in a scatterplot. Strength can be described as strong, moderate, or weak.

DAT-1.A.6

Unusual features of a scatter plot include clusters of points or points with relatively large discrepancies between the value of the response variable and a predicted value for the response variable.

TOPIC 2.5

Correlation

SKILLS
 *Data Analysis*
2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

 *Statistical Argumentation*
4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

Required Course Content

ENDURING UNDERSTANDING

DAT-1

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE

DAT-1.B

Determine the correlation for a linear relationship.
[Skill 2.C]

ESSENTIAL KNOWLEDGE

DAT-1.B.1

The correlation, r , gives the direction and quantifies the strength of the linear association between two quantitative variables.

DAT-1.B.2

The correlation coefficient can be calculated by:

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right).$$

However, the most common way to determine r is by using technology.

DAT-1.B.3

A correlation coefficient close to 1 or -1 does not necessarily mean that a linear model is appropriate.

continued on next page

LEARNING OBJECTIVE

DAT-1.C

Interpret the correlation for a linear relationship. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

DAT-1.C.1

The correlation, r , is unit-free, and always between -1 and 1 , inclusive. A value of $r = 0$ indicates that there is no linear association. A value of $r = 1$ or $r = -1$ indicates that there is a perfect linear association.

DAT-1.C.2

A perceived or real relationship between two variables does not mean that changes in one variable cause changes in the other. That is, correlation does not necessarily imply causation.

TOPIC 2.6

Linear Regression Models

SKILL



Data Analysis

2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

Required Course Content

ENDURING UNDERSTANDING

DAT-1

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE

DAT-1.D

Calculate a predicted response value using a linear regression model. [Skill 2.C]

ESSENTIAL KNOWLEDGE

DAT-1.D.1

A simple linear regression model is an equation that uses an explanatory variable, x , to predict the response variable, y .

DAT-1.D.2

The predicted response value, denoted by $\hat{y}$, is calculated as $\hat{y} = a + bx$, where a is the y -intercept and b is the slope of the regression line, and x is the value of the explanatory variable.

DAT-1.D.3

Extrapolation is predicting a response value using a value for the explanatory variable that is beyond the interval of x -values used to determine the regression line. The predicted value is less reliable as an estimate the further we extrapolate.

SKILLS

 Data Analysis

2.B

Construct numerical or graphical representations of distributions.

2.A

Describe data presented numerically or graphically.

TOPIC 2.7

Residuals

Required Course Content

ENDURING UNDERSTANDING

DAT-1

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE

DAT-1.E

Represent differences between measured and predicted responses using residual plots. [Skill 2.B]

DAT-1.F

Describe the form of association of bivariate data using residual plots. [Skill 2.A]

ESSENTIAL KNOWLEDGE

DAT-1.E.1

The residual is the difference between the actual value and the predicted value:
 $\text{residual} = y - \hat{y}$.

DAT-1.E.2

A residual plot is a plot of residuals versus explanatory variable values or predicted response values.

DAT-1.F.1

Apparent randomness in a residual plot for a linear model is evidence of a linear form to the association between the variables.

DAT-1.F.2

Residual plots can be used to investigate the appropriateness of a selected model.

TOPIC 2.8

Least Squares
Regression

SKILLS

 Data Analysis

2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

Required Course Content

ENDURING UNDERSTANDING

DAT-1

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE

DAT-1.G

Estimate parameters for the least-squares regression line model. [Skill 2.C]

ESSENTIAL KNOWLEDGE

DAT-1.G.1

The least-squares regression model minimizes the sum of the squares of the residuals and contains the point $(\bar{x}, \bar{y})$.

DAT-1.G.2

The slope, b , of the regression line can be calculated as $b = r \left(\frac{s_y}{s_x} \right)$ where r is the

correlation between x and y , s_y is the sample standard deviation of the response variable, y , and s_x is the sample standard deviation of the explanatory variable, x .

DAT-1.G.3

Sometimes, the y -intercept of the line does not have a logical interpretation in context.

DAT-1.G.4

In simple linear regression, r^2 is the square of the correlation, r . It is also called the coefficient of determination. r^2 is the proportion of variation in the response variable that is explained by the explanatory variable in the model.

continued on next page

LEARNING OBJECTIVE

DAT-1.H

Interpret coefficients for the least-squares regression line model. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

DAT-1.H.1

The coefficients of the least-squares regression model are the estimated slope and y -intercept.

DAT-1.H.2

The slope is the amount that the predicted y -value changes for every unit increase in x .

DAT-1.H.3

The y -intercept value is the predicted value of the response variable when the explanatory variable is equal to 0. The formula for the y -intercept, a , is $a = \bar{y} - b\bar{x}$.

TOPIC 2.9

Analyzing Departures
from Linearity

SKILLS

 Data Analysis

2.A

Describe data presented numerically or graphically.

2.C

Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.

Required Course Content

ENDURING UNDERSTANDING

DAT-1

Regression models may allow us to predict responses to changes in an explanatory variable.

LEARNING OBJECTIVE

DAT-1.I

Identify influential points in regression. [Skill 2.A]

ESSENTIAL KNOWLEDGE

DAT-1.I.1

An outlier in regression is a point that does not follow the general trend shown in the rest of the data and has a large residual when the Least Squares Regression Line (LSRL) is calculated.

DAT-1.I.2

A high-leverage point in regression has a substantially larger or smaller x -value than the other observations have.

DAT-1.I.3

An influential point in regression is any point that, if removed, changes the relationship substantially. Examples include much different slope, y -intercept, and/or correlation. Outliers and high leverage points are often influential.*continued on next page*

LEARNING OBJECTIVE

DAT-1.J

Calculate a predicted response using a least-squares regression line for a transformed data set.

[Skill 2.C]

ESSENTIAL KNOWLEDGE

DAT-1.J.1

Transformations of variables, such as evaluating the natural logarithm of each value of the response variable or squaring each value of the explanatory variable, can be used to create transformed data sets, which may be more linear in form than the untransformed data.

DAT-1.J.2

Increased randomness in residual plots after transformation of data and/or movement of r^2 to a value closer to 1 offers evidence that the least-squares regression line for the transformed data is a more appropriate model to use to predict responses to the explanatory variable than the regression line for the untransformed data.

AP STATISTICS

UNIT 3

Collecting Data



12–15%
AP EXAM WEIGHTING



~9–10
CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 3

Multiple-choice: ~20 questions

Free-response: 2 questions

- Exploring Data and Collecting Data
- Collecting Data

Collecting Data



Developing Understanding

BIG IDEA 1 Variation and Distribution **VAR**

- What do our data tell us?

BIG IDEA 3 Data-Based Predictions, Decisions, and Conclusions **DAT**

- Why might the data we collected not be valid for drawing conclusions about an entire population?

Depending on how data are collected, we may or may not be able to generalize findings or establish evidence of causal relationships. For example, if random selection is not used to obtain a sample from a population, bias may result and statistics from the sample cannot be assumed to generalize to the population. For data collected using well-designed experiments, statistically significant differences between or among experimental treatment groups are evidence that the treatments caused the effect. Students learn important principles of sampling and experimental design in this unit; they will learn about statistical inference in Units 6–9.

Building Course Skills

1.B 1.C 4.A 4.B

Statisticians must be adept at determining “What is this question asking?” Students should get into the habit of identifying the task in the given prompt before they begin, then checking that their response addresses that task. For example, when asked if it would be appropriate to generalize the results of a given experiment, students need to provide a clear “yes” or “no” decision in their response, along with an explanation that supports their decision.

Although students may recognize that they need to justify their reasoning, they often struggle to include explicit evidence supporting their claims. For instance, claims about non-response bias should be supported with evidence indicating whether the sample result is likely to be too high or too low compared to the population value that is being estimated. As another example, students need to clearly explain why a particular variable might lead to confounding in a given setting.

Preparing for the AP Exam

Students should continue to practice using precise language in their writing for the AP Exam. For example, some students refer to random selection when they should be writing about random assignment, or vice versa. Students should write about *random selection* when discussing generalizing results of a sample to the population. Students should write about *random assignment* when discussing experiments or their results. Because random assignment to treatments in an experiment tends to balance the effects of uncontrolled variables across groups, researchers can conclude that statistically significant differences in the response are caused by the effects of the treatments. When justifying a claim that experimental treatments cause such statistically significant differences, students should cite the random assignment of treatments and should not refer to other irrelevant elements of well-designed experiments. Students should also continue to practice connecting responses to the specific context of the question.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~9–10 CLASS PERIODS
VAR-1	3.1 Introducing Statistics: Do the Data We Collected Tell the Truth?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
	3.2 Introduction to Planning a Study	1.C Describe an appropriate method for gathering and representing data. 4.A Make an appropriate claim or draw an appropriate conclusion.	
DAT-2	3.3 Random Sampling and Data Collection	1.C Describe an appropriate method for gathering and representing data.	
	3.4 Potential Problems with Sampling	1.C Describe an appropriate method for gathering and representing data.	
VAR-3	3.5 Introduction to Experimental Design	1.C Describe an appropriate method for gathering and representing data. 1.B Identify key and relevant information to answer a question or solve a problem.	
	3.6 Selecting an Experimental Design	1.C Describe an appropriate method for gathering and representing data.	
	3.7 Inference and Experiments	4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
 Go to AP Classroom to assign the Personal Progress Check for Unit 3. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	3.2	Graphic Organizer Provide students with a table listing all possible combinations of whether a study involves random sampling (yes or no) and random assignment (yes or no). Ask them to fill in each cell with both the type of conclusion that is appropriate (association or causation) and the generalizability of the results (to the population or to only those similar to the study participants).
2	3.2	Odd One Out After modeling an odd one out example, have students form groups of four and give each of them a description of a statistical study. Explain that three of the studies are of the same type (observational or experimental) and one is different. Have students work together in their groups to determine which study is the odd one out and explain why.
3	3.2 3.3	Password-Style Games After completing the lessons on sampling and surveying, use the following 10 terms in a password-style game: census, simple random sample, stratified random sample, cluster sample, systematic random sample, bias, voluntary response bias, undercoverage, nonresponse bias, and response bias. The winner is the pair whose partner guesses the most terms correctly from the descriptions given.
4	3.5	Sentence Starters Provide students with the scenario from 2006 Form B FRQ 5 . Have them complete the following sentence starter to explain confounding: "_____ are confounded with _____ because each _____ was used in only one _____. If a difference in the draft is observed, we will not know whether the difference is due to the _____ or the _____."
5	3.5	Think-Pair-Share Provide students with a description of a well-designed experiment (e.g., 2010 FRQ 1) and ask them to individually identify the type of design, the experimental units, the treatments, and how the study addresses the principles of a well-designed experiment (including random assignment, control, blinding, and replication). Then have students share their thoughts with their neighbor.

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

TOPIC 3.1

Introducing Statistics: Do the Data We Collected Tell the Truth?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.E

Identify questions to be answered about data collection methods. **[Skill 1.A]**

ESSENTIAL KNOWLEDGE

VAR-1.E.1

Methods for data collection that do not rely on chance result in untrustworthy conclusions.

TOPIC 3.2

Introduction to Planning a Study

SKILLS

 *Selecting Statistical Methods*

1.C

Describe an appropriate method for gathering and representing data.



Statistical Argumentation

4.A

Make an appropriate claim or draw an appropriate conclusion.

Required Course Content

ENDURING UNDERSTANDING

DAT-2

The way we collect data influences what we can and cannot say about a population.

LEARNING OBJECTIVE

DAT-2.A

Identify the type of a study.
[Skill 1.C]

ESSENTIAL KNOWLEDGE

DAT-2.A.1

A population consists of all items or subjects of interest.

DAT-2.A.2

A sample selected for study is a subset of the population.

DAT-2.A.3

In an observational study, treatments are not imposed. Investigators examine data for a sample of individuals (retrospective) or follow a sample of individuals into the future collecting data (prospective) in order to investigate a topic of interest about the population. A sample survey is a type of observational study that collects data from a sample in an attempt to learn about the population from which the sample was taken.

DAT-2.A.4

In an experiment, different conditions (treatments) are assigned to experimental units (participants or subjects).

DAT-2.B

Identify appropriate generalizations and determinations based on observational studies.
[Skill 4.A]

DAT-2.B.1

It is only appropriate to make generalizations about a population based on samples that are randomly selected or otherwise representative of that population.

continued on next page

LEARNING OBJECTIVE

DAT-2.B

Identify appropriate generalizations and determinations based on observational studies.

[Skill 4.A]

ESSENTIAL KNOWLEDGE

DAT-2.B.2

A sample is only generalizable to the population from which the sample was selected.

DAT-2.B.3

It is not possible to determine causal relationships between variables using data collected in an observational study.

TOPIC 3.3

Random Sampling and Data Collection

SKILL

*Selecting Statistical Methods*

1.C

Describe an appropriate method for gathering and representing data.

Required Course Content

ENDURING UNDERSTANDING

DAT-2

The way we collect data influences what we can and cannot say about a population.

LEARNING OBJECTIVE

DAT-2.C

Identify a sampling method, given a description of a study.

[Skill 1.C]

ESSENTIAL KNOWLEDGE

DAT-2.C.1

When an item from a population can be selected only once, this is called sampling without replacement. When an item from the population can be selected more than once, this is called sampling with replacement.

DAT-2.C.2

A simple random sample (SRS) is a sample in which every group of a given size has an equal chance of being chosen. This method is the basis for many types of sampling mechanisms. A few examples of mechanisms used to obtain SRSs include numbering individuals and using a random number generator to select which ones to include in the sample, ignoring repeats, using a table of random numbers, or drawing a card from a deck without replacement.

DAT-2.C.3

A stratified random sample involves the division of a population into separate groups, called strata, based on shared attributes or characteristics (homogeneous grouping). Within each stratum a simple random sample is selected, and the selected units are combined to form the sample.

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LEARNING OBJECTIVE

DAT-2.C

Identify a sampling method, given a description of a study. **[Skill 1.C]**

DAT-2.D

Explain why a particular sampling method is or is not appropriate for a given situation. **[Skill 1.C]**

ESSENTIAL KNOWLEDGE

DAT-2.C.4

A cluster sample involves the division of a population into smaller groups, called clusters. Ideally, there is heterogeneity within each cluster, and clusters are similar to one another in their composition. A simple random sample of clusters is selected from the population to form the sample of clusters. Data are collected from all observations in the selected clusters.

DAT-2.C.5

A systematic random sample is a method in which sample members from a population are selected according to a random starting point and a fixed, periodic interval.

DAT-2.C.6

A census selects all items/subjects in a population.

DAT-2.D.1

There are advantages and disadvantages for each sampling method depending upon the question that is to be answered and the population from which the sample will be drawn.

TOPIC 3.4

Potential Problems with Sampling

SKILL

*Selecting Statistical Methods*

1.C

Describe an appropriate method for gathering and representing data.

Required Course Content

ENDURING UNDERSTANDING

DAT-2

The way we collect data influences what we can and cannot say about a population.

LEARNING OBJECTIVE

DAT-2.E

Identify potential sources of bias in sampling methods.

[Skill 1.C]

ESSENTIAL KNOWLEDGE

DAT-2.E.1

Bias occurs when certain responses are systematically favored over others.

DAT-2.E.2

When a sample is comprised entirely of volunteers or people who choose to participate, the sample will typically not be representative of the population (voluntary response bias).

DAT-2.E.3

When part of the population has a reduced chance of being included in the sample, the sample will typically not be representative of the population (undercoverage bias).

DAT-2.E.4

Individuals chosen for the sample for whom data cannot be obtained (or who refuse to respond) may differ from those for whom data can be obtained (nonresponse bias).

DAT-2.E.5

Problems in the data gathering instrument or process result in response bias. Examples include questions that are confusing or leading (question wording bias) and self-reported responses.

DAT-2.E.6

Non-random sampling methods (for example, samples chosen by convenience or voluntary response) introduce potential for bias because they do not use chance to select the individuals.

SKILLS

 *Selecting Statistical Methods*

1.C

Describe an appropriate method for gathering and representing data.

1.B

Identify key and relevant information to answer a question or solve a problem.

TOPIC 3.5

Introduction to Experimental Design

Required Course Content

ENDURING UNDERSTANDING

VAR-3

Well-designed experiments can establish evidence of causal relationships.

LEARNING OBJECTIVE

VAR-3.A

Identify the components of an experiment. [Skill 1.C]

ESSENTIAL KNOWLEDGE

VAR-3.A.1

The experimental units are the individuals (which may be people or other objects of study) that are assigned treatments. When experimental units consist of people, they are sometimes referred to as participants or subjects.

VAR-3.A.2

An explanatory variable (or factor) in an experiment is a variable whose levels are manipulated intentionally. The levels or combination of levels of the explanatory variable(s) are called treatments.

VAR-3.A.3

A response variable in an experiment is an outcome from the experimental units that is measured after the treatments have been administered.

VAR-3.A.4

A confounding variable in an experiment is a variable that is related to the explanatory variable and influences the response variable and may create a false perception of association between the two.

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LEARNING OBJECTIVE**VAR-3.B**

Describe elements of a well-designed experiment.

[Skill 1.B]

VAR-3.C

Compare experimental designs and methods.

[Skill 1.C]

ESSENTIAL KNOWLEDGE**VAR-3.B.1**

A well-designed experiment should include the following:

- Comparisons of at least two treatment groups, one of which could be a control group.
- Random assignment/allocation of treatments to experimental units.
- Replication (more than one experimental unit in each treatment group).
- Control of potential confounding variables where appropriate.

VAR-3.C.1

In a completely randomized design, treatments are assigned to experimental units completely at random. Random assignment tends to balance the effects of uncontrolled (confounding) variables so that differences in responses can be attributed to the treatments.

VAR-3.C.2

Methods for randomly assigning treatments to experimental units in a completely randomized design include using a random number generator, a table of random values, drawing chips without replacement, etc.

VAR-3.C.3

In a single-blind experiment, subjects do not know which treatment they are receiving, but members of the research team do, or vice versa.

VAR-3.C.4

In a double-blind experiment neither the subjects nor the members of the research team who interact with them know which treatment a subject is receiving.

VAR-3.C.5

A control group is a collection of experimental units either not given a treatment of interest or given a treatment with an inactive substance (placebo) in order to determine if the treatment of interest has an effect.

VAR-3.C.6

The placebo effect occurs when experimental units have a response to a placebo.

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LEARNING OBJECTIVE

VAR-3.C

Compare experimental designs and methods.
[Skill 1.C]

ESSENTIAL KNOWLEDGE

VAR-3.C.7

For randomized complete block designs, treatments are assigned completely at random within each block.

VAR-3.C.8

Blocking ensures that at the beginning of the experiment the units within each block are similar to each other with respect to at least one blocking variable. A randomized block design helps to separate natural variability from differences due to the blocking variable.

VAR-3.C.9

A matched pairs design is a special case of a randomized block design. Using a blocking variable, subjects (whether they are people or not) are arranged in pairs matched on relevant factors. Matched pairs may be formed naturally or by the experimenter. Every pair receives both treatments by randomly assigning one treatment to one member of the pair and subsequently assigning the remaining treatment to the second member of the pair. Alternately, each subject may get both treatments.

TOPIC 3.6

Selecting an Experimental Design

SKILL

*Selecting Statistical Methods*

1.C

Describe an appropriate method for gathering and representing data.

Required Course Content

ENDURING UNDERSTANDING

VAR-3

Well-designed experiments can establish evidence of causal relationships.

LEARNING OBJECTIVE

VAR-3.D

Explain why a particular experimental design is appropriate. [Skill 1.C]

ESSENTIAL KNOWLEDGE

VAR-3.D.1

There are advantages and disadvantages for each experimental design depending on the question of interest, the resources available, and the nature of the experimental units.

SKILL

 *Statistical
Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

TOPIC 3.7

Inference and Experiments

Required Course Content

ENDURING UNDERSTANDING

VAR-3

Well-designed experiments can establish evidence of causal relationships.

LEARNING OBJECTIVE

VAR-3.E

Interpret the results of a well-designed experiment.
[Skill 4.B]

ESSENTIAL KNOWLEDGE

VAR-3.E.1

Statistical inference attributes conclusions based on data to the distribution from which the data were collected.

VAR-3.E.2

Random assignment of treatments to experimental units allows researchers to conclude that some observed changes are so large as to be unlikely to have occurred by chance. Such changes are said to be statistically significant.

VAR-3.E.3

Statistically significant differences between or among experimental treatment groups are evidence that the treatments caused the effect.

VAR-3.E.4

If the experimental units used in an experiment are representative of some larger group of units, the results of an experiment can be generalized to the larger group. Random selection of experimental units gives a better chance that the units will be representative.

AP STATISTICS

UNIT 4

Probability, Random Variables, and Probability Distributions



10–20%

AP EXAM WEIGHTING



~18–20

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 4

Multiple-choice: ~45 questions

Free-response: 2 questions

- Probability
- Investigative Task

Probability, Random Variables, and Probability Distributions



Developing Understanding

BIG IDEA 1 Variation and Distribution **VAR**

- How can an event be both random and predictable?

BIG IDEA 2 Patterns and Uncertainty **UNC**

- About how many rolls of a fair six-sided die would we anticipate it taking to get three 1s?

Probabilistic reasoning allows statisticians to quantify the likelihood of random events over the long run and to make statistical inferences. Simulations and concrete examples can help students to understand the abstract definitions and calculations of probability. This unit builds on understandings of simulated or empirical data distributions and fundamental principles of probability to represent, interpret, and calculate parameters for theoretical probability distributions for discrete random variables. Interpretations of probabilities and parameters associated with a probability distribution should use appropriate units and relate to the context of the situation.

Building Course Skills

2.B 3.A 3.B 4.B

Probability is a notoriously difficult topic for students to grasp because it's difficult to conceptualize future outcomes in concrete ways. Before introducing new formulas, teachers can help students get an intuitive feel for why the formulas (and related notation) make sense. For example, the probability formulas for $P(A \text{ or } B)$ and for $P(A|B)$ can be presented intuitively with two-way tables. Simulations can also help students internalize what it means to quantify random behavior. To help students understand when to apply different probability rules, teachers can use explicit strategies such as matching verbal scenarios to their corresponding probability formulas.

Students frequently misinterpret probability distributions and parameters for random variables. Teachers can reinforce that a complete interpretation will include context and units. A common misconception later in the course is that every question involving probability requires a significance test. Students should practice making predictions and decisions based on probability alone to avoid this misconception early on.

They should revisit these problems in later units to practice differentiating between inference and probability problems.

Preparing for the AP Exam

To help students prepare for the AP Exam, teachers can model showing all steps in probability calculations and expect students to do the same. Calculations on the AP Exam should include presentation of an appropriate expression that communicates the structure of the formula, substitution of relevant values extracted from the problem, and an answer. In **2017 FRQ 3**, for example, a student who writes " $P(G) = P(G|J) \cdot P(J) + P(G|K) \cdot P(K) = (0.2119)(0.7) + (0.8413)(0.3) = 0.4007$ " has communicated the products in the multiplication rule, the sum in the addition rule, and an understanding that the events are mutually exclusive—all components of a complete response. To avoid a common error, students who present the same work using a tree diagram should practice using probabilities in the diagram correctly. Students importing incorrect solutions from one part of a multipart question to solve another will not be penalized a second time, unless the subsequent result is not a reasonable value (like a probability less than 0 or greater than 1).

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~18–20 CLASS PERIODS
VAR-1	4.1 Introducing Statistics: Random and Non-Random Patterns?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
UNC-2	4.2 Estimating Probabilities Using Simulation	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
VAR-4	4.3 Introduction to Probability	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	4.4 Mutually Exclusive Events	4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	4.5 Conditional Probability	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
	4.6 Independent Events and Unions of Events	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
VAR-5	4.7 Introduction to Random Variables and Probability Distributions	2.B Construct numerical or graphical representations of distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	4.8 Mean and Standard Deviation of Random Variables	3.B Determine parameters for probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	4.9 Combining Random Variables	3.B Determine parameters for probability distributions. 3.C Describe probability distributions.	

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UNIT AT A GLANCE *(cont'd)*

Enduring Understanding	Topic	Skills	Class Periods
			~18–20 CLASS PERIODS
UNC-3	4.10 Introduction to the Binomial Distribution	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
	4.11 Parameters for a Binomial Distribution	3.B Determine parameters for probability distributions.	
		4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	4.12 The Geometric Distribution	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations. 3.B Determine parameters for probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
 Go to AP Classroom to assign the Personal Progress Check for Unit 4. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity																
1	4.3 4.5 4.8	Error Analysis Using 2015 FRQ 3 , provide students with several answers containing errors for each part. Provide some responses with incorrect notation, incorrect work, missing work, work that shows calculator commands only, an incorrect formula or approach, and an incorrect final answer. Then ask students to identify the errors.																
2	4.3 4.5 4.6	Think-Pair-Share Provide students with a set of five probability questions: one for the complement rule, the conditional probability formula, the general multiplication rule, the multiplication rule for independent events, and the general addition rule. Ask students to individually identify the formula needed to solve each problem, without doing the final calculations. Then have them share their thoughts with a partner.																
3	4.5 4.6	Create Representations Provide students with the scenario from 2018 FRQ 3 . Ask them to create a tree diagram to organize the information in the problem. Then ask them to use the information in the problem to set up a hypothetical 100,000 table (to make the decimals easy to work with), such as the one below. Encourage students to try both representations when solving probability questions in the future. <table><tr><th></th><th>Multiple Birth</th><th>Single Birth</th><th>Total</th></tr><tr><td>Left handed</td><td>770</td><td>10,615</td><td>11,385</td></tr><tr><td>Right handed</td><td>2,730</td><td>85,885</td><td>88,615</td></tr><tr><td>Total</td><td>3,500</td><td>96,500</td><td>100,000</td></tr></table>		Multiple Birth	Single Birth	Total	Left handed	770	10,615	11,385	Right handed	2,730	85,885	88,615	Total	3,500	96,500	100,000
	Multiple Birth	Single Birth	Total															
Left handed	770	10,615	11,385															
Right handed	2,730	85,885	88,615															
Total	3,500	96,500	100,000															
4	4.10 4.12	Odd One Out After modeling an odd one out example, have students form groups of four and give each of them a description of either a binomial or a geometric random variable. Explain that three of their variables follow the same probability distribution and one is different. Have students work together in their groups to determine whose is the odd one out and explain why.																
5	4.2 4.12	Predict and Confirm Ask students to consider couples who plan to continue having children until they have one girl and predict how many children they think these couples will have, on average. Then ask each student to perform 10 trials using a coin toss where Heads = Girl and Tails = Boy. A trial is finished once one girl is observed and the number of total children is recorded. Combine the class results and calculate the average. Confirm with the geometric mean formula once it is discussed.																

TOPIC 4.1

Introducing Statistics: Random and Non-Random Patterns?

Required Course Content

ENDURING UNDERSTANDING**VAR-1**

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE**VAR-1.F**

Identify questions suggested by patterns in data. **[Skill 1.A]**

ESSENTIAL KNOWLEDGE**VAR-1.F.1**

Patterns in data do not necessarily mean that variation is not random.

SKILL

Selecting Statistical Methods

1.A

Identify the question to be answered or problem to be solved.

SKILL

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.



AVAILABLE RESOURCES

- Classroom Resources >
 - ♦ [Graphing Calculator Simulations Simplified](#)
 - ♦ [Three Calculator Simulation Activities](#)

ILLUSTRATIVE EXAMPLES

An outcome:

- Rolling a particular value on a six-sided number cube is one of six possible outcomes.

An event:

- When rolling two six-sided number cubes, an event would be a sum of seven. The corresponding collection of outcomes would be (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1), where the ordered pairs indicate (face value on one cube, face value on the other cube).

TOPIC 4.2

Estimating Probabilities Using Simulation

Required Course Content

ENDURING UNDERSTANDING

UNC-2

Simulation allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-2.A

Estimate probabilities using simulation. **[Skill 3.A]**

ESSENTIAL KNOWLEDGE

UNC-2.A.1

A random process generates results that are determined by chance.

UNC-2.A.2

An outcome is the result of a trial of a random process.

UNC-2.A.3

An event is a collection of outcomes.

UNC-2.A.4

Simulation is a way to model random events, such that simulated outcomes closely match real-world outcomes. All possible outcomes are associated with a value to be determined by chance. Record the counts of simulated outcomes and the count total.

UNC-2.A.5

The relative frequency of an outcome or event in simulated or empirical data can be used to estimate the probability of that outcome or event.

UNC-2.A.6

The law of large numbers states that simulated (empirical) probabilities tend to get closer to the true probability as the number of trials increases.

TOPIC 4.3

Introduction to Probability

SKILLS

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

Required Course Content

ENDURING UNDERSTANDING

VAR-4

The likelihood of a random event can be quantified.

LEARNING OBJECTIVE

VAR-4.A

Calculate probabilities for events and their complements. [Skill 3.A]

ESSENTIAL KNOWLEDGE

VAR-4.A.1

The sample space of a random process is the set of all possible non-overlapping outcomes.

VAR-4.A.2

If all outcomes in the sample space are equally likely, then the probability an event E will occur is defined as the fraction:

$$\frac{\text{number of outcomes in event } E}{\text{total number of outcomes in sample space}}$$

VAR-4.A.3

The probability of an event is a number between 0 and 1, inclusive.

VAR-4.A.4

The probability of the complement of an event E , E^c or E^C , (i.e., not E) is equal to $1 - P(E)$.

VAR-4.B

Interpret probabilities for events. [Skill 4.B]

VAR-4.B.1

Probabilities of events in repeatable situations can be interpreted as the relative frequency with which the event will occur in the long run.

SKILL

 *Statistical
Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

TOPIC 4.4

Mutually Exclusive Events

Required Course Content

ENDURING UNDERSTANDING

VAR-4

The likelihood of a random event can be quantified.

LEARNING OBJECTIVE

VAR-4.C

Explain why two events are (or are not) mutually exclusive.
[Skill 4.B]

ESSENTIAL KNOWLEDGE

VAR-4.C.1

The probability that events A and B both will occur, sometimes called the joint probability, is the probability of the intersection of A and B , denoted $P(A \cap B)$.

VAR-4.C.2

Two events are mutually exclusive or disjoint if they cannot occur at the same time. So $P(A \cap B) = 0$.

TOPIC 4.5

Conditional Probability

SKILL

 Using Probability and Simulation

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

Required Course Content

ENDURING UNDERSTANDING

VAR-4

The likelihood of a random event can be quantified.

LEARNING OBJECTIVE

VAR-4.D

Calculate conditional probabilities. [Skill 3.A]

ESSENTIAL KNOWLEDGE

VAR-4.D.1

The probability that event A will occur given that event B has occurred is called a conditional probability and denoted

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

VAR-4.D.2

The multiplication rule states that the probability that events A and B both will occur is equal to the probability that event A will occur multiplied by the probability that event B will occur, given that A has occurred. This is denoted $P(A \cap B) = P(A) \cdot P(B | A)$.

SKILL

 Using Probability and Simulation

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

TOPIC 4.6

Independent Events and Unions of Events

Required Course Content

ENDURING UNDERSTANDING

VAR-4

The likelihood of a random event can be quantified.

LEARNING OBJECTIVE

VAR-4.E

Calculate probabilities for independent events and for the union of two events.
[Skill 3.A]

ESSENTIAL KNOWLEDGE

VAR-4.E.1

Events A and B are independent if, and only if, knowing whether event A has occurred (or will occur) does not change the probability that event B will occur.

VAR-4.E.2

If, and only if, events A and B are independent, then $P(A | B) = P(A)$, $P(B | A) = P(B)$, and $P(A \cap B) = P(A) \cdot P(B)$.

VAR-4.E.3

The probability that event A or event B (or both) will occur is the probability of the union of A and B , denoted $P(A \cup B)$.

VAR-4.E.4

The addition rule states that the probability that event A or event B or both will occur is equal to the probability that event A will occur plus the probability that event B will occur minus the probability that both events A and B will occur. This is denoted $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

TOPIC 4.7

Introduction to Random Variables and Probability Distributions

Required Course Content

ENDURING UNDERSTANDING

VAR-5

Probability distributions may be used to model variation in populations.

LEARNING OBJECTIVE

VAR-5.A

Represent the probability distribution for a discrete random variable. **[Skill 2.B]**

VAR-5.B

Interpret a probability distribution. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

VAR-5.A.1

The values of a random variable are the numerical outcomes of random behavior.

VAR-5.A.2

A discrete random variable is a variable that can only take a countable number of values. Each value has a probability associated with it. The sum of the probabilities over all of the possible values must be 1.

VAR-5.A.3

A probability distribution can be represented as a graph, table, or function showing the probabilities associated with values of a random variable.

VAR-5.A.4

A cumulative probability distribution can be represented as a table or function showing the probability of being less than or equal to each value of the random variable.

VAR-5.B.1

An interpretation of a probability distribution provides information about the shape, center, and spread of a population and allows one to make conclusions about the population of interest.

SKILLS

 *Data Analysis*

2.B

Construct numerical or graphical representations of distributions.



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.


ILLUSTRATIVE EXAMPLES

Outcomes of trials of a random process:

- The sum of the outcomes for rolling two dice
- The number of puppies in a randomly selected litter for a certain breed of dog

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

TOPIC 4.8

Mean and Standard Deviation of Random Variables

Required Course Content

ENDURING UNDERSTANDING

VAR-5

Probability distributions may be used to model variation in populations.

LEARNING OBJECTIVE

VAR-5.C

Calculate parameters for a discrete random variable.

[Skill 3.B]

VAR-5.D

Interpret parameters for a discrete random variable.

[Skill 4.B]

ESSENTIAL KNOWLEDGE

VAR-5.C.1

A numerical value measuring a characteristic of a population or the distribution of a random variable is known as a parameter, which is a single, fixed value.

VAR-5.C.2

The mean, or expected value, for a discrete random variable X is $\mu_X = \sum x_i \cdot P(x_i)$.

VAR-5.C.3

The standard deviation for a discrete random variable X is $\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 \cdot P(x_i)}$.

VAR-5.D.1

Parameters for a discrete random variable should be interpreted using appropriate units and within the context of a specific population.

TOPIC 4.9

Combining Random Variables

SKILLS

 Using Probability and Simulation

3.B

Determine parameters for probability distributions.

3.C

Describe probability distributions.

Required Course Content

ENDURING UNDERSTANDING

VAR-5

Probability distributions may be used to model variation in populations.

LEARNING OBJECTIVE

VAR-5.E

Calculate parameters for linear combinations of random variables. [Skill 3.B]

ESSENTIAL KNOWLEDGE

VAR-5.E.1

For random variables X and Y and real numbers a and b , the mean of $aX + bY$ is $a\mu_x + b\mu_y$.

VAR-5.E.2

Two random variables are independent if knowing information about one of them does not change the probability distribution of the other.

VAR-5.E.3

For independent random variables X and Y and real numbers a and b , the mean of $aX + bY$ is $a\mu_x + b\mu_y$, and the variance of $aX + bY$ is $a^2\sigma_x^2 + b^2\sigma_y^2$.

VAR-5.F

Describe the effects of linear transformations of parameters of random variables. [Skill 3.C]

VAR-5.F.1

For $Y = a + bX$, the probability distribution of the transformed random variable, Y , has the same shape as the probability distribution for X , so long as $a > 0$ and $b > 0$. The mean of Y is $\mu_y = a + b\mu_x$. The standard deviation of Y is $\sigma_y = |b|\sigma_x$.

SKILL

 Using Probability
and Simulation

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

TOPIC 4.10

Introduction to the Binomial Distribution

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.A

Estimate probabilities of binomial random variables using data from a simulation.
[Skill 3.A]

UNC-3.B

Calculate probabilities for a binomial distribution.
[Skill 3.A]

ESSENTIAL KNOWLEDGE

UNC-3.A.1

A probability distribution can be constructed using the rules of probability or estimated with a simulation using random number generators.

UNC-3.A.2

A binomial random variable, X , counts the number of successes in n repeated independent trials, each trial having two possible outcomes (success or failure), with the probability of success p and the probability of failure $1 - p$.

UNC-3.B.1

The probability that a binomial random variable, X , has exactly x successes for n independent trials, when the probability of success is p , is calculated

$$\text{as } P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, 2, \dots, n.$$

This is the binomial probability function.

TOPIC 4.11

Parameters for a Binomial Distribution

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.C

Calculate parameters for a binomial distribution.

[Skill 3.B]

UNC-3.D

Interpret probabilities and parameters for a binomial distribution. [Skill 4.B]

ESSENTIAL KNOWLEDGE

UNC-3.C.1

If a random variable is binomial, its mean, μ_x , is np and its standard deviation, σ_x , is $\sqrt{np(1-p)}$.

UNC-3.D.1

Probabilities and parameters for a binomial distribution should be interpreted using appropriate units and within the context of a specific population or situation.

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

SKILLS

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

3.B

Determine parameters for probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

TOPIC 4.12

The Geometric Distribution

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.E

Calculate probabilities for geometric random variables. **[Skill 3.A]**

ESSENTIAL KNOWLEDGE

UNC-3.E.1

For a sequence of independent trials, a geometric random variable, X , gives the number of the trial on which the first success occurs. Each trial has two possible outcomes (success or failure) with the probability of success p and the probability of failure $1 - p$.

UNC-3.E.2

The probability that the first success for repeated independent trials with probability of success p occurs on trial x is calculated as $P(X = x) = (1 - p)^{x-1}p$, $x = 1, 2, 3, \dots$. This is the geometric probability function.

UNC-3.F

Calculate parameters of a geometric distribution. **[Skill 3.B]**

UNC-3.F.1

If a random variable is geometric, its mean, μ_x , is $\frac{1}{p}$ and its standard deviation, σ_x , is $\frac{\sqrt{1-p}}{p}$.

UNC-3.G

Interpret probabilities and parameters for a geometric distribution. **[Skill 4.B]**

UNC-3.G.1

Probabilities and parameters for a geometric distribution should be interpreted using appropriate units and within the context of a specific population or situation.

QUICK REFERENCE FOR NOTATION AND FORMULAS FOR PROBABILITY DISTRIBUTIONS

Distribution	Notes	Parameter(s)	Random Variable	Conditions	Mean for Distribution	Standard Deviation for Distribution
Probability distribution for a random variable	- Represent discrete random variables using frequency/ relative frequency tables or histograms - Represent continuous random variables with density functions.	μ_x, σ_x	X	- All probabilities must be between 0 and 1. $\sum \text{Probabilities} = 1$.	$\mu_x = \sum x_i \cdot P(x_i)$ "expected value"	$\sigma_x = \sqrt{\sum (x_i - \mu_x)^2 \cdot P(x_i)}$
Sum or difference of independent random variables	See Unit 5 for distributions of other linear transformations of random variables.	$\mu_x, \sigma_x, \mu_y, \sigma_y$	$X + Y$ or $X - Y$	To calculate the variance or standard deviation of the difference, the random variables must be independent.	$\mu_{X+Y} = \mu_x + \mu_y$ or $\mu_{X-Y} = \mu_x - \mu_y$	Variance, $\sigma_{X+Y}^2 = \sigma_x^2 + \sigma_y^2$ Variance, $\sigma_{X-Y}^2 = \sigma_x^2 + \sigma_y^2$
Binomial probability distribution	Binomial probability function: $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ $x = 0, 1, 2, 3, \dots, n$.	n and p	X	n is predetermined. - Binary - Independent p is the same for each trial.	$\mu_x = np$	$\sigma_x = \sqrt{np(1-p)}$
Geometric probability distribution	Geometric probability formula: $P(X = x) = (1-p)^{x-1} p$, $x = 1, 2, 3, \dots$	p	X	n is not predetermined. - Binary - Independent p is the same for each repetition (random).	$\mu = \frac{1}{p}$ expected number of trials to get the first success	$\sigma = \frac{\sqrt{1-p}}{p}$

Note: Other notation could also be correct if properly defined. Incorrect notation will result in lost points on the AP exam.

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AP STATISTICS

UNIT 5

Sampling Distributions



7–12%

AP EXAM WEIGHTING



~10–12

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 5

Multiple-choice: ~35 questions

Free-response: 2 questions

- Probability and Sampling Distributions
- Investigative Task

Sampling Distributions



Developing Understanding

BIG IDEA 1 *Variation and Distribution* **VAR**

- How likely is it to get a value this large just by chance?

BIG IDEA 2 *Patterns and Uncertainty* **UNC**

- How can we anticipate patterns in the values of a statistic from one sample to another?

This unit applies probabilistic reasoning to sampling, introducing students to sampling distributions of statistics they will use when performing inference in Units 6 and 7. Students should understand that sample statistics can be used to estimate corresponding population parameters and that measures of center (mean) and variability (standard deviation) for these sampling distributions can be determined directly from the population parameters when certain sampling criteria are met. For large enough samples from any population, these sampling distributions can be approximated by a normal distribution. Simulating sampling distributions helps students to understand how the values of statistics vary in repeated random sampling from populations with known parameters.

Building Course Skills

3.B 3.C 4.B

The probabilities associated with the normal distribution are what statisticians use to justify claims about populations they'll never be able to measure directly. Revisiting these properties early in Unit 5 will reinforce why sampling distributions allow statisticians to approximate parameters for the population of interest. Sketching, shading, and labeling a normal distribution aids in understanding the probability being calculated. Students should practice creating graphical representations, labeling the mean, and marking off values 1, 2, and 3 standard deviations from the mean.

Students often struggle to interpret parameters of probability distributions in context, simply describing features of the graph rather than explicitly connecting those features to the situation described in the problem. Teachers can remind students that context is about a variable ("tip amounts," for example), not just the units (dollars). It's also critical that students explicitly show that the appropriate conditions have been verified, and that they avoid using nonspecific language

like "it" in their interpretations. Using an error analysis strategy with sample responses can help familiarize students with these issues before they make similar mistakes.

Preparing for the AP Exam

Responses on the AP Exam often uncover gaps in understanding of sampling distributions. Students must clearly communicate whether they are talking about the distribution of a population, a sample of values (heights, for example), or a sample statistic from repeated samples (mean heights, for example). Broad generalizations, such as "larger samples have less variability," leave the exam reader unsure of whether the student is referring to variability within a sample (for which the statement would be false) or a sampling distribution. The word "it" often introduces ambiguity to a response. Students frequently show confusion about what condition to check when asserting that the sampling distribution of a given statistic is approximately normal. Students should support normal probability calculations with a sketch or a calculation of a standardized score (z -score), rather than relying on calculator syntax.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~10–12 CLASS PERIODS
VAR-1	5.1 Introducing Statistics: Why Is My Sample Not Like Yours?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
	5.2 The Normal Distribution, Revisited	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations. 3.C Describe probability distributions.	
UNC-3	5.3 The Central Limit Theorem	3.C Describe probability distributions.	
	5.4 Biased and Unbiased Point Estimates	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 3.B Determine parameters for probability distributions.	
	5.5 Sampling Distributions for Sample Proportions	3.B Determine parameters for probability distributions. 3.C Describe probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	5.6 Sampling Distributions for Differences in Sample Proportions	3.B Determine parameters for probability distributions. 3.C Describe probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	5.7 Sampling Distributions for Sample Means	3.B Determine parameters for probability distributions. 3.C Describe probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
	5.8 Sampling Distributions for Differences in Sample Means	3.B Determine parameters for probability distributions. 3.C Describe probability distributions. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
 Go to AP Classroom to assign the Personal Progress Check for Unit 5. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	5.2	Think Aloud Group students into pairs within a larger group of four. Have each student individually read 2014 FRQ 3 and think aloud with their partner, brainstorming ways to begin each part of the question. Each student then independently completes all parts. Have the pairs compare answers within their groups, improving their individual responses as necessary. Groups can then compare their responses with other groups. Finally, have students score their responses according to the rubric.
2	5.3	Use Manipulatives From a large container of pennies, have each student take two random samples of size 5, two of size 10, and two of size 25, and record the dates on those pennies. Have students calculate the mean of the dates in each sample and then construct four “dotplots” on the floor: one using the pennies, one using nickels placed at the mean of the student’s sample size 5, one using dimes placed at the mean of the sample size 10, and one using quarters placed at the mean of the sample size 25.
3	5.5 5.7	Password-Style Games Have partners sit facing opposite sides of the room. Display vocabulary terms from the unit on the classroom screen. Have the students facing the screen describe the terms to their partner who then tries to guess the terms described. After half of the terms have been used, have students switch roles. Terms to include: parameter, statistic, sampling distribution, distribution of sample data, sample distribution, unbiased estimator, sampling variability of a statistic, bias, sample proportion, sample mean, μ_p , σ_p , $\mu_{\bar{x}}$, $\sigma_{\bar{x}}$, and central limit theorem.

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

TOPIC 5.1

Introducing Statistics: Why Is My Sample Not Like Yours?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.G

Identify questions suggested by variation in statistics for samples collected from the same population. **[Skill 1.A]**

ESSENTIAL KNOWLEDGE

VAR-1.G.1

Variation in statistics for samples taken from the same population may be random or not.

TOPIC 5.2

The Normal Distribution, Revisited

Required Course Content

ENDURING UNDERSTANDING

VAR-6

The normal distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-6.A

Calculate the probability that a particular value lies in a given interval of a normal distribution. [Skill 3.A]

VAR-6.B

Determine the interval associated with a given area in a normal distribution. [Skill 3.A]

ESSENTIAL KNOWLEDGE

VAR-6.A.1

A continuous random variable is a variable that can take on any value within a specified domain. Every interval within the domain has a probability associated with it.

VAR-6.A.2

A continuous random variable with a normal distribution is commonly used to describe populations. The distribution of a normal random variable can be described by a normal, or “bell-shaped,” curve.

VAR-6.A.3

The area under a normal curve over a given interval represents the probability that a particular value lies in that interval.

VAR-6.B.1

The boundaries of an interval associated with a given area in a normal distribution can be determined using z -scores or technology, such as a calculator, a standard normal table, or computer-generated output.

SKILLS

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

3.C

Describe probability distributions.



ILLUSTRATIVE EXAMPLE

Continuous random variable:

If one looks at a clock at a random time, the probability that the minute hand is between the 3 and the 6 is one fourth.

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LEARNING OBJECTIVE

VAR-6.B

Determine the interval associated with a given area in a normal distribution.

[Skill 3.A]

ESSENTIAL KNOWLEDGE

VAR-6.B.2

Intervals associated with a given area in a normal distribution can be determined by assigning appropriate inequalities to the boundaries of the intervals:

- $P(X < x_a) = \frac{p}{100}$ means that the lowest $p\%$ of values lie to the left of x_a .
- $P(x_a < X < x_b) = \frac{p}{100}$ means that $p\%$ of values lie between x_a and x_b .
- $P(X > x_b) = \frac{p}{100}$ means that the highest $p\%$ of values lie to the right of x_b .
- To determine the most extreme $p\%$ of values requires dividing the area associated with $p\%$ into two equal areas on either extreme of the distribution:
 $P(X < x_a) = \frac{1}{2} \frac{p}{100}$ and $P(X > x_b) = \frac{1}{2} \frac{p}{100}$
means that half of the $p\%$ most extreme values lie to the left of x_a and half of the $p\%$ most extreme values lie to the right of x_b .



VAR-6.C

Determine the appropriateness of using the normal distribution to approximate probabilities for unknown distributions.

[Skill 3.C]

VAR-6.C.1

Normal distributions are symmetrical and “bell-shaped.” As a result, normal distributions can be used to approximate distributions with similar characteristics.

TOPIC 5.3

The Central Limit Theorem

SKILL

*Using Probability and Simulation*

3.C

Describe probability distributions.

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.H

Estimate sampling distributions using simulation.
[Skill 3.C]

ESSENTIAL KNOWLEDGE

UNC-3.H.1

A sampling distribution of a statistic is the distribution of values for the statistic for all possible samples of a given size from a given population.

UNC-3.H.2

The central limit theorem (CLT) states that when the sample size is sufficiently large, a sampling distribution of the mean of a random variable will be approximately normally distributed.

UNC-3.H.3

The central limit theorem requires that the sample values are independent of each other and that n is sufficiently large.

UNC-3.H.4

A randomization distribution is a collection of statistics generated by simulation assuming known values for the parameters. For a randomized experiment, this means repeatedly randomly reallocating/reassigning the response values to treatment groups.

UNC-3.H.5

The sampling distribution of a statistic can be simulated by generating repeated random samples from a population.

SKILLS

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

TOPIC 5.4

Biased and Unbiased Point Estimates

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.I

Explain why an estimator is or is not unbiased. **[Skill 4.B]**

UNC-3.J

Calculate estimates for a population parameter.
[Skill 3.B]

ESSENTIAL KNOWLEDGE

UNC-3.I.1

When estimating a population parameter, an estimator is unbiased if, on average, the value of the estimator is equal to the population parameter.

UNC-3.J.1

When estimating a population parameter, an estimator exhibits variability that can be modeled using probability.

UNC-3.J.2

A sample statistic is a point estimator of the corresponding population parameter.

TOPIC 5.5

Sampling Distributions for Sample Proportions

Required Course Content

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE**UNC-3.K**

Determine parameters of a sampling distribution for sample proportions.

[Skill 3.B]

ESSENTIAL KNOWLEDGE**UNC-3.K.1**

For independent samples (sampling with replacement) of a categorical variable from a population with population proportion, p , the sampling distribution of the sample proportion, $\hat{p}$, has a mean, $\mu_{\hat{p}} = p$ and a standard deviation,

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}.$$

UNC-3.K.2

If sampling without replacement, the standard deviation of the sample proportion is smaller than what is given by the formula above. If the sample size is less than 10% of the population size, the difference is negligible.

UNC-3.L

Determine whether a sampling distribution for a sample proportion can be described as approximately normal.

[Skill 3.C]

UNC-3.L.1

For a categorical variable, the sampling distribution of the sample proportion, $\hat{p}$, will have an approximate normal distribution, provided the sample size is large enough: $np \geq 10$ and $n(1-p) \geq 10$

UNC-3.M

Interpret probabilities and parameters for a sampling distribution for a sample proportion. **[Skill 4.B]**

UNC-3.M.1

Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

3.C

Describe probability distributions.



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

**AVAILABLE RESOURCES**

- Classroom Resources >
 - ♦ [Sampling Distributions](#)
 - ♦ [Calculations Aren't Enough! The Importance of Communication in AP Statistics](#)

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

3.C

Describe probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.



AVAILABLE RESOURCES

- Classroom Resource >
 - ♦ [Sampling Distributions](#)
 - ♦ [Calculations Aren't Enough! The Importance of Communication in AP Statistics](#)

TOPIC 5.6

Sampling Distributions for Differences in Sample Proportions

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.N

Determine parameters of a sampling distribution for a difference in sample proportions. **[Skill 3.B]**

ESSENTIAL KNOWLEDGE

UNC-3.N.1

For a categorical variable, when randomly sampling with replacement from two independent populations with population proportions p_1 and p_2 , the sampling distribution of the difference in sample proportions $\hat{p}_1 - \hat{p}_2$ has mean, $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ and standard

$$\text{deviation, } \sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}.$$

UNC-3.N.2

If sampling without replacement, the standard deviation of the difference in sample proportions is smaller than what is given by the formula above. If the sample sizes are less than 10% of the population sizes, the difference is negligible.

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LEARNING OBJECTIVE

UNC-3.O

Determine whether a sampling distribution for a difference of sample proportions can be described as approximately normal.

[Skill 3.C]

UNC-3.P

Interpret probabilities and parameters for a sampling distribution for a difference in proportions. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

UNC-3.O.1

The sampling distribution of the difference in sample proportions $\hat{p}_1 - \hat{p}_2$ will have an approximate normal distribution provided the sample sizes are large enough:
 $n_1 p_1 \geq 10, n_1 (1 - p_1) \geq 10, n_2 p_2 \geq 10, n_2 (1 - p_2) \geq 10$.

UNC-3.P.1

Parameters for a sampling distribution for a difference of proportions should be interpreted using appropriate units and within the context of a specific populations.

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

3.C

Describe probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.



AVAILABLE RESOURCES

- Classroom Resources >
 - ♦ Sampling Distributions
 - ♦ Calculations Aren't Enough! The Importance of Communication in AP Statistics

TOPIC 5.7

Sampling Distributions for Sample Means

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.Q

Determine parameters for a sampling distribution for sample means. [Skill 3.B]

ESSENTIAL KNOWLEDGE

UNC-3.Q.1

For a numerical variable, when random sampling with replacement from a population with mean μ and standard deviation, σ , the sampling distribution of the sample mean has mean $\mu_{\bar{x}} = \mu$ and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

UNC-3.Q.2

If sampling without replacement, the standard deviation of the sample mean is smaller than what is given by the formula above. If the sample size is less than 10% of the population size, the difference is negligible.

UNC-3.R

Determine whether a sampling distribution of a sample mean can be described as approximately normal. [Skill 3.C]

UNC-3.R.1

For a numerical variable, if the population distribution can be modeled with a normal distribution, the sampling distribution of the sample mean, $\bar{x}$, can be modeled with a normal distribution.

continued on next page

LEARNING OBJECTIVE

UNC-3.R

Determine whether a sampling distribution of a sample mean can be described as approximately normal.

[Skill 3.C]

UNC-3.S

Interpret probabilities and parameters for a sampling distribution for a sample mean. [Skill 4.B]

ESSENTIAL KNOWLEDGE

UNC-3.R.2

For a numerical variable, if the population distribution cannot be modeled with a normal distribution, the sampling distribution of the sample mean, $\bar{x}$, can be modeled approximately by a normal distribution, provided the sample size is large enough, e.g., greater than or equal to 30.

UNC-3.S.1

Probabilities and parameters for a sampling distribution for a sample mean should be interpreted using appropriate units and within the context of a specific population.

SKILLS

 *Using Probability and Simulation*

3.B

Determine parameters for probability distributions.

3.C

Describe probability distributions.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.



AVAILABLE RESOURCES

- Classroom Resources >
 - ♦ [Sampling Distributions](#)
 - ♦ [Calculations Aren't Enough! The Importance of Communication in AP Statistics](#)

TOPIC 5.8

Sampling Distributions for Differences in Sample Means

Required Course Content

ENDURING UNDERSTANDING

UNC-3

Probabilistic reasoning allows us to anticipate patterns in data.

LEARNING OBJECTIVE

UNC-3.T

Determine parameters of a sampling distribution for a difference in sample means.
[Skill 3.B]

ESSENTIAL KNOWLEDGE

UNC-3.T.1

For a numerical variable, when randomly sampling with replacement from two independent populations with population means μ_1 and μ_2 and population standard deviations σ_1 and σ_2 , the sampling distribution of the difference in sample means $\bar{x}_1 - \bar{x}_2$ has mean $\mu_{(\bar{x}_1 - \bar{x}_2)} = \mu_1 - \mu_2$ and standard deviation,

$$\sigma_{(\bar{x}_1 - \bar{x}_2)} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

UNC-3.T.2

If sampling without replacement, the standard deviation of the difference in sample means is smaller than what is given by the formula above. If the sample sizes are less than 10% of the population sizes, the difference is negligible.

continued on next page

LEARNING OBJECTIVE

UNC-3.U

Determine whether a sampling distribution of a difference in sample means can be described as approximately normal.

[Skill 3.C]

UNC-3.V

Interpret probabilities and parameters for a sampling distribution for a difference in sample means. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

UNC-3.U.1

The sampling distribution of the difference in sample means $\bar{x}_1 - \bar{x}_2$ can be modeled with a normal distribution if the two population distributions can be modeled with a normal distribution.

UNC-3.U.2

The sampling distribution of the difference in sample means $\bar{x}_1 - \bar{x}_2$ can be modeled approximately by a normal distribution if the two population distributions cannot be modeled with a normal distribution but both sample sizes are greater than or equal to 30.

UNC-3.V.1

Probabilities and parameters for a sampling distribution for a difference of sample means should be interpreted using appropriate units and within the context of a specific populations.

QUICK REFERENCE FOR NOTATION AND FORMULAS FOR SAMPLING DISTRIBUTIONS

Distribution	Notes	Parameter(s)	Statistic	Conditions	Mean for Distribution	Standard Deviation for Distribution
Normal distribution	A continuous random probability distribution	μ and σ			μ	σ
Sampling distribution for a sample proportion	Compare to the mean and standard deviation of a binomial random variable, X	p	$\hat{p}$	- Simple random sample (Random) - Normal or $np \geq 10$ and $n(1-p) \geq 10$, (Large counts) - For standard deviations: population $\geq 10n$ (10% rule)	$\mu_{\hat{p}} = p$	$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$
Sampling distribution for a difference in sample proportions		$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	- Simple random samples (Random) - Large counts 10% rule	$\mu_{(\hat{p}_1 - \hat{p}_2)} = p_1 - p_2$	$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$
Sampling distribution for the sample mean		μ	$\bar{x}$	- SRS (Random) - Normal or sample size ≥ 30 10% rule	$\mu_{\bar{x}} = \mu$	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
Sampling distribution for the difference in sample means		$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	- SRS (Random) - Normal or sample sizes ≥ 30 10% rule	$\mu_{(\bar{x}_1 - \bar{x}_2)} = \mu_1 - \mu_2$	$\sigma_{(\bar{x}_1 - \bar{x}_2)} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
Standard deviation		σ	s			

Note: Other notation could also be correct if properly defined. Incorrect notation will result in lost points on the AP exam.

AP STATISTICS

UNIT 6

Inference for Categorical Data: Proportions



12–15%

AP EXAM WEIGHTING



~16–18

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 6

Multiple-choice: ~55 questions

Free-response: 2 questions

- Inference
- Investigative Task

Inference for Categorical Data: Proportions



Developing Understanding

BIG IDEA 1 Variation and Distribution **VAR**

- When can we use a normal distribution to perform inference calculations involving population proportions?

BIG IDEA 2 Patterns and Uncertainty **UNC**

- How can we narrow the width of a confidence interval?

BIG IDEA 3 Data-Based Predictions, Decisions, and Conclusions **DAT**

- If the proportion of subjects who experience serious side effects when taking a new drug is smaller than the proportion of subjects who experience serious side effects when taking a placebo, how can we determine if the difference is statistically significant?

This unit introduces statistical inference, which will continue through the end of the course. Students will analyze categorical data to make inferences about binomial population proportions. Provided conditions are met, students will use statistical inference to construct and interpret confidence intervals to estimate population proportions and perform significance tests to evaluate claims about population proportions. Students begin by learning inference procedures for one proportion and then examine inference methods for a difference between two proportions. They will also interpret the two types of errors that can be made in a significance test, their probabilities, and possible consequences in context.

Building Course Skills

1.D 3.D 4.D

Unit 6 is a critical transition point in the course, as students begin learning skills that will be applied repeatedly in subsequent units. Students need to familiarize themselves with these procedures so they can build proficiency over time. Applying different inference methods requires fluency with verifying conditions. Students often check conditions superficially (e.g., just listing “SRS”) without explicitly connecting them to the problem. Teachers can make sure students practice verifying conditions in context by providing numerical calculations and explaining how each condition is met.

Precision of language is key. Students often interpret *confidence intervals* and *confidence levels* incorrectly. Providing students with sentence starters or templates can help them learn to generate appropriate responses (e.g., Confidence interval: “We are 95% confident that the interval from ___ to ___ captures the [parameter in context].”). For decisions based on a hypothesis test, students may incorrectly claim that “we can accept” or “have proven” the null. Teachers can reinforce

early and often that statistical tests do not provide evidence for what can be accepted or proved; they only provide evidence for “rejecting” or “failing to reject” the null.

Preparing for the AP Exam

When using statistical inference to construct confidence intervals or perform significance tests, students should identify the appropriate inference method by name or formula. For inference with population proportions, students should verify that the following conditions are met: (1) random sample and (2) large sample (e.g., $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$). When sampling without replacement, students should also verify that the sample size is at most 10% of the population. Verification should be simple and specific.

Next, students should present calculations and then interpret results in the context of the problem. Students often find it beneficial to use language provided in the question. In **2017 FRQ 2**, for example, the response might read “We can be 95% confident that the proportion of *all customers who, having asked for a cup of water when placing an order, will fill the cup with a soft drink* is between 0.1883 and 0.3867.”

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~16–18 CLASS PERIODS
VAR-1	6.1 Introducing Statistics: Why Be Normal?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
UNC-4	6.2 Constructing a Confidence Interval for a Population Proportion	1.D Identify an appropriate inference method for confidence intervals. 4.C Verify that inference procedures apply in a given situation. 3.D Construct a confidence interval, provided conditions for inference are met.	
	6.3 Justifying a Claim Based on a Confidence Interval for a Population Proportion	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.D Justify a claim based on a confidence interval. 4.A Make an appropriate claim or draw an appropriate conclusion.	
VAR-6	6.4 Setting Up a Test for a Population Proportion	1.F Identify null and alternative hypotheses. 1.E Identify an appropriate inference method for significance tests. 4.C Verify that inference procedures apply in a given situation.	
VAR-6, DAT-3	6.5 Interpreting p -Values	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
DAT-3	6.6 Concluding a Test for a Population Proportion	4.E Justify a claim using a decision based on significance tests.	

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UNIT AT A GLANCE *(cont'd)*

Enduring Understanding	Topic	Skills	Class Periods
			~16–18 CLASS PERIODS
UNC-5	6.7 Potential Errors When Performing Tests	1.B Identify key and relevant information to answer a question or solve a problem. 3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations. 4.A Make an appropriate claim or draw an appropriate conclusion. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	
UNC-4	6.8 Confidence Intervals for the Difference of Two Proportions	1.D Identify an appropriate inference method for confidence intervals. 4.C Verify that inference procedures apply in a given situation. 3.D Construct a confidence interval, provided conditions for inference are met.	
	6.9 Justifying a Claim Based on a Confidence Interval for a Difference of Population Proportions	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.D Justify a claim based on a confidence interval.	
VAR-6	6.10 Setting Up a Test for the Difference of Two Population Proportions	1.F Identify null and alternative hypotheses. 1.E Identify an appropriate inference method for significance tests. 4.C Verify that inference procedures apply in a given situation.	
VAR-6, DAT-3	6.11 Carrying Out a Test for the Difference of Two Population Proportions	3.E Calculate a test statistic and find a p-value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	
Go to AP Classroom to assign the Personal Progress Check for Unit 6. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	6.4 6.7 6.8	Error Analysis Give student pairs a worksheet with 20 sets of hypotheses (including hypotheses for a population proportion and for the difference of two proportions), each with a common student mistake. Have students circle the incorrect part, write why the circled component is incorrect, and then write the correct hypotheses. Include errors such as using statistics instead of parameters, and interchanging the = and > in the two hypotheses.
2	6.5 6.6 6.11	Sentence Starters For a given question, provide students with a set of hypotheses, p -value, significance level, and context. Have them compare the p -value to the significance level to determine whether or not to reject the null hypothesis. Using a given sentence starter with blanks to fill in, have students write a sentence in context explaining if they have enough evidence to "reject H_0 ", or if they will "fail to reject H_0 ." Make sure students avoid the common mistake of implying that evidence supports an "accept H_0 " conclusion or a "reject H_a " conclusion.
3	6.2 6.8	The Scribe and the Calculator Have students work with a partner to construct and interpret a confidence interval for a population proportion. Only one partner is allowed to use the calculator, and only the other partner is allowed to write. When a calculation needs to be made, the scribe can only describe to the calculator operator which buttons to push; when writing needs to be done, the calculator operator can only describe to the scribe what needs to be written. Have students switch roles when constructing and interpreting a confidence interval for the difference of two population proportions.

TOPIC 6.1

Introducing Statistics: Why Be Normal?

SKILL

*Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.H

Identify questions suggested by variation in the shapes of distributions of samples taken from the same population.

[Skill 1.A]

ESSENTIAL KNOWLEDGE

VAR-1.H.1

Variation in shapes of data distributions may be random or not.

SKILLS

 *Selecting Statistical Methods*

1.D

Identify an appropriate inference method for confidence intervals.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.

 *Using Probability and Simulation*

3.D

Construct a confidence interval, provided conditions for inference are met.

TOPIC 6.2

Constructing a Confidence Interval for a Population Proportion

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.A

Identify an appropriate confidence interval procedure for a population proportion.
[Skill 1.D]

UNC-4.B

Verify the conditions for calculating confidence intervals for a population proportion.
[Skill 4.C]

ESSENTIAL KNOWLEDGE

UNC-4.A.1

The appropriate confidence interval procedure for a one-sample proportion for one categorical variable is a one sample z-interval for a proportion.

UNC-4.B.1

In order to make assumptions necessary for inference on population proportions, means, and slopes, we must check for independence in data collection methods and for selection of the appropriate sampling distribution.

continued on next page

LEARNING OBJECTIVE

UNC-4.B

Verify the conditions for calculating confidence intervals for a population proportion.

[Skill 4.C]

UNC-4.C

Determine the margin of error for a given sample size and an estimate for the sample size that will result in a given margin of error for a population proportion.

[Skill 3.D]

ESSENTIAL KNOWLEDGE

UNC-4.B.2

In order to calculate a confidence interval to estimate a population proportion, p , we must check for independence and that the sampling distribution is approximately normal.

- To check for independence:
 - Data should be collected using a random sample or a randomized experiment.
 - When sampling without replacement, check that $n \leq 10\%N$, where N is the size of the population.
- To check that the sampling distribution of $\hat{p}$ is approximately normal (shape):
 - For categorical variables, check that both the number of successes, $n\hat{p}$, and the number of failures, $n(1 - \hat{p})$ are at least 10 so that the sample size is large enough to support an assumption of normality.

UNC-4.C.1

Based on sample data, the standard error of a statistic is an estimate for the standard deviation for the statistic. The standard error

$$\text{of } \hat{p} \text{ is } SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

UNC-4.C.2

A margin of error gives how much a value of a sample statistic is likely to vary from the value of the corresponding population parameter.

UNC-4.C.3

For categorical variables, the margin of error is the critical value (z^*) times the standard error (SE) of the relevant statistic, which equals

$$z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \text{ for a one sample proportion.}$$

UNC-4.C.4

The formula for margin of error can be rearranged to solve for n , the minimum sample size needed to achieve a given margin of error. For this purpose, use a guess for $\hat{p}$ or use $\hat{p} = 0.5$ in order to find an upper bound for the sample size that will result in a given margin of error.

continued on next page

LEARNING OBJECTIVE

UNC-4.D

Calculate an appropriate confidence interval for a population proportion.

[Skill 3.D]

UNC-4.E

Calculate an interval estimate based on a confidence interval for a population proportion.

[Skill 3.D]

ESSENTIAL KNOWLEDGE

UNC-4.D.1

In general, an interval estimate can be constructed as point estimate $\pm$ (margin of error). For a one-sample proportion, the

interval estimate is $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

CLARIFYING STATEMENT:

Formulas for interval estimates do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

UNC-4.D.2

Critical values represent the boundaries encompassing the middle C% of the standard normal distribution, where C% is an approximate confidence level for a proportion.

UNC-4.E.1

Confidence intervals for population proportions can be used to calculate interval estimates with specified units.

TOPIC 6.3

Justifying a Claim Based on a Confidence Interval for a Population Proportion

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.F

Interpret a confidence interval for a population proportion.
[Skill 4.B]

ESSENTIAL KNOWLEDGE

UNC-4.F.1

A confidence interval for a population proportion either contains the population proportion or it does not, because each interval is based on random sample data, which varies from sample to sample.

UNC-4.F.2

We are C% confident that the confidence interval for a population proportion captures the population proportion.

UNC-4.F.3

In repeated random sampling with the same sample size, approximately C% of confidence intervals created will capture the population proportion.

UNC-4.F.4

Interpreting a confidence interval for a one-sample proportion should include a reference to the sample taken and details about the population it represents.

SKILLS



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.D

Justify a claim based on a confidence interval.

4.A

Make an appropriate claim or draw an appropriate conclusion.



AVAILABLE RESOURCE

- Classroom Resource > [Calculations Aren't Enough! The Importance of Communication in AP Statistics](#)

ILLUSTRATIVE EXAMPLE

UNC-4.F.4:

For interpreting a 99% confidence interval of (0.268, 0.292), based on the proportion of a nationally representative sample of twelfth-grade students who answered a particular multiple choice question correctly:

"We are 99 percent confident that the interval from 0.268 to 0.292 contains the population proportion of all United States twelfth-grade students who would answer this question correctly" (2011 FRQ 6(a)).

continued on next page

LEARNING OBJECTIVE

UNC-4.G

Justify a claim based on a confidence interval for a population proportion. [Skill 4.D]

UNC-4.H

Identify the relationships between sample size, width of a confidence interval, confidence level, and margin of error for a population proportion. [Skill 4.A]

ESSENTIAL KNOWLEDGE

UNC-4.G.1

A confidence interval for a population proportion provides an interval of values that may provide sufficient evidence to support a particular claim in context.

UNC-4.H.1

When all other things remain the same, the width of the confidence interval for a population proportion tends to decrease as the sample size increases. For a population proportion, the width of the interval is proportional to $\frac{1}{\sqrt{n}}$.

UNC-4.H.2

For a given sample, the width of the confidence interval for a population proportion increases as the confidence level increases.

UNC-4.H.3

The width of a confidence interval for a population proportion is exactly twice the margin of error.

TOPIC 6.4

Setting Up a Test for a Population Proportion

Required Course Content

ENDURING UNDERSTANDING

VAR-6

The normal distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-6.D

Identify the null and alternative hypotheses for a population proportion.

[Skill 1.F]

ESSENTIAL KNOWLEDGE

VAR-6.D.1

The null hypothesis is the situation that is assumed to be correct unless evidence suggests otherwise, and the alternative hypothesis is the situation for which evidence is being collected.

VAR-6.D.2

For hypotheses about parameters, the null hypothesis contains an equality reference ($=$, $\geq$, or $\leq$), while the alternative hypothesis contains a strict inequality ($<$, $>$, or $\neq$). The type of inequality in the alternative hypothesis is based on the question of interest. Alternative hypotheses with $<$ or $>$ are called one-sided, and alternative hypotheses with $\neq$ are called two-sided. Although the null hypothesis for a one-sided test may include an inequality symbol, it is still tested at the boundary of equality.

VAR-6.D.3

The null hypothesis for a population proportion is:

$H_0 : p = p_0$, where p_0 is the null hypothesized value for the population proportion.

VAR-6.D.4

A one-sided alternative hypothesis for a proportion is either $H_a : p < p_0$ or $H_a : p > p_0$. A two-sided alternate hypothesis is $H_a : p_1 \neq p_2$.

SKILLS

 *Selecting Statistical Methods*

1.F

Identify null and alternative hypotheses.

1.E

Identify an appropriate inference method for significance tests.



Statistical Argumentation

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCES

- Classroom Resources >
 - ♦ Inference
 - ♦ **Coke® Versus Pepsi®: An Introductory Activity for Test of Significance**

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LEARNING OBJECTIVE

VAR-6.D

Identify the null and alternative hypotheses for a population proportion.
[Skill 1.F]

VAR-6.E

Identify an appropriate testing method for a population proportion.
[Skill 1.E]

VAR-6.F

Verify the conditions for making statistical inferences when testing a population proportion.
[Skill 4.C]

ESSENTIAL KNOWLEDGE

VAR-6.D.5

For a one-sample z -test for a population proportion, the null hypothesis specifies a value for the population proportion, usually one indicating no difference or effect.

VAR-6.E.1

For a single categorical variable, the appropriate testing method for a population proportion is a one-sample z -test for a population proportion.

VAR-6.F.1

In order to make statistical inferences when testing a population proportion, we must check for independence and that the sampling distribution is approximately normal:

- a. To check for independence:
 - i. Data should be collected using a random sample or a randomized experiment.
 - ii. When sampling without replacement, check that $n \leq 10\%N$.
- b. To check that the sampling distribution of $\hat{p}$ is approximately normal (shape):
 - i. Assuming that H_0 is true ($p = p_0$), verify that both the number of successes, np_0 , and the number of failures, $n(1 - p_0)$ are at least 10 so that the sample size is large enough to support an assumption of normality.

TOPIC 6.5

Interpreting p -Values

SKILLS

 Using Probability and Simulation

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

Required Course Content

ENDURING UNDERSTANDING

VAR-6

The normal distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-6.G

Calculate an appropriate test statistic and p -value for a population proportion.

[Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-6.G.1

The distribution of the test statistic assuming the null hypothesis is true (null distribution) can be either a randomization distribution or when a probability model is assumed to be true, a theoretical distribution (z).

VAR-6.G.2

When using a z -test, the standardized test statistic can be written:

$$\text{test statistic} = \frac{\text{sample statistic} - \text{null value of the parameter}}{\text{standard deviation of the statistic}}$$

This is called a z -statistic for proportions.

VAR-6.G.3

The test statistic for a population proportion is:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

CLARIFYING STATEMENT:

The formulas for test statistics do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

continued on next page

LEARNING OBJECTIVE

VAR-6.G

Calculate an appropriate test statistic and p -value for a population proportion.
[Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-6.G.4

A p -value is the probability of obtaining a test statistic as extreme or more extreme than the observed test statistic when the null hypothesis and probability model are assumed to be true. The significance level may be given or determined by the researcher.

ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.A

Interpret the p -value of a significance test for a population proportion.
[Skill 4.B]

ESSENTIAL KNOWLEDGE

DAT-3.A.1

The p -value is the proportion of values for the null distribution that are as extreme or more extreme than the observed value of the test statistic. This is:

- The proportion at or above the observed value of the test statistic, if the alternative is $>$.
- The proportion at or below the observed value of the test statistic, if the alternative is $<$.
- The proportion less than or equal to the negative of the absolute value of the test statistic plus the proportion greater than or equal to the absolute value of the test statistic, if the alternative is $\neq$.

DAT-3.A.2

An interpretation of the p -value of a significance test for a one-sample proportion should recognize that the p -value is computed by assuming that the probability model and null hypothesis are true, i.e., by assuming that the true population proportion is equal to the particular value stated in the null hypothesis.

TOPIC 6.6

Concluding a Test for a Population Proportion

SKILL

Statistical
Argumentation

4.E

Justify a claim using
a decision based on
significance tests.

AVAILABLE RESOURCES

- Classroom Resource >
 - ♦ Inference
 - ♦ Calculations Aren't Enough! The Importance of Communication in AP Statistics

Required Course Content

ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.B

Justify a claim about the population based on the results of a significance test for a population proportion.

[Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.B.1

The significance level, α , is the predetermined probability of rejecting the null hypothesis given that it is true.

DAT-3.B.2

A formal decision explicitly compares the p -value to the significance level, α . If the p -value $\leq \alpha$, reject the null hypothesis. If the p -value $> \alpha$, fail to reject the null hypothesis.

DAT-3.B.3

Rejecting the null hypothesis means there is sufficient statistical evidence to support the alternative hypothesis. Failing to reject the null means there is insufficient statistical evidence to support the alternative hypothesis.

DAT-3.B.4

The conclusion about the alternative hypothesis must be stated in context.

DAT-3.B.5

A significance test can lead to rejecting or not rejecting the null hypothesis, but can never lead to concluding or proving that the null hypothesis is true. Lack of statistical evidence for the alternative hypothesis is not the same as evidence for the null hypothesis.

continued on next page

LEARNING OBJECTIVE

DAT-3.B

Justify a claim about the population based on the results of a significance test for a population proportion.

[Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.B.6

Small p -values indicate that the observed value of the test statistic would be unusual if the null hypothesis and probability model were true, and so provide evidence for the alternative. The lower the p -value, the more convincing the statistical evidence for the alternative hypothesis.

DAT-3.B.7

p -values that are not small indicate that the observed value of the test statistic would not be unusual if the null hypothesis and probability model were true, so do not provide convincing statistical evidence for the alternative hypothesis nor do they provide evidence that the null hypothesis is true.

DAT-3.B.8

A formal decision explicitly compares the p -value to the significance α . If the p -value $\leq \alpha$, then reject the null hypothesis, $H_0 : p = p_0$. If the p -value $> \alpha$, then fail to reject the null hypothesis.

DAT-3.B.9

The results of a significance test for a population proportion can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

TOPIC 6.7

Potential Errors When Performing Tests

Required Course Content

ENDURING UNDERSTANDING

UNC-5

Probabilities of Type I and Type II errors influence inference.

LEARNING OBJECTIVE

UNC-5.A

Identify Type I and Type II errors. [Skill 1.B]

ESSENTIAL KNOWLEDGE

UNC-5.A.1

A Type I error occurs when the null hypothesis is true and is rejected (false positive).

UNC-5.A.2

A Type II error occurs when the null hypothesis is false and is not rejected (false negative).

Table of Errors

		Actual Population Value	
		H_0 true	H_a true
Decision	Reject H_0	Type I Error	Correct Decision
	Fail to Reject H_0	Correct Decision	Type II Error

continued on next page

SKILLS

 *Selecting Statistical Methods*

1.B

Identify key and relevant information to answer a question or solve a problem.

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

 *Statistical Argumentation*

4.A

Make an appropriate claim or draw an appropriate conclusion.

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

LEARNING OBJECTIVE

UNC-5.B

Calculate the probability of a Type I and Type II errors. [Skill 3.A]

UNC-5.C

Identify factors that affect the probability of errors in significance testing. [Skill 4.A]

UNC-5.D

Interpret Type I and Type II errors. [Skill 4.B]

ESSENTIAL KNOWLEDGE

UNC-5.B.1

The significance level, α , is the probability of making a Type I error, if the null hypothesis is true.

UNC-5.B.2

The power of a test is the probability that a test will correctly reject a false null hypothesis.

UNC-5.B.3

The probability of making a Type II error $= 1 - \text{power}$.

UNC-5.C.1

The probability of a Type II error decreases when any of the following occurs, provided the others do not change:

- Sample size(s) increases.
- Significance level (α) of a test increases.
- Standard error decreases.
- True parameter value is farther from the null.

UNC-5.D.1

Whether a Type I or a Type II error is more consequential depends upon the situation.

UNC-5.D.2

Since the significance level, α , is the probability of a Type I error, the consequences of a Type I error influence decisions about a significance level.

TOPIC 6.8

Confidence Intervals for the Difference of Two Proportions

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.I

Identify an appropriate confidence interval procedure for a comparison of population proportions.

[Skill 1.D]

UNC-4.J

Verify the conditions for calculating confidence intervals for a difference between population proportions.

[Skill 4.C]

ESSENTIAL KNOWLEDGE

UNC-4.I.1

The appropriate confidence interval procedure for a two-sample comparison of proportions for one categorical variable is a two-sample z-interval for a difference between population proportions.

UNC-4.J.1

In order to calculate confidence intervals to estimate a difference between proportions, we must check for independence and that the sampling distribution is approximately normal:

- To check for independence:
 - Data should be collected using two independent, random samples or a randomized experiment.
 - When sampling without replacement, check that $n_1 \leq 10\%N_1$ and $n_2 \leq 10\%N_2$.
- To check that sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal (shape).
 - For categorical variables, check that $n_1\hat{p}_1$, $n_1(1 - \hat{p}_1)$, $n_2\hat{p}_2$, and $n_2(1 - \hat{p}_2)$ are all greater than or equal to some predetermined value, typically either 5 or 10.

SKILLS

 *Selecting Statistical Methods*

1.D

Identify an appropriate inference method for confidence intervals.



Statistical Argumentation

4.C

Verify that inference procedures apply in a given situation.



Using Probability and Simulation

3.D

Construct a confidence interval, provided conditions for inference are met.

continued on next page

LEARNING OBJECTIVE

UNC-4.K

Calculate an appropriate confidence interval for a comparison of population proportions.

[Skill 3.D]

UNC-4.L

Calculate an interval estimate based on a confidence interval for a difference of proportions. **[Skill 3.D]**

ESSENTIAL KNOWLEDGE

UNC-4.K.1

For a comparison of proportions, the interval estimate is

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

CLARIFYING STATEMENT:

Formulas for interval estimates do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

UNC-4.L.1

Confidence intervals for a difference in proportions can be used to calculate interval estimates with specified units.

TOPIC 6.9

Justifying a Claim Based on a Confidence Interval for a Difference of Population Proportions

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.M

Interpret a confidence interval for a difference of proportions. [Skill 4.B]

UNC-4.N

Justify a claim based on a confidence interval for a difference of proportions. [Skill 4.D]

ESSENTIAL KNOWLEDGE

UNC-4.M.1

In repeated random sampling with the same sample size, approximately $C\%$ of confidence intervals created will capture the difference in population proportions.

UNC-4.M.2

Interpreting a confidence interval for difference between population proportions should include a reference to the sample taken and details about the population it represents.

UNC-4.N.1

A confidence interval for difference in population proportions provides an interval of values that may provide sufficient evidence to support a particular claim in context.

SKILLS



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.D

Justify a claim based on a confidence interval.



AVAILABLE RESOURCE

- Classroom Resource > [Calculations Aren't Enough! The Importance of Communication in AP Statistics](#)

SKILLS

 *Selecting Statistical Methods*

1.F

Identify null and alternative hypotheses.

1.E

Identify an appropriate inference method for significance tests.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 6.10

Setting Up a Test for the Difference of Two Population Proportion

Required Course Content

ENDURING UNDERSTANDING

VAR-6

The normal distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-6.H

Identify the null and alternative hypotheses for a difference of two population proportions.

[Skill 1.F]

ESSENTIAL KNOWLEDGE

VAR-6.H.1

For a two-sample test for a difference of two proportions, the null hypothesis specifies a value of 0 for the difference in population proportions, indicating no difference or effect.

VAR-6.H.2

The null hypothesis for a difference in proportions is: $H_0 : p_1 = p_2$, or $H_0 : p_1 - p_2 = 0$.

VAR-6.H.3

A one-sided alternative hypothesis for a difference in proportions is $H_a : p_1 < p_2$, or, $H_a : p_1 > p_2$. A two-sided alternative hypothesis for a difference of proportions is $H_a : p_1 \neq p_2$.

VAR-6.I

Identify an appropriate testing method for the difference of two population proportions.

[Skill 1.E]

VAR-6.I.1

For a single categorical variable, the appropriate testing method for the difference of two population proportions is a two-sample z-test for a difference between two population proportions.

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LEARNING OBJECTIVE

VAR-6.J

Verify the conditions for making statistical inferences when testing a difference of two population proportions.

[Skill 4.C]

ESSENTIAL KNOWLEDGE

VAR-6.J.1

In order to make statistical inferences when testing a difference between population proportions, we must check for independence and that the sampling distribution is approximately normal:

- a. To check for independence:
 - i. Data should be collected using two independent, random samples or a randomized experiment.
 - ii. When sampling without replacement, check that $n_1 \leq 10\%N_1$ and $n_2 \leq 10\%N_2$.

- b. To check that the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal (shape):

- i. For the combined sample, define the combined (or pooled) proportion,

$$\hat{p}_c = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}.$$

Assuming that H_0 is true ($p_1 - p_2 = 0$ or $p_1 = p_2$), check that

$$n_1\hat{p}_c, n_1(1 - \hat{p}_c), n_2\hat{p}_c, \text{ and } n_2(1 - \hat{p}_c)$$

are all greater than or equal to some predetermined value, typically either 5 or 10.

SKILLS

 *Using Probability and Simulation*

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 6.11

Carrying Out a Test for the Difference of Two Population Proportions

Required Course Content

ENDURING UNDERSTANDING

VAR-6

The normal distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-6.K

Calculate an appropriate test statistic for the difference of two population proportions. **[Skill 3.E]**

ESSENTIAL KNOWLEDGE

VAR-6.K.1

The test statistic for a difference in proportions is:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where } \hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

CLARIFYING STATEMENT:

The formulas for test statistics do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the standard error formulas for each of the relevant test statistics that are provided on the formula sheet.

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ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.C

Interpret the p -value of a significance test for a difference of population proportions.

[Skill 4.B]

DAT-3.D

Justify a claim about the population based on the results of a significance test for a difference of population proportions.

[Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.C.1

An interpretation of the p -value of a significance test for a difference of two population proportions should recognize that the p -value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population proportions are equal to each other.

DAT-3.D.1

A formal decision explicitly compares the p -value to the significance α . If the p -value $\leq \alpha$, then reject the null hypothesis, $H_0 : p_1 = p_2$, or $H_0 : p_1 - p_2 = 0$. If the p -value $> \alpha$, then fail to reject the null hypothesis.

DAT-3.D.2

The results of a significance test for a difference of two population proportions can serve as the statistical reasoning to support the answer to a research question about the two populations that were sampled.

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AP STATISTICS

UNIT 7

Inference for Quantitative Data: Means



10–18%

AP EXAM WEIGHTING



~14–16

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 7

Multiple-choice: ~50 questions

Free-response: 2 questions

- Inference and Collecting Data
- Investigative Task

Inference for Quantitative Data: Means



Developing Understanding

BIG IDEA 1 Variation and Distribution **VAR**

- How do we know whether to use a t -test or a z -test for inference with means?

BIG IDEA 2 Patterns and Uncertainty **UNC**

- How can we make sure that samples are independent?

BIG IDEA 3 Data-Based Predictions, Decisions, and Conclusions **DAT**

- Why is it inappropriate to accept a hypothesis as true based on the results of statistical inference testing?

In this unit, students will analyze quantitative data to make inferences about population means. Students should understand that t^* and t -tests are used for inference with means when the population standard deviation, σ , is not known. Using s for σ in the formula for z gives a slightly different value, t , whose distribution, which depends on sample size, has more area in the tails than a normal distribution. The boundaries for rejecting a null hypothesis using a t -distribution tend to be further from the mean than for a normal distribution. Students should understand how and why conditions for inference with proportions and means are similar and different.

Building Course Skills

1.E 1.F 4.C 4.E

Unit 7 focuses on means, which has many similarities to the conditions and procedures for proportions. Since students sometimes confuse t -tests with z -tests, it will help to review the underlying rationales each time conditions come up. This will help students develop understanding through repeated practice in new situations. Teachers can encourage students to be mindful of notation and use the formula sheet as a reference.

Teachers can reinforce that inference testing requires careful selection of a procedure based on specific conditions for a given problem. Common errors include mislabeling conditions (e.g., incorrectly associating the large sample condition with independence), relying upon vague references to the normal distribution, or applying an inappropriate large sample condition. The null and alternative hypotheses must be clearly stated in terms of population parameters, not sample statistics. A formal decision compares the p -value to the level of significance. Students should also practice providing a

numerical reference to support their claim (e.g., “Because $p < 0.05$, we reject the null hypothesis.”) and interpreting findings within the context of the question.

Preparing for the AP Exam

It is critical for students to recognize that free-response questions asking whether data provide convincing evidence of some finding are asking for a significance test, not just a descriptive analysis. When using statistical inference for significance tests, students should identify the correct parameter and hypotheses, identify an appropriate test procedure and check conditions, calculate a test statistic and p -value, and provide a conclusion in context, along with a justification based on linkage between the p -value and the conclusion. For inference with means, the appropriate test will often be a t -test, but if σ is known, a z -test would be appropriate (see **2018 FRQ 6(a)**). For a t -test, conditions are (1) random sample and (2) large sample (e.g., $n > 30$). When sampling without replacement, students should also verify that the sample size is at most 10% of the population.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~14–16 CLASS PERIODS
VAR-1	7.1 Introducing Statistics: Should I Worry About Error?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
VAR-7, UNC-4	7.2 Constructing a Confidence Interval for a Population Mean	3.C Describe probability distributions. 1.D Identify an appropriate inference method for confidence intervals. 4.C Verify that inference procedures apply in a given situation. 3.D Construct a confidence interval, provided conditions for inference are met.	
UNC-4	7.3 Justifying a Claim About a Population Mean Based on a Confidence Interval	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.D Justify a claim based on a confidence interval. 4.A Make an appropriate claim or draw an appropriate conclusion.	
VAR-7	7.4 Setting Up a Test for a Population Mean	1.E Identify an appropriate inference method for significance tests. 1.F Identify null and alternative hypotheses. 4.C Verify that inference procedures apply in a given situation.	
VAR-7, DAT-3	7.5 Carrying Out a Test for a Population Mean	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	

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UNIT AT A GLANCE *(cont'd)*

Enduring Understanding	Topic	Skills	Class Periods
			~14–16 CLASS PERIODS
UNC-4	7.6 Confidence Intervals for the Difference of Two Means	1.D Identify an appropriate inference method for confidence intervals. 4.C Verify that inference procedures apply in a given situation. 3.D Construct a confidence interval, provided conditions for inference are met.	
	7.7 Justifying a Claim About the Difference of Two Means Based on a Confidence Interval	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.D Justify a claim based on a confidence interval. 4.A Make an appropriate claim or draw an appropriate conclusion.	
VAR-7	7.8 Setting Up a Test for the Difference of Two Population Means	1.E Identify an appropriate inference method for significance tests. 1.F Identify null and alternative hypotheses. 4.C Verify that inference procedures apply in a given situation.	
VAR-7, DAT-3	7.9 Carrying Out a Test for the Difference of Two Population Means	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	
	7.10 Skills Focus: Selecting, Implementing, and Communicating Inference Procedures	N/A	
 Go to AP Classroom to assign the Personal Progress Check for Unit 7. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	7.2	Predict and Confirm After introducing the confidence interval formula for a population mean when sigma is known, $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$, have students discuss in small groups what will happen if we substitute the sample standard deviation s into the formula when σ is unknown. Have students use the Simulating Confidence Intervals for Population Parameter applet to test their conjectures (see link for Rossman/Chance Applets on page 210).
2	7.2 7.3 7.5	Team Challenge Give each team of three to four students a copy of 2004 FRQ 6 , which focuses on the connection between a one-sample t -interval, a one-sample t -test, and the unfamiliar concept of a one-sided confidence interval. Challenge teams to collaboratively produce a model solution in 30 minutes.
3	7.4	Discussion Groups Ask each group of three to four students to identify the conditions for performing a test about a population mean. For each condition, have them explain why the condition is required and what would go wrong with the test if the condition were violated. Have groups pair up and compare answers.
4	7.9	Team FRQ Give each team of four students copies of a free-response question that involves performing a two-sample t -test (e.g., 2011 FRQ 4). Have each team member take responsibility for writing one part of the model solution (hypotheses, procedure and conditions, calculations, conclusion) with group input.
5	7.10	Graphic Organizer Have students work in teams of two to three to develop a flowchart for determining which inference procedure from Units 6 and 7 to use in a given setting.

TOPIC 7.1

Introducing Statistics: Why Should I Worry About Error?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.I

Identify questions suggested by probabilities of errors in statistical inference. **[Skill 1.A]**

ESSENTIAL KNOWLEDGE

VAR-1.I.1

Random variation may result in errors in statistical inference.

SKILL

Selecting Statistical Methods

1.A

Identify the question to be answered or problem to be solved.

SKILLS

 *Using Probability and Simulation*

3.C

Describe probability distributions.

3.D

Construct a confidence interval, provided conditions for inference are met.

 *Selecting Statistical Methods*

1.D

Identify an appropriate inference method for confidence intervals.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.

TOPIC 7.2

Constructing a Confidence Interval for a Population Mean

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.A

Describe t -distributions.
[Skill 3.C]

ESSENTIAL KNOWLEDGE

VAR-7.A.1

When s is used instead of σ to calculate a test statistic, the corresponding distribution, known as the t -distribution, varies from the normal distribution in shape, in that more of the area is allocated to the tails of the density curve than in a normal distribution.

VAR-7.A.2

As the degrees of freedom increase, the area in the tails of a t -distribution decreases.

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ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.O

Identify an appropriate confidence interval procedure for a population mean, including the mean difference between values in matched pairs. [Skill 1.D]

UNC-4.P

Verify the conditions for calculating confidence intervals for a population mean, including the mean difference between values in matched pairs. [Skill 4.C]

ESSENTIAL KNOWLEDGE

UNC-4.O.1

Because σ is typically not known for distributions of quantitative variables, the appropriate confidence interval procedure for estimating the population mean of one quantitative variable for one sample is a one-sample t -interval for a mean.

UNC-4.O.2

For one quantitative variable, X , that is normally distributed, the distribution of $t = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}}$ is a t -distribution with $n - 1$ degrees of freedom.

UNC-4.O.3

Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for confidence intervals proceeds as for a population mean.

UNC-4.P.1

In order to calculate confidence intervals to estimate a population mean, we must check for independence and that the sampling distribution is approximately normal:

- a. To check for independence:
 - i. Data should be collected using a random sample or a randomized experiment.
 - ii. When sampling without replacement, check that $n \leq 10\%N$, where N is the size of the population.
- b. To check that the sampling distribution of $\bar{x}$ is approximately normal (shape):
 - i. If the observed distribution is skewed, n should be greater than 30.
 - ii. If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

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LEARNING OBJECTIVE

UNC-4.Q

Determine the margin of error for a given sample size for a one-sample t -interval.

[Skill 3.D]

UNC-4.R

Calculate an appropriate confidence interval for a population mean, including the mean difference between values in matched pairs.

[Skill 3.D]

ESSENTIAL KNOWLEDGE

UNC-4.Q.1

The critical value t^* with $n - 1$ degrees of freedom can be found using a table or computer-generated output.

UNC-4.Q.2

The standard error for a sample mean is given by $SE = \frac{s}{\sqrt{n}}$, where s is the sample standard deviation.

UNC-4.Q.3

For a one-sample t -interval for a mean, the margin of error is the critical value (t^*) times the standard error (SE), which equals $t^* \left(\frac{s}{\sqrt{n}} \right)$.

UNC-4.R.1

The point estimate for a population mean is the sample mean, $\bar{x}$.

UNC-4.R.2

For the population mean for one sample with unknown population standard deviation, the confidence interval is $\bar{x} \pm t^* \frac{s}{\sqrt{n}}$.

CLARIFYING STATEMENT:

Formulas for interval estimates do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

TOPIC 7.3

Justifying a Claim About a Population Mean Based on a Confidence Interval

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.S

Interpret a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.B]

UNC-4.T

Justify a claim based on a confidence interval for a population mean, including the mean difference between values in matched pairs. [Skill 4.D]

ESSENTIAL KNOWLEDGE

UNC-4.S.1

A confidence interval for a population mean either contains the population mean or it does not, because each interval is based on data from a random sample, which varies from sample to sample.

UNC-4.S.2

We are C% confident that the confidence interval for a population mean captures the population mean.

UNC-4.S.3

An interpretation of a confidence interval for a population mean includes a reference to the sample taken and details about the population it represents.

UNC-4.T.1

A confidence interval for a population mean provides an interval of values that may provide sufficient evidence to support a particular claim in context.

SKILLS



Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.D

Justify a claim based on a confidence interval.

4.A

Make an appropriate claim or draw an appropriate conclusion.



ILLUSTRATIVE EXAMPLE

UNC-4.S.3:

For interpreting a 96% confidence interval for mean foot length for all footprints found in a cave based on a particular randomly selected sample of footprints in the cave:

"We are 96% confident that the mean foot length for all footprints found in the cave falls within the confidence interval" (based on [2000 FRQ 2](#)).

continued on next page

LEARNING OBJECTIVE**UNC-4.U**

Identify the relationships between sample size, width of a confidence interval, confidence level, and margin of error for a population mean.

[Skill 4.A]

ESSENTIAL KNOWLEDGE**UNC-4.U.1**

When all other things remain the same, the width of a confidence interval for a population mean tends to decrease as the sample size increases.

UNC-4.U.2

For a single mean, the width of the interval is proportional to $\frac{1}{\sqrt{n}}$.

UNC-4.U.3

For a given sample, the width of the confidence interval for a population mean increases as the confidence level increases.

TOPIC 7.4

Setting Up a Test for a Population Mean

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.B

Identify an appropriate testing method for a population mean with unknown σ , including the mean difference between values in matched pairs.

[Skill 1.E]

VAR-7.C

Identify the null and alternative hypotheses for a population mean with unknown σ , including the mean difference between values in matched pairs.

[Skill 1.F]

ESSENTIAL KNOWLEDGE

VAR-7.B.1

The appropriate test for a population mean with unknown σ is a one-sample t -test for a population mean.

VAR-7.B.2

Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for significance testing proceeds as for a population mean.

VAR-7.C.1

The null hypothesis for a one-sample t -test for a population mean is $H_0: \mu = \mu_0$, where μ_0 is the hypothesized value. Depending upon the situation, the alternative hypothesis is $H_a: \mu < \mu_0$, or $H_a: \mu > \mu_0$, or $H_a: \mu \neq \mu_0$.

VAR-7.C.2

When finding the mean difference, μ_d , between values in a matched pair, it is important to define the order of subtraction.

SKILLS

 *Selecting Statistical Methods*

1.E

Identify an appropriate inference method for significance tests.

1.F

Identify null and alternative hypotheses.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

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LEARNING OBJECTIVE

VAR-7.D

Verify the conditions for the test for a population mean, including the mean difference between values in matched pairs. **[Skill 4.C]**

ESSENTIAL KNOWLEDGE

VAR-7.D.1

In order to make statistical inferences when testing a population mean, we must check for independence and that the sampling distribution is approximately normal:

- a. To check for independence:
 - i. Data should be collected using a random sample or a randomized experiment.
 - ii. When sampling without replacement, check that $n \leq 10\%N$.
- b. To check that the sampling distribution of $\bar{x}$ is approximately normal (shape):
 - i. If the observed distribution is skewed, n should be greater than 30.
 - ii. If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

TOPIC 7.5

Carrying Out a Test
for a Population Mean

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.E

Calculate an appropriate test statistic for a population mean, including the mean difference between values in matched pairs. [Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-7.E.1

For a single quantitative variable when random sampling with replacement from a population that can be modeled with a normal distribution with mean μ and standard deviation σ , the sampling distribution of $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ has a

t -distribution with $n - 1$ degrees of freedom.

CLARIFYING STATEMENT:

The formulas for test statistics do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

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SKILLS

 Using Probability and Simulation

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.E

Interpret the p -value of a significance test for a population mean, including the mean difference between values in matched pairs.

[Skill 4.B]

DAT-3.F

Justify a claim about the population based on the results of a significance test for a population mean.

[Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.E.1

An interpretation of the p -value of a significance test for a population mean should recognize that the p -value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population mean is equal to the particular value stated in the null hypothesis.

DAT-3.F.1

A formal decision explicitly compares the p -value to the significance α . If the p -value $\leq \alpha$, then reject the null hypothesis, $H_0 : \mu = \mu_0$. If the p -value $> \alpha$, then fail to reject the null hypothesis.

DAT-3.F.2

The results of a significance test for a population mean can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

TOPIC 7.6

Confidence Intervals for the Difference of Two Means

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

Identify an appropriate confidence interval procedure for a difference of two population means.

[Skill 1.D]

ESSENTIAL KNOWLEDGE

UNC-4.V.1

Consider a simple random sample from population 1 of size n_1 , mean μ_1 , and standard deviation σ_1 and a second simple random sample from population 2 of size n_2 , mean μ_2 , and standard deviation σ_2 . If the distributions of populations 1 and 2 are normal or if both n_1 and n_2 are greater than 30, then the sampling distribution of the difference of means, $\bar{x}_1 - \bar{x}_2$, is also normal. The mean for the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is $\mu_1 - \mu_2$. The standard

deviation of $\bar{x}_1 - \bar{x}_2$ is $\sqrt{\frac{(\sigma_1)^2}{n_1} + \frac{(\sigma_2)^2}{n_2}}$.

UNC-4.V.2

The appropriate confidence interval procedure for one quantitative variable for two independent samples is a two-sample t -interval for a difference between population means.

SKILLS

 *Selecting Statistical Methods*

1.D

Identify an appropriate inference method for confidence intervals.



Statistical Argumentation

4.C

Verify that inference procedures apply in a given situation.



Using Probability and Simulation

3.D

Construct a confidence interval, provided conditions for inference are met.

continued on next page

LEARNING OBJECTIVE

UNC-4.W

Verify the conditions to calculate confidence intervals for the difference of two population means.

[Skill 4.C]

UNC-4.X

Determine the margin of error for the difference of two population means. **[Skill 3.D]**

UNC-4.Y

Calculate an appropriate confidence interval for a difference of two population means. **[Skill 3.D]**

ESSENTIAL KNOWLEDGE

UNC-4.W.1

In order to calculate confidence intervals to estimate a difference of population means, we must check for independence and that the sampling distribution is approximately normal:

- To check for independence:
 - Data should be collected using two independent, random samples or a randomized experiment.
 - When sampling without replacement, check that $n_1 \leq 10\%N_1$ and $n_2 \leq 10\%N_2$.
- To check that the sampling distribution of $(\bar{x}_1 - \bar{x}_2)$ should be approximately normal (shape):
 - If the observed distributions are skewed, both n_1 and n_2 should be greater than 30.

UNC-4.X.1

For the difference of two sample means, the margin of error is the critical value (t^*) times the standard error (SE) of the difference of two means.

UNC-4.X.2

The standard error for the difference in two sample means with sample standard

deviations, s_1 and s_2 , is $\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}$.

UNC-4.Y.1

The point estimate for the difference of two population means is the difference in sample means, $\bar{x}_1 - \bar{x}_2$.

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LEARNING OBJECTIVE

UNC-4.Y

Calculate an appropriate confidence interval for a difference of two population means. [Skill 3.D]

ESSENTIAL KNOWLEDGE

UNC-4.Y.2

For a difference of two population means where the population standard deviations are not known, the confidence interval is

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \text{ where } \pm t^* \text{ are the critical}$$

values for the central C% of a t -distribution with appropriate degrees of freedom that can be found using technology.

CLARIFYING STATEMENT:

Formulas for interval estimates do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the relevant standard error formulas that are provided on the formula sheet.

SKILLS

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.D

Justify a claim based on a confidence interval.

4.A

Make an appropriate claim or draw an appropriate conclusion.



ILLUSTRATIVE EXAMPLE

UNC-4.Z.2:

For interpreting a confidence interval for a difference between mean response times for two fire stations (northern – southern): “Based on these samples, one can be 95 percent confident that the difference in the population mean response times (northern – southern) is between –2.37 minutes and 0.37 minutes” (**2009 FRQ 4**).

TOPIC 7.7

Justifying a Claim About the Difference of Two Means Based on a Confidence Interval

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.Z

Interpret a confidence interval for a difference of population means. **[Skill 4.B]**

ESSENTIAL KNOWLEDGE

UNC-4.Z.1

In repeated random sampling with the same sample size, approximately C% of confidence intervals created will capture the difference of population means.

UNC-4.Z.2

An interpretation for a confidence interval for the difference of two population means should include a reference to the samples taken and details about the populations they represent.

UNC-4.AA

Justify a claim based on a confidence interval for a difference of population means. **[Skill 4.D]**

UNC-4.AA.1

A confidence interval for a difference of population means provides an interval of values that may provide sufficient evidence to support a particular claim in context.

UNC-4.AB

Identify the effects of sample size on the width of a confidence interval for the difference of two means. **[Skill 4.A]**

UNC-4.AB.1

When all other things remain the same, the width of the confidence interval for the difference of two means tends to decrease as the sample sizes increase.

TOPIC 7.8

Setting Up a Test for the Difference of Two Population Means

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.F

Identify an appropriate selection of a testing method for a difference of two population means. **[Skill 1.E]**

VAR-7.G

Identify the null and alternative hypotheses for a difference of two population means. **[Skill 1.F]**

ESSENTIAL KNOWLEDGE

VAR-7.F.1

For a quantitative variable, the appropriate test for a difference of two population means is a two-sample t -test for a difference of two population means.

VAR-7.G.1

The null hypothesis for a two-sample t -test for a difference of two population means, μ_1 and μ_2 , is: $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$. The alternative hypothesis is $H_a: \mu_1 - \mu_2 < 0$, or $H_a: \mu_1 - \mu_2 > 0$, or $H_a: \mu_1 - \mu_2 \neq 0$, or $H_a: \mu_1 > \mu_2$, or $H_a: \mu_1 < \mu_2$, or $H_a: \mu_1 \neq \mu_2$.

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SKILLS

 *Selecting Statistical Methods*

1.E

Identify an appropriate inference method for significance tests.

1.F

Identify null and alternative hypotheses.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.

LEARNING OBJECTIVE

VAR-7.H

Verify the conditions for the significance test for the difference of two population means. [Skill 4.C]

ESSENTIAL KNOWLEDGE

VAR-7.H.1

In order to make statistical inferences when testing a difference between population means, we must check for independence and that the sampling distribution is approximately normal:

- a. Individual observations should be independent:
 - i. Data should be collected using simple random samples or a randomized experiment.
 - ii. When sampling without replacement, check that $n_1 \leq 10\%N_1$ and $n_2 \leq 10\%N_2$.
- b. The sampling distribution of $\bar{x}_1 - \bar{x}_2$ should be approximately normal (shape).
 - i. If the observed distribution is skewed, both n_1 and n_2 should be greater than 30.
 - ii. If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers. This should be checked for BOTH samples.

TOPIC 7.9

Carrying Out a Test for the Difference of Two Population Means

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.I

Calculate an appropriate test statistic for a difference of two means. [Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-7.I.1

For a single quantitative variable, data collected using independent random samples or a randomized experiment from two populations, each of which can be modeled with a normal distribution, the sampling distribution of

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

is an approximate

t -distribution with degrees of freedom that can be found using technology. The degrees of freedom fall between the smaller of $n_1 - 1$ and $n_2 - 1$ and $n_1 + n_2 - 2$.

CLARIFYING STATEMENT:

The formulas for test statistics do not appear explicitly on the AP Statistics Formula Sheet provided with the AP Statistics Exam. However, these formulas do not need to be memorized, as they can be constructed based on the general test statistic formula and the standard error formulas for each of the relevant test statistics that are provided on the formula sheet.

SKILLS

 *Using Probability and Simulation*

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

ILLUSTRATIVE EXAMPLE

VAR-7.I.1:

In a study comparing mean recovery times for two surgical procedures to repair a torn anterior cruciate ligament (ACL), the group receiving one procedure had a sample size of 110, while the group receiving the other procedure had a sample size of 100. The degrees of freedom fall between 100 (the smaller of 110 and 100) and 208 ($110 + 100 - 2$). The degrees of freedom may be determined using technology. If the test statistic for this study is $t \approx 7.13$, then the p -value is the area greater than 7.13 for a t -distribution with $df = 207.18$ (2018 FRQ 4).

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ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.G

Interpret the p -value of a significance test for a difference of population means. [Skill 4.B]

DAT-3.H

Justify a claim about the population based on the results of a significance test for a difference of two population means in context. [Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.G.1

An interpretation of the p -value of a significance test for a two-sample difference of population means should recognize that the p -value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population means are equal to each other.

DAT-3.H.1

A formal decision explicitly compares the p -value to the significance α . If the p -value $\leq \alpha$, then reject the null hypothesis, $H_0 : \mu_1 - \mu_2 = 0$, or $H_0 : \mu_1 = \mu_2$. If the p -value $> \alpha$, then fail to reject the null hypothesis.

DAT-3.H.2

The results of a significance test for a two-sample test for a difference between two population means can serve as the statistical reasoning to support the answer to a research question about the populations that were sampled.

TOPIC 7.10

Skills Focus: Selecting, Implementing, and Communicating Inference Procedures

Required Course Content

This topic is intended to focus on the skill of selecting an appropriate inference procedure, now that students have a range of options. Students should be given opportunities to practice when and how to apply all learning objectives relating to inference involving proportions or means.

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AP STATISTICS

UNIT 8

Inference for Categorical Data: Chi-Square



2–5%

AP EXAM WEIGHTING



~10–11

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 8

Multiple-choice: ~30 questions

Free-response: 2 questions

- Inference
- Inference and Exploring Data/
Collecting Data

Inference for Categorical Data: Chi-Square



Developing Understanding

BIG IDEA 1 Variation and Distribution **VAR**

- How does increasing the degrees of freedom influence the shape of the chi-square distribution?

BIG IDEA 3 Data-Based Predictions, Decisions, and Conclusions **DAT**

- Why is it inappropriate to use statistical inference to justify a claim that there is *no association* between variables?

Unit 6 introduced inference for proportions of categorical data. Unit 8 introduces chi-square tests, which can be used when there are two or more categories. Students need to understand how to select from the following tests: the chi-square test for goodness of fit (for a distribution of proportions of one categorical variable in a population), the chi-square test for independence (for associations between categorical variables within a single population), or the chi-square test for homogeneity (for comparing distributions of a categorical variable across populations or treatments). To integrate conceptual understanding, teachers can make connections between frequency tables, conditional probability, and calculating expected counts. The chi-square statistic is introduced to measure the distance between observed and expected counts relative to expected counts.

Building Course Skills

1.E 3.E 4.C 4.E

In Unit 8, students should continue applying the same problem-solving structure to chi-square significance testing: State the hypotheses in words, explicitly identify the correct procedure, verify conditions, calculate the test statistic and the p -value, and then draw a conclusion in context that is directly linked to the p -value. Students should have opportunities to practice the distinctive elements for each type of chi-square test, such as analysis of expected counts, degrees of freedom, verbally stated hypotheses, and two-way tables.

When the p -value is large, drawing an appropriate conclusion is challenging for students. Saying there is “no association” between two variables is equivalent to incorrectly “accepting the null hypothesis.” Instead, teachers can teach students to use nondeterministic language in their conclusions, that is, “The data do not provide strong enough evidence to conclude that the variables are associated.” Students should have frequent opportunities to practice writing, with detailed feedback to help them improve.

Preparing for the AP Exam

When writing hypotheses, students should refer to the population, using language from the question. For example, “The null hypothesis is that the age group at diagnosis and gender are independent (i.e., they are not associated) for the population of people currently being treated for schizophrenia” (see [Scoring Guidelines for 2017 FRQ 5](#)). As always, students should name the test and provide evidence verifying appropriate conditions. For chi-square tests, the conditions are (1) random selection or randomized experiment and (2) large counts. Students should be sure to say that all expected counts (rather than actual counts) are at least 5. Students need to clearly present calculations and state the conclusion in context with linkage to p -values. Students should avoid tacitly accepting the null hypothesis. If the p -value is greater than conventional significance levels, the correct conclusion of a chi-square test for independence would be that there is insufficient evidence that there is an association.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~10–11 CLASS PERIODS
VAR-1	8.1 Introducing Statistics: Are My Results Unexpected?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
VAR-8	8.2 Setting Up a Chi-Square Goodness of Fit Test	3.C Describe probability distributions. 1.F Identify null and alternative hypotheses. 1.E Identify an appropriate inference method for significance tests. 3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations. 4.C Verify that inference procedures apply in a given situation.	
VAR-8, DAT-3	8.3 Carrying Out a Chi-Square Test for Goodness of Fit	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	
VAR-8	8.4 Expected Counts in Two-Way Tables	3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	
	8.5 Setting Up a Chi-Square Test for Homogeneity or Independence	1.F Identify null and alternative hypotheses. 1.E Identify an appropriate inference method for significance tests. 4.C Verify that inference procedures apply in a given situation.	
VAR-8, DAT-3	8.6 Carrying Out a Chi-Square Test for Homogeneity or Independence	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	
	8.7 Skills Focus: Selecting an Appropriate Inference Procedure for Categorical Data	N/A	
 Go to AP Classroom to assign the Personal Progress Check for Unit 8. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity
1	8.1	Simulation Prepare several bags with an identical mix of at least 250 chips or beads of three colors in different proportions (e.g., red = 0.5, white = 0.3, blue = 0.2). Have each student take a random sample of 25 chips or beads from the bag, calculate $\sum \frac{(Observed\ count - Expected\ count)^2}{Expected\ count}$, and plot their value on a class dotplot. Use this graph to introduce the chi-square distribution with $df = 2$.
2	8.5	Discussion Groups Give each group of three to four students an example of a chi-square test involving a two-way table. Have students work together to state appropriate hypotheses, describe a Type 1 and Type 2 error in context, and give a possible consequence of each of those errors.
3	8.7	Graphic Organizer Have students work in teams of two to three to develop a chart that summarizes the three types of chi-square tests, including when each is appropriate, as well as the hypotheses, conditions, and degrees of freedom.

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

TOPIC 8.1

Introducing Statistics: Are My Results Unexpected?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.J

Identify questions suggested by variation between observed and expected counts in categorical data.

[Skill 1.A]

ESSENTIAL KNOWLEDGE

VAR-1.J.1

Variation between what we find and what we expect to find may be random or not.

TOPIC 8.2

Setting Up a Chi-Square Goodness of Fit Test

Required Course Content

ENDURING UNDERSTANDING

VAR-8

The chi-square distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-8.A

Describe chi-square distributions. [Skill 3.C]

VAR-8.B

Identify the null and alternative hypotheses in a test for a distribution of proportions in a set of categorical data. [Skill 1.F]

ESSENTIAL KNOWLEDGE

Expected counts of categorical data are counts consistent with the null hypothesis. In general, an expected count is a sample size times a probability.

The chi-square statistic measures the distance between observed and expected counts relative to expected counts.

Chi-square distributions have positive values and are skewed right. Within a family of density curves, the skew becomes less pronounced with increasing degrees of freedom.

For a chi-square goodness-of-fit test, the null hypothesis specifies null proportions for each category, and the alternative hypothesis is that at least one of these proportions is not as specified in the null hypothesis.

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SKILLS

 Using Probability and Simulation

3.C

Describe probability distributions.

 Selecting Statistical Methods

1.F

Identify null and alternative hypotheses.

1.E

Identify an appropriate inference method for significance tests.

 Using Probability and Simulation

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

 Statistical Argumentation

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

LEARNING OBJECTIVE

VAR-8.C

Identify an appropriate testing method for a distribution of proportions in a set of categorical data.

[Skill 1.E]

VAR-8.D

Calculate expected counts for the chi-square test for goodness of fit. [Skill 3.A]

VAR-8.E

Verify the conditions for making statistical inferences when testing goodness of fit for a chi-square distribution.

[Skill 4.C]

ESSENTIAL KNOWLEDGE

VAR-8.C.1

When considering a distribution of proportions for one categorical variable, the appropriate test is the chi-square test for goodness of fit.

VAR-8.D.1

Expected counts for a chi-square goodness-of-fit test are (sample size) (null proportion).

VAR-8.E.1

In order to make statistical inferences for a chi-square test for goodness of fit we must check the following:

- a. To check for independence:
 - i. Data should be collected using a random sample or randomized experiment.
 - ii. When sampling without replacement, check that $n \leq 10\%N$.
- b. The chi-square test for goodness of fit becomes more accurate with more observations, so large counts should be used (shape).
 - i. A conservative check for large counts is that all expected counts should be greater than 5.

TOPIC 8.3

Carrying Out a Chi-Square Test for Goodness of Fit

Required Course Content

ENDURING UNDERSTANDING

VAR-8

The chi-square distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-8.F

Calculate the appropriate statistic for the chi-square test for goodness of fit.

[Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-8.F.1

The test statistic for the chi-square test for goodness of fit is

$$\chi^2 = \sum \frac{(\text{Observed count} - \text{Expected count})^2}{\text{Expected count}},$$

with
degrees of freedom = *number of categories* – 1.

VAR-8.F.2

The distribution of the test statistic assuming the null hypothesis is true (null distribution) can be either a randomization distribution or, when a probability model is assumed to be true, a theoretical distribution (chi-square).

VAR-8.G

Determine the p -value for chi-square test for goodness of fit significance test.

[Skill 3.E]

VAR-8.G.1

The p -value for a chi-square test for goodness of fit for a number of degrees of freedom is found using the appropriate table or computer generated output.

continued on next page

SKILLS

 *Using Probability and Simulation*

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.


AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.I

Interpret the p -value for the chi-square test for goodness of fit. [Skill 4.B]

DAT-3.J

Justify a claim about the population based on the results of a chi-square test for goodness of fit. [Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.I.1

An interpretation of the p -value for the chi-square test for goodness of fit is the probability, given the null hypothesis and probability model are true, of obtaining a test statistic as, or more, extreme than the observed value.

DAT-3.J.1

A decision to either reject or fail to reject the null hypothesis is based on comparison of the p -value to the significance level, α .

DAT-3.J.2

The results of a chi-square test for goodness of fit can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

TOPIC 8.4

Expected Counts in Two-Way Tables

SKILL

 *Using Probability and Simulation*

3.A

Determine relative frequencies, proportions, or probabilities using simulation or calculations.

Required Course Content

ENDURING UNDERSTANDING**VAR-8**

The chi-square distribution may be used to model variation.

LEARNING OBJECTIVE**VAR-8.H**

Calculate expected counts for two-way tables of categorical data. **[Skill 3.A]**

ESSENTIAL KNOWLEDGE**VAR-8.H.1**

The expected count in a particular cell of a two-way table of categorical data can be calculated using the formula:

$$\text{expected count} = \frac{(\text{row total})(\text{column total})}{\text{table total}}$$

SKILLS

 *Selecting Statistical Methods*

1.F

Identify null and alternative hypotheses.

1.E

Identify an appropriate inference method for significance tests.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 8.5

Setting Up a Chi-Square Test for Homogeneity or Independence

Required Course Content

ENDURING UNDERSTANDING

VAR-8

The chi-square distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-8.1

Identify the null and alternative hypotheses for a chi-square test for homogeneity or independence. **[Skill 1.F]**

ESSENTIAL KNOWLEDGE

VAR-8.1.1

The appropriate hypotheses for a chi-square test for homogeneity are:

H_0 : There is no difference in distributions of a categorical variable across populations or treatments.

H_a : There is a difference in distributions of a categorical variable across populations or treatments.

VAR-8.1.2

The appropriate hypotheses for a chi-square test for independence are:

H_0 : There is no association between two categorical variables in a given population or the two categorical variables are independent.

H_a : Two categorical variables in a population are associated or dependent.

continued on next page

LEARNING OBJECTIVE

VAR-8.J

Identify an appropriate testing method for comparing distributions in two-way tables of categorical data. **[Skill 1.E]**

VAR-8.K

Verify the conditions for making statistical inferences when testing a chi-square distribution for independence or homogeneity. **[Skill 4.C]**

ESSENTIAL KNOWLEDGE

VAR-8.J.1

When comparing distributions to determine whether proportions in each category for categorical data collected from different populations are the same, the appropriate test is the chi-square test for homogeneity.

VAR-8.J.2

To determine whether row and column variables in a two-way table of categorical data might be associated in the population from which the data were sampled, the appropriate test is the chi-square test for independence.

VAR-8.K.1

In order to make statistical inferences for a chi-square test for two-way tables (homogeneity or independence), we must verify the following:

- a. To check for independence:
 - i. For a test for independence: Data should be collected using a simple random sample.
 - ii. For a test for homogeneity: Data should be collected using a stratified random sample or randomized experiment.
 - iii. When sampling without replacement, check that $n \leq 10\%N$.
- b. The chi-square tests for independence and homogeneity become more accurate with more observations, so large counts should be used (shape).
 - i. A conservative check for large counts is that all expected counts should be greater than 5.

SKILLS

 *Using Probability and Simulation*

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 8.6

Carrying Out a Chi-Square Test for Homogeneity or Independence

Required Course Content

ENDURING UNDERSTANDING

VAR-8

The chi-square distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-8.L

Calculate the appropriate statistic for a chi-square test for homogeneity or independence. **[Skill 3.E]**

VAR-8.M

Determine the p -value for a chi-square significance test for independence or homogeneity. **[Skill 3.E]**

ESSENTIAL KNOWLEDGE

VAR-8.L.1

The appropriate test statistic for a chi-square test for homogeneity or independence is the chi-square statistic:

$$\chi^2 = \sum \frac{(\text{Observed count} - \text{Expected count})^2}{\text{Expected count}}$$

with degrees of freedom equal to:
 $(\text{number of rows} - 1)(\text{number of columns} - 1)$.

VAR-8.M.1

The p -value for a chi-square test for independence or homogeneity for a number of degrees of freedom is found using the appropriate table or technology.

VAR-8.M.2

For a test of independence or homogeneity for a two-way table, the p -value is the proportion of values in a chi-square distribution with appropriate degrees of freedom that are equal to or larger than the test statistic.

continued on next page

ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.K

Interpret the p -value for the chi-square test for homogeneity or independence. [Skill 4.B]

DAT-3.L

Justify a claim about the population based on the results of a chi-square test for homogeneity or independence. [Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.K.1

An interpretation of the p -value for the chi-square test for homogeneity or independence is the probability, given the null hypothesis and probability model are true, of obtaining a test statistic as, or more, extreme than the observed value.

DAT-3.L.1

A decision to either reject or fail to reject the null hypothesis for a chi-square test for homogeneity or independence is based on comparison of the p -value to the significance level, α .

DAT-3.L.2

The results of a chi-square test for homogeneity or independence can serve as the statistical reasoning to support the answer to a research question about the population that was sampled (independence) or the populations that were sampled (homogeneity).

TOPIC 8.7

**Skills Focus: Selecting
an Appropriate Inference
Procedure for Categorical Data**

AVAILABLE RESOURCE

- Classroom Resource >
[Inference](#)

Required Course Content

This topic is intended to focus on the skill of selecting an appropriate inference procedure now that students have a range of options. Students should be given opportunities to practice when and how to apply all learning objectives relating to inference for categorical data.

AP STATISTICS

UNIT 9

Inference for Quantitative Data: Slopes



2–5%

AP EXAM WEIGHTING



~7–8

CLASS PERIODS



Remember to go to [AP Classroom](#) to assign students the online **Personal Progress Check** for this unit.

Whether assigned as homework or completed in class, the **Personal Progress Check** provides each student with immediate feedback related to this unit's topics and skills.

Personal Progress Check 9

Multiple-choice: ~25 questions

Free-response: 1 question

- Inference and Exploring Data

Inference for Quantitative Data: Slopes



Developing Understanding

BIG IDEA 1 *Variation and Distribution* **VAR**

- How can there be variability in slope if the slope statistic is uniquely determined for a line of best fit?

BIG IDEA 2 *Patterns and Uncertainty* **UNC**

- When is it appropriate to perform inference about the slope of a population regression line based on sample data?

BIG IDEA 3 *Data-Based Predictions, Decisions, and Conclusions* **DAT**

- Why do we not conclude that there is no correlation between two variables based on the results of a statistical inference for slopes?

Students may be surprised to learn that there is variability in slope. In their experience in previous courses, the slope of the line of best fit does not vary for a particular set of bivariate quantitative data. However, suppose that every student in a university physics course collects data on spring length for 10 different hanging masses and calculates the least-squares regression line for their sample data. The students' slopes would likely vary as part of an approximately normal sampling distribution centered at the (true) slope of the population regression line relating spring length to hanging mass. In this unit, students will learn how to construct confidence intervals for and perform significance tests about the slope of a population regression line when appropriate conditions are met.

Building Course Skills

1.A 4.B 4.E

In Unit 9, students should have multiple opportunities to practice interpreting the slope, y -intercept, r^2 , standard deviation of the residual s , and standard error of the slope in context from computer output. They should refrain from using deterministic language such as “a 1-foot increase in X is associated with a 0.445-point increase in Y ,” instead framing the association in terms of potential outcomes (i.e., “a *predicted* 0.445-point increase”). Students should also practice writing “increase” or “additional” for both variables, not just the dependent variable.

Students should practice identifying what the question is asking or what needs to be solved. Without careful reading, students often provide answers that are not relevant or required, for example, conducting a significance test when the question does not call for one, or giving the expected number of successes or failures when asked to calculate a probability. Teachers can have

them practice identifying the task before they begin, then checking that the response they've provided addresses the task.

Preparing for the AP Exam

Students should pay attention to timing as they work through full-length sections of past exams in order to leave enough time to complete the investigative task, which is weighted more heavily than the other free-response questions. The investigative task includes both familiar course content and questions requiring extended reasoning. As an example of a straightforward application of a topic from this unit, **2007 Form B FRQ 6 part b** asks students to find a 95% confidence interval for the slope of a regression line. This familiar task gives students an opportunity to gain confidence and earn some credit, and it serves as an entry to subsequent parts of the question. Although the investigative task will require students to transfer course skills to unfamiliar settings, students who understand course content will have everything they need to complete the task.

UNIT AT A GLANCE

Enduring Understanding	Topic	Skills	Class Periods
			~7–8 CLASS PERIODS
VAR-1	9.1 Introducing Statistics: Do Those Points Align?	1.A Identify the question to be answered or problem to be solved (<i>not assessed</i>).	
UNC-4	9.2 Confidence Intervals for the Slope of a Regression Model	1.D Identify an appropriate inference method for confidence intervals. 4.C Verify that inference procedures apply in a given situation. 3.D Construct a confidence interval, provided conditions for inference are met.	
	9.3 Justifying a Claim About the Slope of a Regression Model Based on a Confidence Interval	4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.D Justify a claim based on a confidence interval. 4.A Make an appropriate claim or draw an appropriate conclusion.	
VAR-7	9.4 Setting Up a Test for the Slope of a Regression Model	1.E Identify an appropriate inference method for significance tests. 1.F Identify null and alternative hypotheses. 4.C Verify that inference procedures apply in a given situation.	
VAR-7, DAT-3	9.5 Carrying Out a Test for the Slope of a Regression Model	3.E Calculate a test statistic and find a p -value, provided conditions for inference are met. 4.B Interpret statistical calculations and findings to assign meaning or assess a claim. 4.E Justify a claim using a decision based on significance tests.	
	9.6 Skills Focus: Selecting an Appropriate Inference Procedure	N/A	
 Go to AP Classroom to assign the Personal Progress Check for Unit 9. Review the results in class to identify and address any student misunderstandings.			

SAMPLE INSTRUCTIONAL ACTIVITIES

The sample activities on this page are optional and are offered to provide possible ways to incorporate various instructional approaches into the classroom. They were developed in partnership with teachers from the AP community to share ways that they approach teaching some of the topics in this unit. Please refer to the Instructional Approaches section beginning on p. 207 for more examples of activities and strategies.

Activity	Topic	Sample Activity										
1	9.2	Note-Taking Begin by having students use a chart to record the symbols for statistics and parameters that have been used previously to construct confidence intervals: <table><thead><tr><th>Statistic</th><th>Parameter</th></tr></thead><tbody><tr><td>$\hat{p}$</td><td>p</td></tr><tr><td>$\bar{x}$</td><td>μ</td></tr><tr><td>s</td><td>σ</td></tr></tbody></table> Then, when constructing a confidence interval for the population slope parameter, have students add a new row for the symbols for the sample slope and population slope: b and β, respectively. This will reinforce that the slope of the least-squares regression line is a sample statistic and can be used to estimate the population parameter slope.	Statistic	Parameter	$\hat{p}$	p	$\bar{x}$	μ	s	σ		
Statistic	Parameter											
$\hat{p}$	p											
$\bar{x}$	μ											
s	σ											
2	9.3	Error Analysis Give students some raw data on the distance and cost to fly from their hometown to various major cities. For example: <table><thead><tr><th colspan="2">Flying from ____ to ____</th></tr><tr><th>Distance</th><th>Cost</th></tr></thead><tbody><tr><td>512 miles</td><td>\$179</td></tr><tr><td>1256 miles</td><td>\$257</td></tr><tr><td>3256 miles</td><td>\$387</td></tr></tbody></table> Then introduce some questions justifying a claim and error analysis. For example, how could you refute a claim that the average cost per mile (the population slope) is \$0.50 per mile if you believe it to be false?	Flying from ____ to ____		Distance	Cost	512 miles	\$179	1256 miles	\$257	3256 miles	\$387
Flying from ____ to ____												
Distance	Cost											
512 miles	\$179											
1256 miles	\$257											
3256 miles	\$387											
3	9.5	Notation Read Aloud Have students read AP Exam questions aloud (e.g., 2011 FRQ 5, 2010 Form B FRQ 6, 2005 Form B FRQ 5, and 2001 FRQ 6), including the given notation. Remind students that the computer output provides the two-sided p-value, and that there are two different p-values in the chart: The top p-value is for the intercept, and the bottom p-value is for the slope. Then have students discuss each of the values in the computer output and carry out a test for the slope of a regression model.										

SKILL

 *Selecting Statistical Methods*

1.A

Identify the question to be answered or problem to be solved.

TOPIC 9.1

Introducing Statistics: Do Those Points Align?

Required Course Content

ENDURING UNDERSTANDING

VAR-1

Given that variation may be random or not, conclusions are uncertain.

LEARNING OBJECTIVE

VAR-1.K

Identify questions suggested by variation in scatter plots.
[Skill 1.A]

ESSENTIAL KNOWLEDGE

VAR-1.K.1

Variation in points' positions relative to a theoretical line may be random or non-random.

TOPIC 9.2

Confidence Intervals for the Slope of a Regression Model

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.AC

Identify an appropriate confidence interval procedure for a slope of a regression model. [Skill 1.D]

ESSENTIAL KNOWLEDGE

UNC-4.AC.1

Consider a response variable, y , that is linearly related to an explanatory variable, x . For a simple random sample of n observations, the sample regression line, $\hat{y} = a + bx$, is an estimate of the population regression line $\mu_y = \alpha + \beta x$. For a particular observation, (x_i, y_i) , the residual from the sample regression line, $y_i - \hat{y}_i = y_i - (a + bx_i)$, is an estimate of $y_i - (\alpha + \beta x_i)$, the deviation of the response variable from the population regression line. For all points (x, y) in the population, the standard deviation of all of the deviations of the response variable from the population regression line, σ , can be estimated by the standard deviation of the residuals from the sample regression line,

$$s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}. \text{ (Note: This formula}$$

uses $n - 2$ in the denominator instead of $n - 1$ because two parameters, α and β , must be estimated to obtain the predicted values from the least-squares regression line.)

UNC-4.AC.2

For a simple random sample of n observations, let b represent the slope of a sample regression line. Then the mean of the sampling distribution for b equals the population slope: $\mu_b = \beta$. The standard deviation of the sampling distribution for b is $\sigma_b = \frac{\sigma}{\sigma_x \sqrt{n}}$, where

$$\sigma_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}.$$

UNC-4.AC.3

The appropriate confidence interval for the slope of a regression model is a t -interval for the slope.

SKILLS

 *Selecting Statistical Methods*

1.D

Identify an appropriate inference method for confidence intervals.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.

 *Using Probability and Simulation*

3.D

Construct a confidence interval, provided conditions for inference are met.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

continued on next page

LEARNING OBJECTIVE

UNC-4.AD

Verify the conditions to calculate confidence intervals for the slope of a regression model. [Skill 4.C]

UNC-4.AE

Determine the given margin of error for the slope of a regression model. [Skill 3.D]

UNC-4.AF

Calculate an appropriate confidence interval for the slope of a regression model. [Skill 3.D]

ESSENTIAL KNOWLEDGE

UNC-4.AD.1

In order to calculate a confidence interval to estimate the slope of a regression line, we must check the following:

- The true relationship between x and y is linear. Analysis of residuals may be used to verify linearity.
- The standard deviation for y , σ_y , does not vary with x . Analysis of residuals may be used to check for approximately equal standard deviations for all x .
- To check for independence:
 - Data should be collected using a random sample or a randomized experiment.
 - When sampling without replacement, check that $n \leq 10\%N$.
- For a particular value of x , the responses (y -values) are approximately normally distributed. Analysis of graphical representations of residuals may be used to check for normality.
 - If the observed distribution is skewed, n should be greater than 30.

UNC-4.AE.1

For the slope of a regression line, the margin of error is the critical value (t^*) times the standard error (SE) of the slope.

UNC-4.AE.2

The standard error for the slope of a regression line with sample standard deviation, s , is

$$SE = \frac{s}{s_x \sqrt{n-1}}, \text{ where } s \text{ is the estimate of } \sigma$$

and s_x is the sample standard deviation of the x values.

UNC-4.AF.1

The point estimate for the slope of a regression model is the slope of the line of best fit, b .

UNC-4.AF.2

For the slope of a regression model, the interval estimate is $b \pm t^* (SE_b)$.

TOPIC 9.3

Justifying a Claim About the Slope of a Regression Model Based on a Confidence Interval

Required Course Content

ENDURING UNDERSTANDING

UNC-4

An interval of values should be used to estimate parameters, in order to account for uncertainty.

LEARNING OBJECTIVE

UNC-4.AG

Interpret a confidence interval for the slope of a regression model. **[Skill 4.B]**

UNC-4.AH

Justify a claim based on a confidence interval for the slope of a regression model. **[Skill 4.D]**

UNC-4.AI

Identify the effects of sample size on the width of a confidence interval for the slope of a regression model. **[Skill 4.A]**

ESSENTIAL KNOWLEDGE

UNC-4.AG.1

In repeated random sampling with the same sample size, approximately $C\%$ of confidence intervals created will capture the slope of the regression model, i.e., the true slope of the population regression model.

UNC-4.AG.2

An interpretation for a confidence interval for the slope of a regression line should include a reference to the sample taken and details about the population it represents.

UNC-4.AH.1

A confidence interval for the slope of a regression model provides an interval of values that may provide sufficient evidence to support a particular claim in context.

UNC-4.AI.1

When all other things remain the same, the width of the confidence interval for the slope of a regression model tends to decrease as the sample size increases.

SKILLS


Statistical Argumentation

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.D

Justify a claim based on a confidence interval.

4.A

Make an appropriate claim or draw an appropriate conclusion.


AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

SKILLS

 *Selecting Statistical Methods*

1.E

Identify an appropriate inference method for significance tests.

1.F

Identify null and alternative hypotheses.

 *Statistical Argumentation*

4.C

Verify that inference procedures apply in a given situation.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 9.4

Setting Up a Test for the Slope of a Regression Model

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.J

Identify the appropriate selection of a testing method for a slope of a regression model. [Skill 1.E]

VAR-7.K

Identify appropriate null and alternative hypotheses for a slope of a regression model. [Skill 1.F]

ESSENTIAL KNOWLEDGE

VAR-7.J.1

The appropriate test for the slope of a regression model is a t -test for a slope.

VAR-7.K.1

The null hypothesis for a t -test for a slope is: $H_0 : \beta = \beta_0$, where β_0 is the hypothesized value from the null hypothesis. The alternative hypothesis is $H_0 : \beta < \beta_0$ or $H_0 : \beta > \beta_0$, or $H_0 : \beta \neq \beta_0$.

continued on next page

LEARNING OBJECTIVE

VAR-7.L

Verify the conditions for the significance test for the slope of a regression model.

[Skill 4.C]

ESSENTIAL KNOWLEDGE

VAR-7.L.1

In order to make statistical inferences when testing for the slope of a regression model, we must check the following:

- The true relationship between x and y is linear. Analysis of residuals may be used to verify linearity.
- The standard deviation for y , σ_y , does not vary with x . Analysis of residuals may be used to check for approximately equal standard deviations for all x .
- To check for independence:
 - Data should be collected using a random sample or a randomized experiment.
 - When sampling without replacement, check that $n \leq 10\%N$.
- For a particular value of x , the responses (y -values) are approximately normally distributed. Analysis of graphical representations of residuals may be used to check for normality.
 - If the observed distribution is skewed, n should be greater than 30.
 - If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

SKILLS

 *Using Probability and Simulation*

3.E

Calculate a test statistic and find a p -value, provided conditions for inference are met.

 *Statistical Argumentation*

4.B

Interpret statistical calculations and findings to assign meaning or assess a claim.

4.E

Justify a claim using a decision based on significance tests.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 9.5

Carrying Out a Test for the Slope of a Regression Model

Required Course Content

ENDURING UNDERSTANDING

VAR-7

The t -distribution may be used to model variation.

LEARNING OBJECTIVE

VAR-7.M

Calculate an appropriate test statistic for the slope of a regression model. [Skill 3.E]

ESSENTIAL KNOWLEDGE

VAR-7.M.1

The distribution of the slope of a regression model assuming all conditions are satisfied and the null hypothesis is true (null distribution) is a t -distribution.

VAR-7.M.2

For simple linear regression when random sampling from a population for the response that can be modeled with a normal distribution for each value of the explanatory variable,

the sampling distribution of $t = \frac{b - \beta}{SE_b}$ has a

t -distribution with degrees of freedom equal to $n - 2$. When testing the slope in a simple linear regression model with one parameter, the slope, the test for the slope has $df = n - 1$.

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ENDURING UNDERSTANDING

DAT-3

Significance testing allows us to make decisions about hypotheses within a particular context.

LEARNING OBJECTIVE

DAT-3.M

Interpret the p -value of a significance test for the slope of a regression model.

[Skill 4.B]

DAT-3.N

Justify a claim about the population based on the results of a significance test for the slope of a regression model. [Skill 4.E]

ESSENTIAL KNOWLEDGE

DAT-3.M.1

An interpretation of the p -value of a significance test for the slope of a regression model should recognize that the p -value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population slope is equal to the particular value stated in the null hypothesis.

DAT-3.N.1

A formal decision explicitly compares the p -value to the significance α . If the p -value $\leq \alpha$, then reject the null hypothesis, $H_0 : \beta = \beta_0$. If the p -value $> \alpha$, then fail to reject the null hypothesis.

DAT-3.N.2

The results of a significance test for the slope of a regression model can serve as the statistical reasoning to support the answer to a research question about that sample.



AVAILABLE RESOURCE

- Classroom Resource > [Inference](#)

TOPIC 9.6

Skills Focus: Selecting an Appropriate Inference Procedure

Required Course Content

This topic is intended to focus on the skill of selecting an appropriate inference procedure now that students have a range of options. Students should be given opportunities to practice when and how to apply all learning objectives relating to inference.

AP STATISTICS

Instructional Approaches



Selecting and Using Course Materials

This section includes descriptions of available resources and materials that many AP Statistics teachers have found helpful. Teachers are encouraged to get involved in some of the online communities to ask questions, discuss ideas, and trade thoughts with other teachers.

Textbooks

The foundational component for most classes is still the textbook, and there are many excellent textbooks to choose from. Most current textbooks come with an array of resources designed to provide reinforcement for the students and facilitate the assessment process for the teacher. For example, some textbooks provide an accompanying pacing guide, a bank of practice questions, activities for each unit, video tutorials, or online homework problems that coincide with each chapter.

While the College Board provides [examples of textbooks](#) on AP Central to help determine whether a text is considered appropriate in meeting the AP Statistics Course Audit curricular requirement, teachers select textbooks locally. Also note that many publishers offer teachers the opportunity to request a sample copy of a textbook they are interested in adopting for classroom use. This process allows teachers to review materials in order to determine what book will work best for them and their students.

The Use of Technology

The AP Statistics course depends heavily on the availability of technology suitable for the interactive, investigative aspects of data analysis. Ideally, students should have access to both computers and calculators for work in and outside the classroom.

A computer does more than eliminate the drudgery of hand computation and graphing—it is an essential tool for structured inquiry. First, it produces graphs that are specifically designed for data analysis. These graphical displays make it easier to observe patterns in data, to identify important subgroups of the data, and to locate any unusual data points. Second, the computer allows the student to fit complex mathematical models

to the data and to assess how well the model fits the data by examining the residuals. Finally, the computer is helpful in identifying an observation that has an undue influence on the analysis and in isolating its effects.

In addition to its use in data analysis, the computer facilitates the simulation approach to probability that is emphasized in the AP Statistics course. Probabilities of random events, probability distributions of random variables, and sampling distributions of statistics can be studied conceptually, using simulation. This frees the student and teacher from a narrow approach that depends on a few simple probabilistic models.

Because the computer is central to what statisticians do, it is considered essential for teaching the AP Statistics course. However, it is not yet possible for students to have access to a computer during the AP Statistics Exam. Without a computer and under the conditions of a timed exam, students cannot be asked to perform the amount of computation that is needed for many statistical investigations. Consequently, standard computer output will be provided as necessary and students will be expected to interpret it. For an example of computer output, see sample multiple-choice question #16 on p. 247.

Currently, the graphing calculator is the only computational aid that is available to students for use as a tool for data analysis on the AP Statistics Exam. Computational capabilities of graphing calculators used in the course and on the exam should include standard statistical univariate and bivariate summaries through linear regression. Graphical capabilities should include common univariate and bivariate displays such as histograms, boxplots, and scatterplots. Students find calculators where data are entered into a spreadsheet format particularly easy to use.

Students may not use minicomputers, pocket organizers, electronic writing pads, or calculators with QWERTY (i.e., typewriter) keyboards on the AP Statistics Exam. For a complete list of calculators eligible for use on the exam, see [Calculator Policy for AP Statistics](#), found on the AP Statistics course page on [apstudent.collegeboard.org](#). The calculator memory will not

be cleared, but students may only use the memory to store programs, not notes. For the exam, students are not allowed to access any information in their graphing calculators or elsewhere if it's not directly related to upgrading the statistical functionality of older graphing calculators to make them comparable to statistical features found on newer models. The only acceptable upgrades are those that improve the computational functionalities and/or graphical functionalities for data they key into the calculator while taking the examination. Unacceptable enhancements include, but are not limited to, keying or scanning text or response templates into the calculator. Students attempting to augment the capabilities of their graphing calculators in any way other than for the purpose of upgrading features as described above will be considered to be cheating on the exam.

Online Tools and Resources

In addition to the resources offered through a teacher's choice of textbook, many great resources exist online—and are often free. These include sample test questions, applets, simulations, question banks, and best practices. Teachers can sample these resources to find which ones can most benefit their students.

APPLETS AND SIMULATIONS

The use of internet-based applets or simulation sites is fast becoming a critical component of the AP Statistics pedagogy. Interpreting data procured through a simulation process is an important skill not only for the AP Exam but also for sound statistical reasoning in other contexts. Thus, it is important that students have experience with producing and interpreting data. Below are a few examples of free applets and simulation programs that are easily incorporated into classroom activities and instruction. This is not a comprehensive list, nor is it an endorsement of any of these resources.

- **StatKey**
lock5stat.com/StatKey
Designed by the authors of *Statistics: Unlocking the Power of Data* (published by John Wiley & Sons) "to help introductory students understand and easily implement bootstrap intervals and randomization tests."¹
- **StatCrunch**
statcrunch.com/explore.php
This web-based statistical software allows users to share data sets, results, and reports. In conjunction with Pearson publishers.

- **Rossman/Chance Applet Collection**
rossmanchance.com/applets
Designed by a former AP Statistics Chief Reader, this site has lots of applets with simulations for use in the classroom (in conjunction with *Workshop Statistics: Discovery with Data*, published by John Wiley & Sons).
- **Stapplet**
stapplet.com
This online tool has a simple interface organized around different topics in the AP Statistics curriculum (in conjunction with *The Practice of Statistics*, published by W.H. Freeman & Co).
- **Art of Stat**
artofstat.com
Designed by a former Chief Reader, this site has interactive web apps, downloadable data sets, and user-friendly how-to guides (in conjunction with *Statistics: The Art and Science of Learning from Data*, published by Pearson).

STUDENT PRACTICE

Even with the best lessons in place, students will still need additional reinforcement with particular concepts from time to time. These sites provide additional practice for students who want to review specific content areas on their own time.

- **STATS4STEM AP Review Assignments**
stats4stem.org
Supported by the National Science Foundation (NSF), this site provides a collection of learning, assessment, tutoring, data, and computing resources for statistics educators and their students. The assessment and tutoring features allow for instantaneous student feedback with question-specific hints for students who are struggling.
- **LOCUS**
locus.statisticseducation.org
LOCUS is a project funded by the National Science Foundation (NSF), focused on developing assessments of statistical understanding. They provide assessments that measure students' understanding across levels of development as identified in the Guidelines for Assessment and Instruction in Statistics Education (GAISE) Report. The items and resources can be viewed according to grade level or by components of the statistical problem-solving process. Their content has been reviewed and endorsed by the American Statistical Association (ASA) and National Council of Teachers of Mathematics (NCTM).

¹ "StatKey: Online Tools for Bootstrap Intervals and Randomization Tests," iase-web.org/icots/9/proceedings/pdfs/ICOTS9_9B2_LOCKMORGAN.pdf

CONTENT REINFORCEMENT

If a teacher is looking for resources or activities to use for a particular unit, there are many sites offering activities that can be used throughout the course. Some of these sites offer a listing of best practices. Others will offer guidance on how to present or explain certain content that may be useful.

- **Stats Medic**
statsmedic.com
This site includes pacing guides, daily schedules, and activities for both AP Statistics and Intro Statistics courses.
- **StatsMonkey**
apstatsmonkey.com
This site provides additional notes and resources for AP Statistics topics. It also contains dozens of references and activities.
- **mathcoachblog**
mathcoachblog.com/ap-statistics-reading
This site contains best practices presentation files created by AP Statistics readers as well as many examples of how to use desmos.com in class.

ONLINE COMMUNITIES

Online communities provide opportunities for collaboration. While some teachers are the only AP Statistics teacher in their school or district, online communities offer a place to discuss strategies, plan timelines, or share insights. Students can also benefit from interactive online discussions and statistics activities.

- **What's Going On in This Graph**
nytimes.com/column/whats-going-on-in-this-graph
These activities are designed by The Learning Network and hosted by *The New York Times*. Students interpret graphs based on their

statistical knowledge and participate in live discussions with other students and teachers from across the country. New graphs are posted on a regular basis throughout the school year.

- **Twitter**
twitter.com
There are many AP Statistics teachers on Twitter who share ideas and insights into teaching the course. Search hashtags like #edchat and #statschat to find the latest educational conversations.
- **Facebook - AP Statistics Groups**
facebook.com
There are multiple Facebook groups set up for AP Statistics teachers to share ideas, swap resources, and offer advice and other support. Groups include AP Stats Survival, AP Stats Teachers Support Group, and the AP Statistics Readers Group (this group is limited to readers of the AP Statistics Exam).
- **Census at School**
ww2.amstat.org/censusatschool
Census at School is an international classroom project that engages students in grades 4–12 in statistical problem solving. Students complete a brief online survey, analyze their class census results, and compare their class with random samples of students in the United States and other countries.
- **This is Statistics**
thisisstatistics.org
Sponsored by the American Statistical Association (ASA), this site contains many resources for teachers, students, and counselors for use in the classroom. It also highlights the opportunities and advantages gained from enrolling in statistics courses in high school and college. This is a great source to help recruit students to take AP Statistics.

Hands-on Manipulatives

Since a major component of the AP Statistics course deals with the proper gathering and interpreting of data, it is important that students spend at least part of the year gathering data of their own.

Activities that involve gathering data through simulation exercises should be incorporated throughout the year. Computer programs are useful for working in large scales of data and running greater numbers of simulations, but the initial process of simulating a chance event and generating data from that event is more valuable to students if they are actively generating that data themselves.

For that reason, running an exercise that simulates the event in a convenient way for students to do in class is strongly encouraged. Often this can be done with simple everyday items, including dice, cards, coins, beads, and bead counters. Or, there are many activities online that simulate the use of candy, beads, or other simple items. Large post-it easel paper can be used to display the results around the classroom, summarize the work, and keep it visible to refer to later in the year.

Professional Organizations

Professional organizations also serve as excellent resources for best practices and professional development opportunities. Below is a list of prominent organizations that serve the statistics and math education communities.

- **American Statistical Association (ASA)**

amstat.org

The American Statistical Association is the world's largest community of statisticians, supporting excellence in the development, application, and

dissemination of statistical science through meetings, publications, membership services, education, accreditation, and advocacy.

- **National Math and Science Initiative (NMSI)**

nms.org

NMSI is a nonprofit organization whose mission is to improve student performance in the subjects of science, technology, engineering, and math (STEM) in the United States. It employs experienced AP teachers to train students and teachers in the STEM courses.

- **Consortium for the Advancement of Undergraduate Statistics Education (CAUSE)**

causeweb.org/cause

CAUSE is a national organization whose mission is to support and advance undergraduate statistics education, in four target areas: resources, professional development, outreach, and research. The site includes simulations, data sets, sample lectures, fun activities, and cartoons to use in the classroom.

- **National Council of Teachers of Mathematics (NCTM)**

nctm.org

Founded in 1920, NCTM is the world's largest mathematics education organization, advocating for high-quality mathematics teaching and learning for each and every student.

Instructional Strategies

The AP Statistics course framework outlines the concepts and skills students must master in order to be successful on the AP Exam. In order to address those concepts and skills effectively, it helps to incorporate a variety of instructional approaches into daily lessons and activities. Important components of the course should include the use of technology, projects and laboratories, cooperative group problem-solving, and writing.

Note that the AP Statistics course lends itself naturally to a mode of teaching that engages students in constructing their own knowledge. For example, students working

individually or in small groups can plan and perform data collection and analyses where the teacher serves in the role of a facilitator rather than a lecturer. This gives students ample opportunity to think through problems, make decisions, and share questions and conclusions with other students as well as with the teacher.

These approaches, along with targeted strategies for teaching AP Statistics, will allow students to build interdisciplinary connections with other subjects and with their world outside of school. The following table presents strategies that can help students apply their understanding of course concepts.

Strategy	Definition	Purpose	Example
<i>Ask the Expert</i>	Students are assigned as “experts” on problems they have mastered; groups rotate through the expert stations to learn about problems they have not yet mastered.	Provides opportunities for students to share their knowledge and learn from one another.	When practicing probability problems, assign students as “experts” on mutually exclusive events, conditional probability, independent events, and unions of events. Have students rotate through stations in groups, working with the station expert to complete a small batch of problems that apply a particular probability rule.
<i>Build the Model Solution</i>	Students work in pairs to assemble a sentence representing a model solution to a free-response question.	Helps students build proficiency with precise language and communication, especially for topics that involve interpretations (e.g., confidence intervals and correlation coefficients).	Take a printed version of a model solution and cut it apart into sections (e.g., grouped by phrases). Print and cut out additional words that do not belong in the response (e.g., include the word “mean” if the response is supposed to be about a “proportion”). Distribute sets of these strips to student pairs and have them select which strips are relevant and assemble those into a model solution. They should set aside the strips that are not relevant to the solution.

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Strategy	Definition	Purpose	Example
Create Representations	Students create pictures, tables, graphs, lists, equations, models, and/or verbal expressions to interpret text or data.	Helps organize information using multiple ways to present data and to answer a question or show a problem solution.	Pair students up and give one partner a graphical display and the other student a blank piece of paper. Without sharing their papers with their partners, the student with the graph should describe their distribution to the other student. The other student should draw the graph based solely on the description they were given. Next, have students compare graphs and discuss how the instructions could have been clearer.
Debriefing	Students discuss their understanding of a concept to lead to consensus on its meaning.	Helps clarify misconceptions and deepen understanding of content.	Provide students with a dotplot of 50 individuals' heights, as well as a numbered list of those heights (from 01 to 50). First, ask each student to randomly select two heights from the list using a random number generator or a random number table and then find the mean of these two heights. Have all students collectively create a dotplot of their sample means. Inform students that this is called a sampling distribution with a sample size of 2. Next, have students discuss with a partner and develop an explanation of what a sampling distribution is and why it is utilized.
Discussion Groups	Students work within groups to discuss content, to create problem solutions, and to explain and justify a solution.	Aids understanding through the sharing of ideas, interpretation of concepts, and analysis of problem scenarios.	Give each group of students an example of a statistical test. Have students work together to write out the null and alternative hypothesis in words. They should then work together to determine Type I and Type II errors as well as potential consequences of those errors. Different groups would then compare answers, and if there is any disagreement, they will need to rationalize their decisions.
Error Analysis	Students analyze an existing solution to determine whether (or where) errors have occurred.	Allows students to troubleshoot existing errors so they can solve correctly when they do the same types of problems on their own.	Provide students a sample response that implicitly accepts the null hypothesis and ask them to explain the faulty reasoning.

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Strategy	Definition	Purpose	Example
<i>FRQ Partner Quiz</i> <i>(also see The Scribe and the Calculator)</i>	Students work with a partner to solve a problem. However, only one of the partners is allowed to write on the paper, and only the other is allowed to use the calculator.	Students must communicate and discuss how to do the problem more so than in a typical group setting.	Have students work with partners on a statistical situation (test or interval). Have them discuss which test or interval they need to do. The student with the calculator will find the test statistic and p -value (or will find the interval). The student with the paper will write down the hypothesis (if a test) and assumptions/conditions. Once they get the test statistic and p -value (or the interval), have them write up the decision and/or conclusion. Have students switch jobs for the next problem.
<i>Gallery Walk</i>	Student pairs write their solution on a large piece of paper and post on a wall in the room. Students stand in front of their solutions and then move to the right to the next solution. Using sticky notes, students write a positive comment about the response; then move to the right again and continue the process until they've visited all posted responses. Pairs then return to their original solution and read the feedback.	Provides feedback from multiple perspectives, is more immediate than teacher feedback on students' writing, and gives students practice reading and doing error analysis on existing responses.	This strategy can be used anytime students are practicing free-response questions (FRQs), especially when asked to justify their answer. It also serves well as a method for end-of-unit or end-of-course review.
<i>Graphic Organizer</i>	Students represent ideas and information visually (e.g., Venn diagrams, flowcharts).	Provides students a visual system for organizing multiple ideas and details related to a particular concept.	Have students create their own flowchart for determining which inference procedure applies for a given situation.
<i>Look for a Pattern</i>	Students observe information or create visual representations to find a trend.	Helps to identify patterns that may be used to make predictions.	Give partners a tape measure and have them measure each other's heights and arm spans. On a large group scatterplot, have students plot the data point that represents their measurements. Then have them describe to their partners what trends or patterns they see. Once every student has added their point to the scatterplot, ask them to predict arm spans based on different heights that you give them, including at least one instance where they are asked to extrapolate.

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Strategy	Definition	Purpose	Example
Marking the Text	Students highlight, underline, and/or annotate text to focus on key information to help understand the text or solve the problem.	Helps the student identify important information in the text and make notes about the interpretation of tasks required and concepts to apply to reach a solution.	Give students examples of statistical situations. Have students highlight or underline the key phrases to help them determine which statistical test to use for each situation. For example, does it require a significance test or a confidence interval? Does it involve means or proportions? Is there only one variable or are there two or more variables?
Match Mine	Students work in pairs to describe the layout of cards on their game board, without being able to see each other's boards.	Emphasizes the use of appropriate vocabulary and allows students to practice accurate communication around different representations.	Create a blank 3x3 grid as a "game board" and 9 index cards depicting distributions of differing shapes, center, skewness, clustering, etc. Pair up students and insert a folder between the partners so they can't see each other's boards. Distribute a blank grid and a set of cards to each partner. Student A arranges the cards however they choose on their grid then describes that arrangement so that Student B can attempt to match it on their own board. When the pair thinks that they have correctly made all matches, have them compare their arrangements to see how well they did.
Model Questions	Students answer questions from released AP Statistics Exams.	Provides rigorous practice and assesses students' ability to apply multiple skills on content being presented in either a multiple choice or a free-response question.	After learning the three different chi-square tests, have students practice by completing free-response questions in which they are asked to determine which chi-square test to use and then carry out the test to find a solution.
Notation Read Aloud	Students read symbols and notational representations aloud.	Helps students to accurately interpret symbolic representations.	This strategy can be used to introduce new symbols and statistical notation to ensure that students learn proper terminology from the start. For example, after introducing symbolic notation for conditional probability, ask students to write or say aloud the verbal translation of a given problem that uses that notation.

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Strategy	Definition	Purpose	Example
Note-Taking	Students create a record of information while reading a text or listening to a speaker.	Helps in organizing ideas and processing information.	Have students write down the conditions to be verified when calculating confidence intervals for a population proportion so they can refer to those conditions at a later point in time.
Notice and Wonder	Students are shown a data display on its own with little or no context around it. Students comment on what they notice or wonder about it.	Allows students to access the relevant information being conveyed by the graph without worrying about having to answer something "correctly."	This strategy works equally well for a 5-minute warm-up or for a lengthier, in-depth discussion. Select an interesting graph, or one they will be seeing in an upcoming lesson, and ask students to write down three things they "notice" or "wonder" as they look at it. Then have student volunteers share their observations in groups or with the class.
Odd One Out	In groups of four, students are given four problems, images, or graphs. Three of the items should have something in common. Each student in the group works with one of the four items and must decide individually whether their item fits with the other three items and then write a reason why. Students then respond within their groups.	Allows students to explore relationships and graphical features that are similar to or different from one another.	Begin by modeling an example: three images that are four-legged animals and one that's an inanimate object. Explain why the object is the "odd one out." Then provide each student in the group with one of four images displaying distribution graphs. Three of the graphs, for instance, could be approximately normal, while the fourth graph is bimodal. Have students examine their graph and write on mini whiteboards "Mine is in because . . ." or "Mine is out because . . ." Have each group discuss and signal when they've reached consensus. Then reveal the answer and explain what the three similar graphs have in common and why the other is the odd one out.
Paraphrasing	Students restate in their own words the essential information in a text or problem description.	Assists with comprehension, recall of information, and problem solving.	After reading a free-response question, have students re-express the scenario in their own words. For example, on an experimental design question, have them restate the important facets of the experiment (control, randomization, replication, the variable being measured, comparison) before reading what the question itself asks.

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Strategy	Definition	Purpose	Example
Password-Style Games	Students are in pairs with one student facing the teacher and the other with their back to the teacher. The teacher projects a word and the student facing the teacher describes the word to their partner. This repeats every 10 seconds for six words. Students then switch roles.	Reinforces understanding of key vocabulary terms by having students use different approaches for describing those terms.	Have students arrange themselves into partners (or triads), with one partner seated so their back is to the teacher. When projecting the term, "normal distribution," the students facing the teacher provide clues to try to get their partner to write the term on their mini whiteboard. After 10 seconds, project a new term: "chi-square test." Continue the process for as many terms as there are for that activity.
Peer Critique	Students work in teams or with partners to critique student solutions and responses to free-response questions.	Helps to establish criteria for a model response based on content and communication.	Project an anonymous student solution to a homework question (e.g., a description of the shape, center, spread, and outliers for a histogram in context). Have students discuss/critique in their groups and share with the whole class.
Predict and Confirm	Students make conjectures about what results will develop in an activity, confirming or modifying the conjectures based on outcomes.	Stimulates thinking by making, checking, and correcting predictions based on evidence from the outcome.	Before performing a class simulation, have students discuss with their neighbor what values the distribution may take and how often they expect/predict those values to occur. For instance, what might the distribution look like for a binomial setting with $p = 0.5$ and $n = 16$, when we perform 20 trials? Have students perform the simulation, then confirm their predictions.
Quickwrite	Students write for a short, specific amount of time about a designated topic.	Helps generate ideas in a short time.	Provide students with a prompt, such as "Describe how to construct a sampling distribution." Then allow 1–2 minutes for them to write a response (e.g., "obtain sample, collect responses, calculate statistic, plot statistic, repeat"). Allow more writing time for more difficult prompts.

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Strategy	Definition	Purpose	Example
Quiz-Quiz-Trade	Students answer a question independently and then quiz a partner on the same question.	Allows students to talk to multiple peers and solve many problems while coaching each other and hearing different ways of solving or arriving at a solution.	Give students a card containing a question and have them write the answer on the back. Then have them stand up and find a partner. One student quizzes the other and then they reverse roles. Have them switch cards, find a new partner, and the process repeats.
Relief Pitcher	In teams of four, students rotate from "bull pen" to "pitcher's mound" to answer questions and earn points for the team.	Allows students to work together toward a common goal as they practice applying their statistics knowledge and skills.	This strategy works well to review multiple concepts at the end of a unit or to spiral review at the end of the course. For instance, when reviewing for a unit test, students can earn 5 points if they get a question correct when up at the "pitcher's mound" and 3 points if they don't know the answer but their team in the "bull pen" helps them answer it.
Reversing Interpretations	Students are provided with an interpretation and then they generate the value and a possible scenario rather than providing students the value and having them generate the interpretation.	Helps students explore the same concept from multiple approaches and deepens their understanding of why a certain interpretation aligns with a particular value in context.	Instead of only asking students to interpret the slope and y -intercept of a least-squares regression line, provide them with an interpretation and ask them to unpack it. For example, "Weight increases by 35.02 kg for each 1-meter increase in length, on average. What does this describe?"
Round Table	Students are given a worksheet with multiple problems, and work individually to solve, then rotate after each problem to allow other group members to check their work.	Allows students the chance to analyze each other's work and coach their peers, if necessary.	In groups of four, each student has an identical paper with four different problems on it. Have students complete the first one on their paper and then pass the paper clockwise to another member in their group. That student checks the first problem and then completes the second problem on the paper. Have students rotate again, and the process continues until each student has their original paper back.

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Strategy	Definition	Purpose	Example										
Scavenger Hunt	Provide students with a “starter” question. Then place solution cards around the room—each card should contain the solution to a previous problem, along with the next problem in the “scavenger hunt.” Students walk around answering each new problem and finding the corresponding solution around the room.	Allows students to self-correct and analyze their own work if they get an answer they don’t see posted.	When practicing probability calculations, place a card with a starter question somewhere in the classroom, for example: Calculate the expected number of tardy students for the probability distribution: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$P(X)$</td><td>0.5</td><td>0.3</td><td>0.1</td><td>0.1</td></tr></table> Place another card somewhere in the room with the solution to that card, plus another question, for example: “Solution: 0.8 Next problem: Jeff’s bowling scores are roughly normal with mean 120 and a standard deviation of 20. Jared’s bowling scores are roughly normal with mean 180 and a standard deviation of 10. If their scores are independent, what is their combined standard deviation of their team score?” The answer is written on the next card (“Solution: 22.4”). Continue posting solution cards with new problems until the final card presents a problem whose solution is on the original starter card (note that this solution would be added to the starter card above). Students can begin the scavenger hunt at any question, then search around the room for the solution they found, as this will lead them to the next problem they need to solve, until they end up at the first card they started with.	X	0	1	2	3	$P(X)$	0.5	0.3	0.1	0.1
X	0	1	2	3									
$P(X)$	0.5	0.3	0.1	0.1									

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Strategy	Definition	Purpose	Example
<i>Sentence Starters</i>	Students respond to a prompt by filling in the missing parts of a given sentence template.	Helps students practice communication skills by providing a starting point and modeling a sentence structure that would apply for a particular type of problem.	For linear regression models, provide a scenario that includes a least-squares equation. Then have students use a given sentence starter to interpret the meaning of the slope in context. For example, students could be given the scenario in 2017 FRQ 1 , including the least-squares equation $\hat{y} = -16.46 + 35.02x$. To help students begin writing their interpretations, provide a sentence starter such as "The least-squares regression line predicts that the _____ would increase by approximately _____ for each _____." The sentence starter provided could become more open ended as students become more comfortable with writing interpretations in context.
<i>Simulation</i>	Students develop a "best guess" for a parameter by collecting data through simulation, then investigating the distribution of data collected by the class.	Helps students understand what sampling distribution is by crowdsourcing data from the class; also helps make abstract problems more accessible and reinforces the role of probability in inference.	When estimating a population proportion of red cards in a standard deck, put students in pairs and have them draw n cards and then record the proportion that are red. Repeat a large number of times and then create a class dotplot and estimate the parameter value based on the center of the dotplot.
<i>Sketch and Switch</i>	Students create a sketch of a distribution that has a given set of characteristics then switch papers to review each other's work.	Reinforces key vocabulary, provides exposure to varying solutions based on the same characteristics, and allows students to give and receive feedback on each other's work.	For distributions of a single variable, have students sketch a "bimodal, left-skewed" distribution or a "symmetric, bimodal distribution with two outliers." For scatterplots, have them sketch a "weak negative non-linear association."
<i>Team Challenge</i>	Students work in teams of three or four to solve an unfamiliar problem that requires transfer of knowledge and skills from multiple areas of the course.	Builds confidence in approaching complex questions by sharing ideas and providing feedback to peers.	To introduce the investigative task, have students work in teams to develop a model solution to a question that is beyond most students' ability to fully complete on their own.

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Strategy	Definition	Purpose	Example
Team FRQ	Each student in a group of four completes one step of the four-step process for inference with the rest of the group's input.	Allows students to integrate their own knowledge with that of their teammates in order to create a comprehensive solution to a free-response question.	Give student groups a free-response question (FRQ) that requires students to identify, set up, perform, and interpret the results of an appropriate significance test to answer a particular question. Student A states a pair of hypotheses with group input. Student B identifies a test procedure while the group checks that the conditions have been verified. Student C finds the value of the test statistic while others find the p -value. Student D states the conclusion while the group checks for linkage to p -value and context. When groups are finished, have them score their responses as a class using the free-response question scoring guidelines.
The Scribe and the Calculator (also see FRQ Partner Quiz)	Students do problems together in pairs; only one student is allowed to use the calculator, and only the other student is allowed to write on the paper.	Allows students to practice communication and use of technology with a peer.	Group students into pairs and present them with the following scenario: "A basketball player makes 80% of her free throws. Assuming her shots are independent and she takes 20 shots, calculate the probability that at least 15 of the shots are made." Instruct students to begin working on a solution, noting that one student can only write (but can't touch the calculator), while the other student can only use the calculator (but can't write anything). Consider assigning the "calculator" role to the student who is weaker on technology to help them practice.
Think Aloud	Students talk through a difficult problem by describing what the text means.	Helps in comprehending the text, understanding the components of a problem, and thinking about possible paths to a solution.	When students have learned the different types of inference tests and intervals, have them talk through key questions, such as "How many groups?" "Is this a categorical or quantitative variable?" "How was the data collected?" or "Am I testing a claim or estimating an amount?"

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Strategy	Definition	Purpose	Example
<i>Think-Pair-Share (or Wait, Turn, and Talk)</i>	Students are given time to think through a problem on their own, followed by sharing their ideas with a partner and then concluding by sharing results with the class.	Allows students uninterrupted time to process their initial ideas and then revise those ideas with a partner in preparation for sharing them with a larger group.	When looking at a dotplot or boxplot, have students first think individually about what features are the most interesting to them (or about the center, shape, and variability) and then have them share their thoughts with their neighbor. Call on pairs of students to share their thoughts.
<i>Two Wrongs Make a Right</i>	Students list all the reasonably likely ways in which a student could answer a question incorrectly.	Helps students to recognize the common errors and misconceptions that occur so they are less likely to make those errors themselves.	When comparing distributions, have students provide examples that do <i>not</i> describe all relevant features (shape, center, spread, and unusual points).
<i>Use Manipulatives</i>	Students use objects to examine relationships between the information given.	Supports comprehension by providing a tangible or visual representation of a problem or concept.	When constructing the sampling distribution of the sample mean, have students put dot stickers with an $\bar{x}$ written on them to reinforce the idea that the dots on the dotplot are means, not single observations. When performing simulations, use concrete objects to help students understand how often certain values of a variable would occur just by chance. For example, if a simulation has 10 freshmen and six seniors, use 10 blue beads to represent “freshmen” and six red beads to represent “seniors.”

Developing Course Skills

Throughout the AP Statistics course, students will develop skills that are fundamental to the discipline of statistics. Students will benefit from multiple opportunities to develop these skills in a scaffolded manner.

The tables that follow look at each of the course skills and provide examples of questions or tasks for each skill, along with sample activities and strategies for incorporating that skill into the course.

Skill Category 1: Select methods for collecting and/or analyzing data for statistical inference

The table that follows provides examples of questions, activities, and suggested strategies for teaching students to successfully select statistical methods for different topics throughout the course.

Skill Category 1: *Selecting Statistical Methods*

Skill	Key Questions	Sample Activities	Instructional Strategies
1.A <i>Identify the question to be answered or problem to be solved.</i>	- What is this question asking? - How will we know when we've answered the question? - Does this response answer the question that was asked?	Have students examine free-response questions and underline or highlight the parts where they are being asked to answer a question or provide an explanation for something.	- FRQ Partner Quiz - Marking the Text - Model Questions
1.B <i>Identify key and relevant information to answer a question or solve a problem.</i>	- What are the key words in this problem that are important for answering the question? - Is there any information in this problem that we don't need? - How can we use . . . to answer this question?	Create a T-chart poster with "Type I Error" in one column and "Type II Error" in the other column, then distribute cards with different Type I or Type II error scenarios on them. Have students work with a partner to determine which column their scenario card belongs to. [Topic 6.7]	- Marking the Text - Model Questions
1.C <i>Describe an appropriate method for gathering and representing data.</i>	- Is the data qualitative or quantitative? - How many people (or subjects) were included? - How do I get an unbiased sample? - Is one sampling method better than another in this context? In what way is it better? - Is a random sample better than a biased one? How so? - Which types of displays would work best to show the distribution of the data? Explain. - Would a different display tell a different story with this data?	- Have students review a brief newspaper article and identify how the data was (or could have been) gathered. [Topics 1.4, 1.5, 3.3] - Have students illustrate (draw) the difference between a stratified, cluster, and simple random sample. - Have students collect data (1) by choosing the values themselves (2) by a random process and then describe the pros/cons of each approach.	- Graphic Organizer - Odd One Out

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Skill Category 1: Selecting Statistical Methods (continued)

Skill	Key Questions	Sample Activities	Instructional Strategies
INFERENCE			
1.D Identify an appropriate inference method for confidence intervals.	- Were the data gathered from one sample or two samples? - Is the population standard deviation known? - Are the data in counts or in proportions? - Is the most appropriate summary statistic a sample mean or a sample proportion? - What formula on the formula sheet could help with the construction of the interval? - What is the name of the confidence interval procedure? - How many samples/groups are there? - Do I want to estimate an unknown quantity (interval) or do I want to test a claim (test)? - What type of data will I obtain/am I given: categorical or quantitative?	- Given summary statistics based on data gathered in class (hours of sleep, drive to school or not, etc.), have students determine what confidence interval could be constructed. [Topics 6.2, 6.8, 7.2, 7.6, 9.2] - Have students develop a flowchart of questions that leads to each inference method. - Have students use Categorizing Statistics to practice identifying the appropriate inference procedure for a given scenario.	- Graphic Organizer - Peer Critique
1.E Identify an appropriate inference method for significance tests.	- How many samples were taken? - What procedure would be needed to test this? - What procedure would be needed to see if a certain percentage has changed?	Given summary statistics based on data gathered in class (hours of sleep, drive to school or not, etc.), have students create a question and determine what inference procedure could be used. [Topics 6.4, 6.10, 7.4, 7.8, 8.2, 8.5, 9.4]	- Graphic Organizer - Peer Critique
1.F Identify null and alternative hypotheses.	- Are we looking for there to be a difference from the hypothesized value (two-tailed test) or are we looking for a change in a specific direction (one-tailed test)? - What is our parameter(s) of interest and what symbol(s) do we need to use to represent our parameter(s) of interest? - Are the data paired? If I shuffled the data, would we lose information? (If so, then it's paired.) - Do I have two independent samples? - Is it a matched pairs design or a completely randomized design?	- Present students with a given scenario and a guiding question, for example, "We suspect the average amount of sleep of all students is less than eight hours. What would be the null and alternative hypotheses?" [Topics 6.4, 6.10, 7.4, 7.8, 8.2, 8.5, 9.4] - Give students a context and ask them to write hypotheses (1) for a paired test and (2) a two-sample test and then ask which is the better design.	- Quickwrite - Quiz-Quiz-Trade

Skill Category 2: Describe patterns, trends, associations, and relationships in data

The table that follows provides examples of questions, activities, and suggested strategies for teaching students to analyze data successfully.

Skill Category 2: Data Analysis

Skill	Key Questions	Sample Activities	Instructional Strategies
2.A Describe data presented numerically or graphically.	- Are the distributions symmetric or skewed? - How do the centers of the distributions compare? - How should I talk about the “tendency”? - How should I talk about the “variability”? What are the best statistics to use in each case? - Where does the mean fall relative to the medians in each of the distributions?	Given a pair of parallel boxplots, have students describe how the distributions are similar and how they are different.	- Match Mine - Notice and Wonder - Sentence Starters - Think-Pair-Share - Quickwrite
2.B Construct numerical or graphical representations of distributions.	Does this graphical representation tell an honest, authentic, and complete story about the situation?	Have students create various graphical displays (stemplot, dotplot, boxplot, histogram) for the time it takes to get to school.	- Create Representations - Look for a Pattern - Sketch and Switch
2.C Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.	Is this quantitative or categorical data?	Have students use technology to calculate summary statistics and then remove/change/add data to see how each statistic is affected.	- Round Table - Scavenger Hunt

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Skill Category 2: Data Analysis (continued)

Skill	Key Questions	Sample Activities	Instructional Strategies
2.D Compare distributions or relative positions of points within a distribution.	▪ If there is a difference, is it significant? ▪ Is this quantitative or categorical data? ▪ Is this a small data set? (If so, use a dotplot or stemplot.) Or is this a large data set? (If so, use a histogram or boxplot.) ▪ What is the most interesting similarity (or difference) between these graphs? ▪ Describe the context for each of these distributions.	▪ Give students the same data set but with different graphs so they can explain the pros/cons of each graphical display (boxplot vs. histogram vs. stemplot). Technology can make this exercise more efficient (like stapplet.com). ▪ With a focus on real-world data, give students scatterplots or dotplots that are similar and different and have them describe the similarities and differences. ▪ Provide students with a set of cards containing different representations of data (e.g., summary statistics, boxplots, histograms, and context descriptions) and then have students match the ones that came from the same data set.	▪ Discussion Groups ▪ Quickwrite ▪ Think-Pair-Share ▪ Two Wrongs Make a Right
INFERENCE <i>not applicable</i>			

Skill Category 3: Explore random phenomena

The table that follows provides examples of questions, activities, and suggested strategies for helping students to develop the skill of using probability and simulation.

Skill Category 3: *Using Probability and Simulation*

Skill	Key Questions	Sample Activities	Instructional Strategies
3.A Determine relative frequencies, proportions, or probabilities using simulation or calculations.	- How can I use a random process to model this scenario? - What population does this dotplot represent? - If your dotplot is not approximately normal, why not? - How is a population distribution different from a sampling distribution? - How is a sampling distribution of sample size 2 similar to and different from a sampling distribution of sample size 5 from the same population?	- Before doing a simulation to find a probability (p-value), ask students what the initial evidence is for the claim. Then calculate the p-value to see how plausible/strong this evidence is. - Have students shoot free throws or toss bean bags to determine the frequency of shots made. - Have students compare and contrast two sampling distributions of different sample sizes to explore the effect of increasing sample size on a distribution. - Present students with a probability scenario with guiding questions, for example, "What was the success rate after 2 free throws of a basketball? After 4? 10? 30? What does this mean?"	- Debriefing - Simulation - The Scribe and the Calculator - Use Manipulatives
3.B Determine parameters for probability distributions.	Why might the mean of a probability distribution for a discrete random variable be less than (or greater than) the average of possible values?	As an introduction, provide students with estimates of probabilities of litter size for black bears: $P(1 \text{ cub}) = 0.06$, $P(2 \text{ cubs}) = 0.23$, $P(3 \text{ cubs}) = 0.63$, $P(4 \text{ cubs}) = 0.08$, and $P(5 \text{ or more cubs}) < 0.01$. Ask student groups to represent this probability distribution for a sample of 30 litters in a table to predict the mean and standard deviation of the number of cubs in a litter for black bears. Students should explain their reasoning in groups. Class discussion should lead to the correct formulas. Ask students to consider interesting questions a scientist studying black bears might ask (see North American Bear Center).	- Debriefing - Predict and Confirm

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Skill Category 3: Using Probability and Simulation (continued)

Skill	Key Questions	Sample Activities	Instructional Strategies
3.B Determine parameters for probability distributions. (continued)	Why might the mean of a probability distribution for a discrete random variable be less than (or greater than) the average of possible values?	Ask students to determine the following: for a bat colony in a given year, if X_1 equals the number of bats born, X_2 equals the number of bats who die, X_3 equals the number of bats who move in, and X_4 equals the number of bats who move out, what do each of the following represent: $X_1 + X_3$, $X_1 - X_2 + X_3 - X_4$? Have them represent the net change in population owing to immigration/emigration. For a scenario in which the death rate is doubled and the number of bats moving out is halved, have students write a linear combination of X_1, X_2, X_3, and X_4 that gives the net population change in bats.	- Debriefing - Predict and Confirm
3.C Describe probability distributions.	When rolling two six-sided dice, why are the outcomes of the sum of the dice not equally likely?	Have students conduct a simulation in which they roll two six-sided dice, record the sum of the values, and then create a table representing the probability distribution from their simulation.	- Predict and Confirm - Sentence Starters - Simulation
INFERENCE			
3.D Construct a confidence interval, provided conditions for inference are met.	- Are the conditions for inference met? How do I check the "normal" condition for this context? - Is there random assignment or random selection? - Is this quantitative or categorical data? Are there one or two samples?	- Give students a large population from which to sample. Have each student group collect a different random sample, calculate statistic, then calculate interval. Students then compare their intervals (sampling variability) and the class calculates their "capture rate" (confidence level). - Give students multiple-choice questions with intervals already constructed and ask them to identify the correct interval.	- Error Analysis - Model Questions - Think Aloud
3.E Calculate a test statistic and find a p -value, provided conditions for inference are met.	- How do we know which test statistic, t or z, is appropriate for a test for a population mean? - In terms of probability, what does it mean to say that a p-value is less than 0.05?	- Give each student a pair of hypotheses for significance testing. Ask each student to identify an appropriate test statistic and formula. Have groups share findings with each other and check notation. - Use the same hypotheses to follow up when learning to draw conclusions: (1) Identify at least one conclusion for $p = 0.12$ that would be incorrect because it would be equivalent to accepting the null hypothesis and (2) write an appropriate conclusion for $p = 0.12$.	- Ask the Expert - Error Analysis - Gallery Walk - Notation Read Aloud - The Scribe and the Calculator - Think Aloud

Skill Category 4: Develop an explanation or justify a conclusion using evidence from data, definitions, or statistical inference

Students consistently struggle with communicating their arguments on the AP Exam. Students often need targeted support to develop this skill, so teachers should remind students that communicating reasoning is at least as important as finding a solution. Well-communicated reasoning validates solutions.

Teachers can also reinforce that when students are asked to provide reasoning or a justification for their solution, a successful response will include:

- A logical sequence of steps
- An argument that explains why those steps are appropriate
- An accurate interpretation of the solution (with units), always in the context of the situation

In order to help students develop these communication skills, teachers can:

- Have students practice explaining their solutions orally to a small group or to the class.
- Present an incomplete argument or explanation and have students supplement it for greater clarity or completeness.
- Provide sentence starters, template guides, and communication tips to help scaffold the writing process.

Skill Category 4: *Statistical Argumentation*

Skill	Key Questions	Sample Activities	Instructional Strategies
4.A <i>Make an appropriate claim or draw an appropriate conclusion.</i>	▪ What do you expect to be true about the number of red cards in a fair deck of cards? What seems to be true? ▪ How would proportions be different when drawing cards from a deck without replacement, as opposed to with replacement? ▪ What are your assumptions about this scenario? ▪ What, if anything, can we conclude about causation in this scenario? ▪ Does our conclusion imply that we have “accepted” the null, that we are “certain” about something, or that we have “proven” something?	▪ Manipulate a deck of cards by stacking it with more reds, for example (the class is not aware of this). Have one student pick cards one at a time, with replacement, and shuffling between each pick. Have another student tally the number of reds/blacks. Have students discuss expectations (50% reds) and what they’re seeing. This can be used as an introduction to the concept of significance tests. [Topic 6.4] ▪ Have students look at student samples of inappropriate claims or drawing inappropriate conclusions from released FRQs. Then have them compare these samples to the scoring guidelines or notes from the Chief Reader Report. Finally, have students correct a set of new incorrect samples so that the samples are making appropriate claims or drawing appropriate conclusions. ▪ After displaying scatterplots with strong correlation, ask students to make predictions using extrapolation.	▪ Build the Model Solution ▪ Error Analysis ▪ Gallery Walk ▪ Sentence Starters

continued on next page

Skill Category 4: Statistical Argumentation (continued)

Skill	Key Questions	Sample Activities	Instructional Strategies
4.B Interpret statistical calculations and findings to assign meaning or assess a claim.	- At what point did you become suspicious? - How can we tell whether the results should be acceptable?	See activity in Skill 4.A. If the deck is very unfairly stacked, the class will eventually become suspicious and start questioning whether it is actually a fair deck. This can lead to a discussion of how likely it is to see the results assuming we're dealing with a fair deck.	- Quickwrite - Reversing Interpretations - Sentence Starters
INFERENCE			
4.C Verify that inference procedures apply in a given situation.	- Are the assumptions met? - How many cards must be selected in order to meet the conditions? - Why is this condition necessary? - What would go wrong with the inference procedure if this condition is violated?	- See activity in Skill 4.A. Have students discuss the assumptions required for a one-proportion significance test (or for a confidence interval). - After students have studied all of the inference procedures in the course, have them make a chart that summarizes the conditions required for each procedure to be valid.	- Think Aloud - Think-Pair-Share
4.D Justify a claim based on a confidence interval.	- Is P_0 in the interval? - How can you be more confident about your conclusion?	- See activity in Skill 4.A. Have students identify the confidence interval and state their conclusion. [Topic 6.2, 6.3] - Use examples of confidence intervals with P_0 in, not in, at the center of, and near the edges of the confidence interval.	Error Analysis
4.E Justify a claim using a decision based on significance tests.	- To what populations can I infer these claims (based on how the data was collected)? - Did we use language that implies absolute knowledge (i.e., that we've "proven" something)?	- See activity in Skill 4.A. Have students conclude the significance test. Have them determine the test statistic, obtain p-value, make their decision, and state their conclusion. - Present students with a scenario involving a significance test and ask students to be explicit about three things: How does the p-value compare to alpha? What do we say about the null? What do we say about the alternative?	- Error Analysis - Two Wrongs Make a Right

AP STATISTICS

Exam Information



Exam Overview

The AP Statistics Exam assesses student understanding of the skills and learning objectives outlined in the course framework. The exam is 3 hours long and includes 40 multiple-choice questions and 6 free-response questions. The details of the exam, including exam weighting and timing, can be found below:

Section	Question Type	Number of Questions	Exam Weighting	Timing
I	Multiple-choice questions	40	50%	90 minutes
II	Free-response questions	6		
	Part A: Questions 1–5		37.5%	65 minutes
	Part B: Question 6: Investigative task		12.5%	25 minutes

Formulas and tables are furnished to students taking the AP Statistics Exam (see p. 259). Each student is expected to bring a graphing calculator with statistical capabilities to the exam. Students may not use minicomputers, pocket organizers, electronic writing pads, or calculators with QWERTY (i.e., typewriter) keyboards on the AP Statistics Exam. See the [Calculator Policy for AP Statistics](#) on apstudent.collegeboard.org for a complete list of eligible calculators and policies concerning enhancements to eligible calculators.

The exams assess content from the three big ideas of the course.
Big Idea 1: Variation and Distribution
Big Idea 2: Patterns and Uncertainty
Big Idea 3: Data-Based Predictions, Decisions, and Conclusions

The multiple-choice section of the AP Exam assesses the nine units of the course with the following exam weighting:

Exam Weighting for the Multiple-Choice Section of the AP Exam

Units	Exam Weighting
Unit 1: Exploring One-Variable Data	15–23%
Unit 2: Exploring Two-Variable Data	5–7%
Unit 3: Collecting Data	12–15%
Unit 4: Probability, Random Variables, and Probability Distributions	10–20%
Unit 5: Sampling Distributions	7–12%
Unit 6: Inference for Categorical Data: Proportions	12–15%
Unit 7: Inference for Quantitative Data: Means	10–18%
Unit 8: Inference for Categorical Data: Chi-Square	2–5%
Unit 9: Inference for Quantitative Data: Slopes	2–5%

How Student Learning Is Assessed on the AP Exam

The four AP Statistics skill categories are assessed on the AP Exam as detailed below.

Skill Category	Multiple-Choice Section	Free-Response Section
Skill 1: Selecting Statistical Methods	Multiple-choice questions assess students' ability to select methods for collecting and/or analyzing data for statistical inference. Students will need to identify key and relevant information to answer a question or solve a problem (including appropriate inference methods), describe appropriate methods for gathering and representing data, and identify null and alternative hypotheses.	One free-response question will primarily focus on collecting data, assessing Skills 1.B and/or 1.C. Other free-response questions may also assess this skill category.
Skill 2: Data Analysis	Multiple-choice questions will assess students' ability to describe patterns, trends, associations, and relationships in data. Students will need to describe data presented numerically or graphically, perform statistical calculations, and compare distributions.	One free-response question will primarily focus on exploring data, assessing some combination of Skills 2.A, 2.B, 2.C, and/or 2.D. Other free-response questions may also assess this skill category.
Skill 3: Using Probability and Simulation	Multiple-choice questions will assess students' ability to explore random phenomena. Students will need to determine relative frequencies, proportions, or probabilities using simulation or calculations. Additionally, students will need to describe and determine parameters for probability distributions. Students will also need to construct confidence intervals and calculate test statistics.	One free-response question will primarily focus on probability and sampling distributions, assessing some combination of Skills 3.A, 3.B, and/or 3.C. Other free-response questions may also assess this skill category.
Skill 4: Statistical Argumentation	Multiple-choice questions will assess students' ability to develop an explanation or justify a conclusion using evidence from data, definitions, or statistical inference. Students will need to make appropriate claims, interpret statistical calculations, verify application of inference procedures, and justify statistical claims.	At least three of the first five free-response questions, as well as the investigative task, will assess statistical argumentation.

Section I: Multiple-Choice

All four AP Statistics skill categories are assessed in the multiple-choice section with the following weighting:

Exam Weighting for the Multiple-Choice Section of the AP Exam

Skill Categories	Exam Weighting
Skill 1: Selecting Statistical Methods	15–23%
Skill 2: Data Analysis	15–23%
Skill 3: Using Probability and Simulation	30–40%
Skill 4: Statistical Argumentation	25–35%

Section II: Free-Response

The second section of the AP Statistics Exam includes six free-response questions, including one investigative task.

The five free-response questions in Part A include the following:

- One multi-part question with a primary focus on Collecting Data, assessing Skill Category 1: Selecting Statistical Methods.
- One multi-part question with a primary focus on Exploring Data, assessing Skill Category 2: Data Analysis.
- One multi-part question with a primary focus on Probability and Sampling Distributions, assessing Skill Category 3: Using Probability and Simulation.

- One question with a primary focus on Inference, assessing the inference skills associated with Skill Categories 1, 3, and 4.
- One question that focuses on two or more skill categories.

The investigative task (Part B) assesses multiple skill categories and content areas, focusing on the application of skills and content in new contexts or in non-routine ways.

Task Verbs Used in Free-Response Questions

The following **task verbs** are commonly used in the free-response questions:

Calculate: Perform mathematical steps to arrive at a final answer (e.g., algebraic expressions or diagrams with properly substituted numbers and correct labeling). Calculate tasks are also phrased with “Find” or interrogatory questions such as “How many?”, “What is?”, “What values?”, “How likely?”, or “How often?”

Compare: Provide a description or explanation of similarities and/or differences.

Construct/Complete: Represent data in graphical or numerical form.

Describe: Provide the relevant characteristics of representations, distributions, or methods.

Determine: Apply an appropriate definition or perform calculations to identify values, intervals, or solutions. Determine tasks are also phrased with interrogatory questions such as “Do the data support?”, “Do the data provide?”, “Is there evidence?”, “Which is better?”, “Does your answer match?”, or “Can it be assumed?”

Estimate: Use models or representations to find approximate values for functions.

Explain: Provide information about how or why a relationship, process, pattern, position, situation, or outcome occurs, using evidence and/or reasoning to support or qualify a claim. “Explain” tasks may also be phrased as “Give a reason for . . .”

Give a point estimate or interval estimate: Use models or representations to find approximate values for uncertain figures.

Give examples: Provide a specific example that meets given criteria.

Identify/Indicate/Circle: Indicate or provide information about a specified topic in words or by circling, shading, or marking given information, without elaboration or explanation. Also phrased as “What is?” or “Which?”

Interpret: Describe the connection between a mathematical expression, representation, or solution and its meaning within the realistic context of a problem, sometimes including consideration of units.

Justify: Provide evidence to support, qualify, or defend a claim and/or provide statistical reasoning to explain how that evidence supports or qualifies the claim.

Verify: Confirm that the conditions of a particular definition, distribution, or inference method are met in order to verify that it is applicable in a given situation. Verify tasks may also be phrased as “Have the conditions been met?” or “Can it be assumed?”

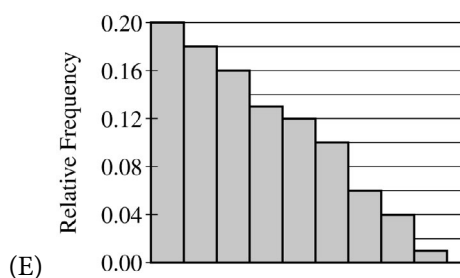
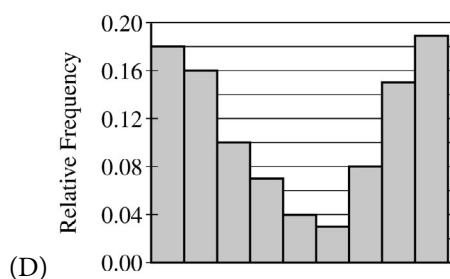
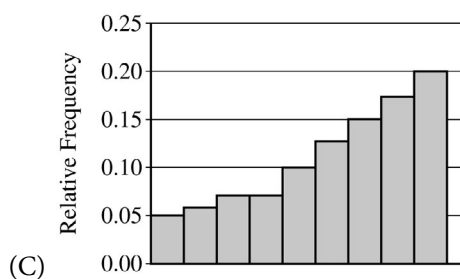
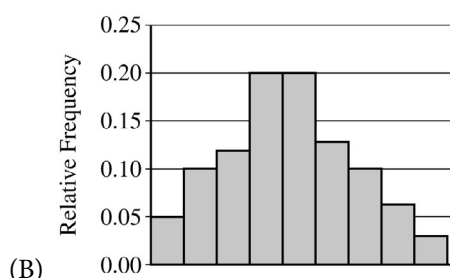
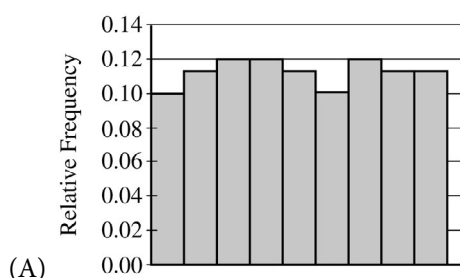
Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and AP Statistics Exam and serve as examples of the types of questions that appear on the exam. After the sample questions is a table that shows which skill, learning objective(s), and unit each question relates to. The table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice

The following are examples of the kinds of multiple-choice questions found on the exam.

1. Which of the following histograms has a shape that is approximately uniform?



2. A recent study reported that 45 percent of adults in the United States now get all their news online. A random sample of 8 adults in the United States will be selected. What is the probability that 6 of the selected adults get all their news online?

(A) $\binom{6}{2}(0.45)^8(0.55)^2$

(B) $\binom{6}{2}(0.45)^6(0.55)^2$

(C) $\binom{8}{6}(0.45)^2(0.55)^8$

(D) $\binom{8}{6}(0.45)^6(0.55)^2$

(E) $\binom{8}{6}(0.45)^8(0.55)^6$

3. In 2011, 17 percent of a random sample of 200 adults in the United States indicated that they consumed at least 3 pounds of bacon that year. In 2016, 25 percent of a random sample of 600 adults in the United States indicated that they consumed at least 3 pounds of bacon that year.

Assuming all conditions for inference are met, which of the following is the most appropriate test statistic to use to investigate whether the proportion of all adults in the United States who consume at least 3 pounds of bacon in 2016 is different from that of 2011?

(A) $z = \frac{0.17 - 0.25}{\sqrt{\frac{(0.17)(0.83)}{200} + \frac{(0.25)(0.75)}{600}}}$

(B) $z = \frac{0.17 - 0.25}{\sqrt{\frac{(0.17)(0.83)}{600} + \frac{(0.25)(0.75)}{200}}}$

(C) $z = \frac{0.17 - 0.25}{\sqrt{(0.21)(0.79)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

(D) $z = \frac{0.17 - 0.25}{\sqrt{(0.23)(0.77)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

(E) $z = \frac{0.17 - 0.25}{\sqrt{(0.50)(0.50)\left(\frac{1}{200} + \frac{1}{600}\right)}}$

4. In a recent year, the distribution of age for senators in the United States Senate was unimodal and roughly symmetric with mean 65 years and standard deviation 10.6 years. Consider a simulation with 200 trials in which, for each trial, a random sample of 5 senators' ages is selected and the mean age calculated.

Which of the following best describes the distribution of the 200 sample mean ages?

- (A) Approximately normal with mean 65 years and standard deviation 10.6 years
- (B) Approximately normal with mean 65 years and standard deviation $\frac{10.6}{\sqrt{5}}$ years
- (C) Approximately normal with mean 65 years and standard deviation $\frac{10.6}{\sqrt{200}}$ years
- (D) Approximately uniform with mean 65 years and standard deviation $\frac{10.6}{\sqrt{5}}$ years
- (E) Approximately uniform with mean 65 years and standard deviation $\frac{10.6}{\sqrt{200}}$ years

5. An athletic director believes that more than 50 percent of the students in a certain school district exercise at least 3.5 hours per week. To investigate the belief, the director selected a random sample of 40 students in the school district and found that 60 percent of the students in the sample exercise at least 3.5 hours per week. Let p represent the proportion of all students in the school district who exercise at least 3.5 hours per week.

Which of the following is the most appropriate alternative hypothesis to test the director's claim?

- (A) $H_a : p = 0.5$
- (B) $H_a : p > 0.5$
- (C) $H_a : p < 0.5$
- (D) $H_a : p = 0.6$
- (E) $H_a : p > 0.6$

6. For a certain population of penguins, the distribution of weight is approximately normal with mean 15.1 kilograms (kg) and standard deviation 2.2 kg. Approximately what percent of the penguins from the population have a weight between 13.0 kg and 16.5 kg?

- (A) 17%
- (B) 34%
- (C) 57%
- (D) 68%
- (E) 74%

7. A researcher collected data on the latitude, in degrees north of the equator, and the average low temperature, in degrees Fahrenheit, for a random sample of cities in Europe. The data were used to create the following equation for the least-squares regression line.

$$\text{predicted average low temperature} = 65.5 - 0.70(\text{latitude})$$

Which of the following is the best interpretation of the slope of the line?

- (A) For each one degree north of the equator increase, the predicted average low temperature increases on average by 0.70 degree Fahrenheit.
 - (B) For each one degree north of the equator increase, the predicted average low temperature decreases on average by 0.70 degree Fahrenheit.
 - (C) For each 0.7 degree north of the equator increase, the predicted average low temperature decreases on average by 1 degree Fahrenheit.
 - (D) For each one degree Fahrenheit increase in average low temperature, the predicted latitude increases on average by 0.70 degree north of the equator.
 - (E) For each one degree Fahrenheit increase in average low temperature, the predicted latitude decreases on average by 0.70 degree north of the equator.
8. An independent polling agency was hired to track the preferences of registered voters in a district for an upcoming election. The polling agency divided the district into twenty regions and believes that the regions are similar to one another in their composition. The agency then randomly selected two of the regions and surveyed all registered voters in both regions.

Which of the following best describes the sampling method used by the polling agency?

- (A) Convenience sampling
 - (B) Simple random sampling
 - (C) Stratified random sampling
 - (D) Systematic sampling
 - (E) Cluster sampling
9. A sports physician conducted a study to investigate whether there is an association between running experience and the occurrence of a certain sport injury for marathon runners while training for a marathon. Data were collected on a random sample of 51 marathon runners. Each runner from the sample was categorized by running experience (low, medium, high) and whether or not the runner experienced the sport injury while training for a marathon. The conditions for inference were met, and a χ^2 test statistic of approximately 8.12 was calculated.

Which of the following describes the p -value of the test?

- (A) $p\text{-value} > 0.25$
- (B) $0.10 < p\text{-value} < 0.25$
- (C) $0.05 < p\text{-value} < 0.10$
- (D) $0.01 < p\text{-value} < 0.05$
- (E) $p\text{-value} < 0.01$

10. An agricultural engineer selected a random sample of 30 farms in the United States to construct a 95 percent confidence interval for the mean size, in acres, of farms in the United States. The resulting interval was (367, 558).

Which of the following is an appropriate interpretation of the 95 percent confidence level?

- (A) Approximately 95% of the farm sizes in the sample are between 367 acres and 558 acres.
 - (B) Approximately 95% of all farm sizes in the United States are between 367 acres and 558 acres.
 - (C) Approximately 95% of all random samples of size 30 from the population will have a mean farm size between 367 acres and 558 acres.
 - (D) Approximately 95% of all random samples of size 30 from the population will produce intervals that contain the mean size of farms in the United States.
 - (E) Approximately 95% of all random samples of size 30 from the population will produce intervals that contain the sample mean.
11. An experiment was designed to investigate the relationship between the dosage of a certain migraine medication and the amount of time until relief from the headache was experienced. Four dosages in milligrams (mg)—2.5, 5.0, 7.5, and 10—of the medication were used. The 20 participants in the experiment were known to experience migraines. Each participant was randomly assigned one of the four dosages. When the participants experienced a migraine, they took the assigned medication and recorded the number of minutes it took to experience relief from the headache. The mean number of minutes it took each group to experience relief was compared.
- What were the experimental units in the experiment?
- (A) The number of minutes to experience relief
 - (B) The mean number of minutes it took each group to experience relief
 - (C) The four dosages
 - (D) The 20 participants
 - (E) The relationship between the dosage and the amount of time
12. The recommended dietary allowance of iron for women under the age of 51 is 18 milligrams (mg) per day. A medical researcher studying women living in a certain region suspected that the women were getting less than the daily allowance of iron, on average. The researcher took a random sample of women under the age of 51 from the region and measured their daily iron intakes. The following hypotheses were tested at the significance level of $\alpha = 0.05$ for the population mean μ of the daily iron intake for women in the region.

$$H_0 : \mu = 18$$

$$H_a : \mu < 18$$

All conditions for inference were met, and the resulting p -value was 0.031.

Which of the following is an appropriate conclusion?

- (A) The p -value is less than α , and the null hypothesis should be rejected. There is convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (B) The p -value is less than α , and the null hypothesis should be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (C) The p -value is less than α , and the null hypothesis should not be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (D) The p -value is greater than α , and the null hypothesis should be rejected. There is convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.
- (E) The p -value is greater than α , and the null hypothesis should not be rejected. There is not convincing statistical evidence that the mean daily intake of iron for women in the region is less than the recommended 18 mg.

13. At a small company, 40 percent of the employees are classified as junior level, and 32 percent of the employees work in department A. Of those classified as junior level, 55 percent work in department A. One employee will be selected at random.

What is the probability that the selected employee works in department A and is not classified as junior level?

- (A) 0.10
- (B) 0.19
- (C) 0.22
- (D) 0.32
- (E) 0.50

14. A one-sample z -test for a proportion at the significance level of $\alpha = 0.10$ will be conducted for the following set of hypotheses.

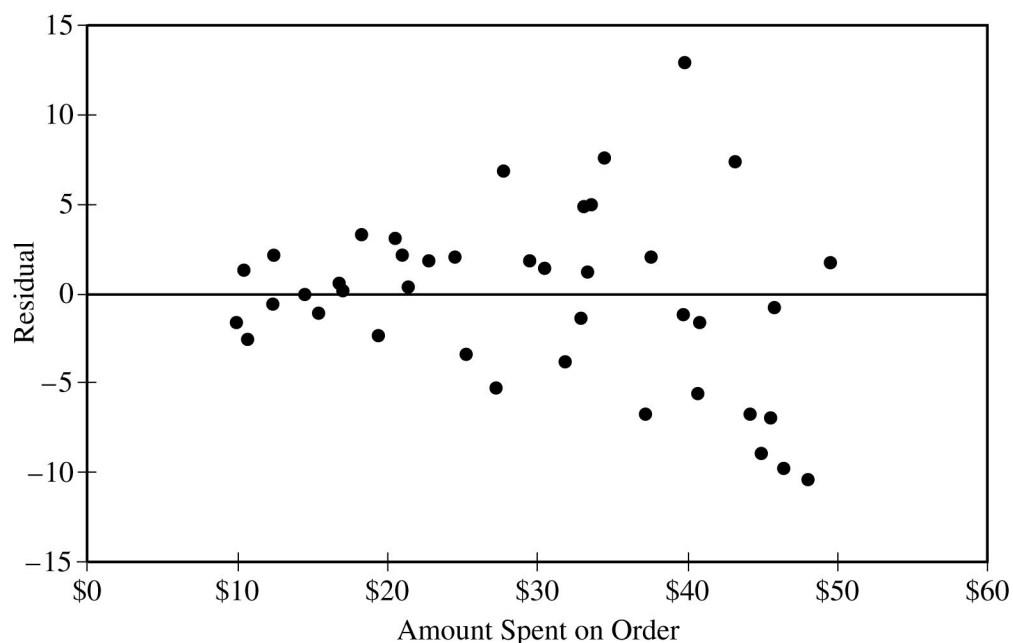
$$H_0 : p = 0.50$$

$$H_a : p < 0.50$$

For which of the following is the probability of a Type II error the greatest?

- (A) A sample size of 200 and a true proportion of 0.45
- (B) A sample size of 200 and a true proportion of 0.40
- (C) A sample size of 200 and a true proportion of 0.35
- (D) A sample size of 100 and a true proportion of 0.45
- (E) A sample size of 100 and a true proportion of 0.35

15. A small-business owner collected data from a random sample on the amount spent on an order and the time it takes to fill the order. The resulting data were used to complete a linear regression analysis of time, in minutes, versus amount spent, in dollars, and to create the following residual plot.



Based on the residual plot, which condition for inference for the slope of the regression line does not appear to be satisfied?

- (A) The true relationship between the number of items and the amount spent on an order is linear.
- (B) For each amount spent on an order, the distribution of time to fill the order is approximately normal.
- (C) The standard deviation of the distribution of time to fill an order is the same for each amount spent on the order.
- (D) The data are collected using a random sample or randomized experiment.
- (E) The sample size is less than 10% of the population size.

16. An athletic trainer wanted to investigate the relationship between an athlete's resting heart rate and the heart rate after exercise. From a sample of 12 athletes, the athletic trainer recorded each athlete's resting heart rate, in beats per minute, and heart rate, in beats per minute, after 5 minutes of moderate exercise. The results of a regression analysis of the data are shown in the following computer output.

Predictor	Coef	SE Coef	T	P
Constant	17.5	29.2	0.60	0.563
Resting Heart Rate	1.786	0.453	3.94	0.003
S = 9.9702		R-Sq = 60.88%		

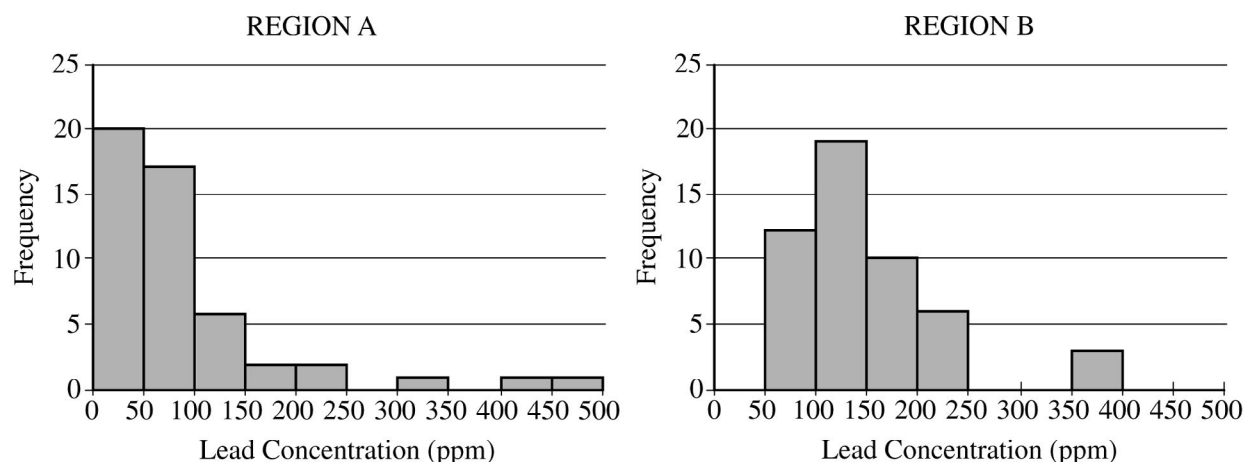
Assume the conditions for inference are met. Which of the following represents a 95 percent confidence for the slope of the population regression line?

- (A) $17.5 \pm 1.96(29.2)$
 (B) $1.786 \pm 1.96(0.453)$
 (C) $1.786 \pm 1.96\left(\frac{0.453}{\sqrt{12}}\right)$
 (D) $1.786 \pm 2.228\left(\frac{0.453}{\sqrt{12}}\right)$
 (E) $1.786 \pm 2.228(0.453)$

Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there are six free-response questions.

1. A geologist studying lead concentration in soil selected random samples of soil from two regions: region A and region B. The following histograms show the distribution of lead concentration, in parts per million (ppm), for the two samples.



- (A) Write a few sentences comparing the distributions of lead concentration in the two samples.
 - (B) To investigate whether the mean lead concentration is different in region A than in region B, the geologist conducted the appropriate test. All conditions for inference are met, and the p -value of the test is 0.007. Based on the p -value, is there convincing statistical evidence, at a level of significance of $\alpha = 0.05$, that there is a difference between the mean lead concentration of region A and the mean lead concentration of region B? Justify your answer.
2. Past records from a certain movie theater indicate that 80 percent of moviegoers make a purchase at the movie theater's snack bar. A random sample of 3 moviegoers from the theater will be selected at random. Let the random variable S represent the number of moviegoers who make a purchase at the snack bar from the 3 selected.
 - (A) Complete the following table to create the probability distribution of S .

s	0	1	2	3
$P(s)$	0.008			0.512

- (B) Calculate and interpret the expected value of S .

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Unit
1	A	2.A	UNC-1.H	1
2	D	3.A	UNC-3.B	4
3	D	3.E	VAR-6.K	6
4	B	3.C	UNC-3.R	5
5	B	1.F	VAR-6.D	6
6	C	3.A	VAR-2.B	1
7	B	4.B	DAT-1.H	2
8	E	1.C	DAT-2.C	3
9	D	3.E	VAR-8.M	8
10	D	4.B	UNC-4.S	7
11	D	1.C	VAR-3.A	3
12	A	4.E	DAT-3.F	7
13	A	3.A	VAR-4.D	4
14	D	4.A	UNC-5.C	6
15	C	4.C	VAR-7.L	9
16	E	3.D	UNC-4.AF	9

Free-Response Question	Skill	Learning Objective	Unit
1	2.A, 2.D, 4.E	UNC-1.H, UNC-1.M, UNC-1.O, DAT-3.H	1, 7
2	2.B, 3.A, 3.B, 4.B	UNC-3.B, VAR-5.A, VAR-5.C, UNC-3.C, VAR-5.D	4

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Statistics Exam Page](#) on AP Central.

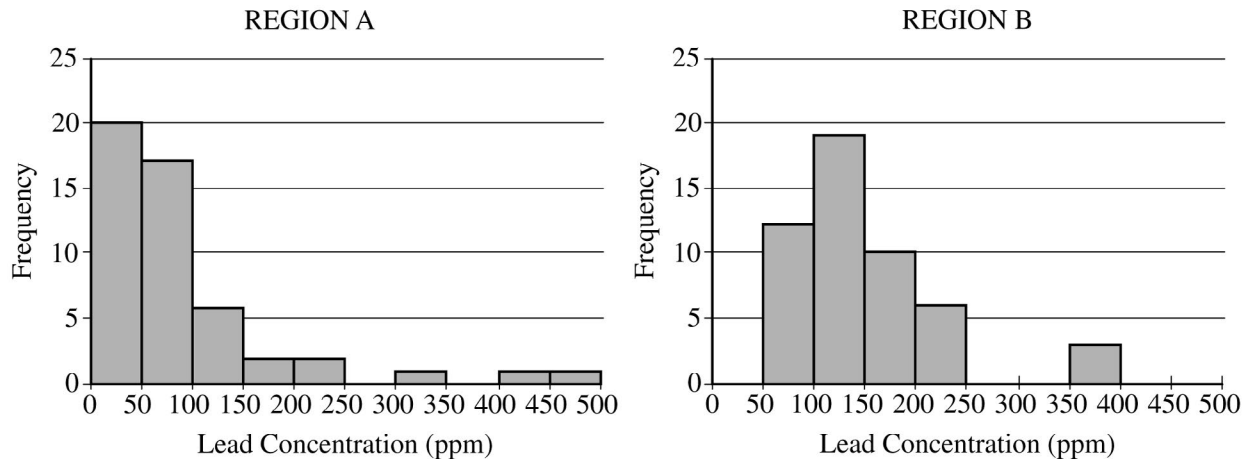
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Scoring Guidelines

Question 1: Focus on Exploring Data

1. A geologist studying lead concentration in soil selected random samples of soil from two regions: region A and region B. The following histograms show the distribution of lead concentration, in parts per million (ppm), for the two samples.



- (A) Write a few sentences comparing the distributions of lead concentration in the two samples.
- (B) To investigate whether the mean lead concentration is different in region A than in region B, the geologist conducted the appropriate test. All conditions for inference are met, and the p -value of the test is 0.007. Based on the p -value, is there convincing statistical evidence, at a level of significance of $\alpha = 0.05$, that there is a difference between the mean lead concentration of region A and the mean lead concentration of region B? Justify your answer.

Scoring Guidelines for Question 1: Focus on Exploring Data

4 points

General Scoring Notes

Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct, partially correct, or incorrect. The response is then categorized based on the scores assigned to each letter part and awarded a score between 0 and 4 (see the table at the end of each question).

	Model Solution	Scoring
Part (A) 2.A 2.D UNC-1.H UNC-1.M UNC-1.O	The distributions of lead concentration for both Region A and Region B are skewed to the right. The median lead concentration of Region B (between 100 ppm and 150 ppm) is greater than the median lead concentration of Region A (between 50 ppm and 100 ppm). The range of lead concentration of Region A (between 400 ppm and 500 ppm) is greater than the range of lead concentration of Region B (between 250 ppm and 350 ppm)	Essentially correct (E) if the response addresses the following five components: 1. Shape (right skewed) 2. Center (e.g., correct location of mean and/or median) 3. Variability (e.g., correct range and/or IQR) 4. Comparison between regions of at least one of the following: shapes, centers, or variability 5. Context (lead concentration) Partially correct (P) if the response includes only three or four of the five components listed above. Incorrect (I) if the response does not satisfy the criteria for E or P.
Part (B) 4.E DAT-3.H	Because the p-value of 0.007 is less than the level of significance of 0.05, the data provide convincing statistical evidence that there is a difference between the mean lead concentration in the two regions.	Essentially correct (E) if the response provides both of the following: 1. Correct conclusion (Yes, <i>the data provide convincing statistical evidence</i>) in context (<i>lead concentration</i>) 2. Justification based on comparing the p-value to the significance level, α Partially correct (P) Option 1: The response provides both of the following: 1. Correct conclusion (Yes) but without context 2. Justification based on comparing the p-value to the significance level, α Partially correct (P) Option 2: The response provides both of the following: 1. Correct conclusion (Yes) in context (<i>lead concentration</i>) 2. Justification based on the p-value, but without comparison of the p-value to the significance level, α Incorrect (I) if the response does not satisfy the criteria for E or P.

Scoring for Question 1	Score
Complete Response Both parts essentially correct	4
Substantial Response One part essentially correct and one part partially correct	3
Developing response Both parts partially correct OR one part essentially correct and one part incorrect	2
Minimal Response One part partially correct and one part incorrect	1

Question 2: Focus on Probability and Sampling Distributions

2. Past records from a certain movie theater indicate that 80 percent of moviegoers make a purchase at the movie theater's snack bar. A random sample of 3 moviegoers from the theater will be selected at random. Let the random variable S represent the number of moviegoers who make a purchase at the snack bar from the 3 selected.

(A) Complete the following table to create the probability distribution of S .

s	0	1	2	3
$P(s)$	0.008			0.512

(B) Calculate and interpret the expected value of S .

Scoring Guidelines for Question 2: Focus on Probability and Sampling Distributions

4 points

General Scoring Notes

Each part of the question (indicated by a letter) is initially scored by determining if it meets the criteria for essentially correct, partially correct, or incorrect. The response is then categorized based on the scores assigned to each letter part and awarded a score between 0 and 4 (see the table at the end of each question).

Learning Objectives: UNC-3.B VAR-5.A VAR-5.C UNC-3.C VAR-5.D

	Model Solution	Scoring										
Part (A) 2.B 3.A UNC-3.B VAR-5.A	Random variable S has a binomial distribution with $n = 3$ and $p = 0.8$. $P(s = 1) = \binom{3}{1}(0.8)^1(0.2)^2 = 0.096$ $P(s = 2) = \binom{3}{2}(0.8)^2(0.2)^1 = 0.384$ <table><tr><td>s</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(s)</td><td>0.008</td><td>0.096</td><td>0.384</td><td>0.512</td></tr></table>	s	0	1	2	3	P(s)	0.008	0.096	0.384	0.512	Essentially correct (E) if the response includes the following three components. - Shows evidence that the calculation of at least one probability necessary to complete the table is based on a binomial with $n = 3$ and $p = 0.8$, either by using the binomial formula with correct values substituted in, or by using the notation $B(3, 0.8)$, or by using a calculator command, such as binompdf ($n = 3, p = 0.8, s = 2$). However, an unlabeled calculator command such as binompdf(3, 0.8, 2) does not satisfy component 1. Note: - A normal approximation is not appropriate because the sample size is too small. - Correct calculation of $P(s = 1)$ or $P(s = 2)$. - The probabilities of the probability distribution sum to 1. Partially correct (P) if the response includes only two of the three components. Incorrect (I) if the response does not satisfy the criteria for E or P.
s	0	1	2	3								
P(s)	0.008	0.096	0.384	0.512								
Part (B) 3.B 4.B VAR-5.C UNC-3.C VAR-5.D	Calculation, Option 1: The expected value of S is $E(S) = \sum sP(s)$ $= 0(0.008) + 1(0.096) + 2(0.384) + 3(0.512)$ $= 2.4$ Calculation, Option 2: Because S is a binomial random variable, $E(S) = np = 3(0.8) = 2.4$. Interpretation: If the process of randomly selecting 3 moviegoers is repeated many times, the average number of moviegoers who make a purchase at the snack bar will be very close to 2.4.	Essentially correct (E) if the response satisfies the following four components. - Calculates the expected value correctly. A correct expected value calculation that is consistent with incorrect probabilities reported in part (a) satisfies component 1. If the expected value calculation is rounded to an integer, the response does not satisfy component 1. - Interpretation—includes reference to long run, many samples or repeated sampling. - Interpretation—includes reference of average. - Context. Partially correct (P) if the response includes only three of the four components. Incorrect (I) if the response does not satisfy the criteria for E or P.										

Scoring for question 2	Score
Complete Response Both parts essentially correct	4
Substantial Response One part essentially correct and one part partially correct	3
Developing response Both parts partially correct OR one part essentially correct and one part incorrect	2
Minimal Response One part partially correct and one part incorrect	1

AP STATISTICS

Appendix



AP STATISTICS

Formula Sheet and Tables

Formulas for AP Statistics

I. Descriptive Statistics

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum x_i = \frac{\sum x_i}{n} & s_x &= \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \\ \hat{y} &= a + bx & \bar{y} &= a + b\bar{x} \\ r &= \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) & b &= r \frac{s_y}{s_x}\end{aligned}$$

II. Probability and Distributions

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability Distribution	Mean	Standard Deviation
Discrete random variable, X	$\mu_X = E(X) = \sum x_i \cdot P(x_i)$	$\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 \cdot P(x_i)}$
If X has a binomial distribution with parameters n and p , then: $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ where $x = 0, 1, 2, 3, \dots, n$	$\mu_X = np$	$\sigma_X = \sqrt{np(1-p)}$
If X has a geometric distribution with parameter p , then: $P(X = x) = (1-p)^{x-1} p$ where $x = 1, 2, 3, \dots$	$\mu_X = \frac{1}{p}$	$\sigma_X = \frac{\sqrt{1-p}}{p}$

III. Sampling Distributions and Inferential Statistics

Standardized test statistic: $\frac{\text{statistic} - \text{parameter}}{\text{standard error of the statistic}}$
Confidence interval: $\text{statistic} \pm (\text{critical value})(\text{standard error of statistic})$
Chi-square statistic: $\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$

III. Sampling Distributions and Inferential Statistics (*continued*)

Sampling distributions for proportions:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For one population: $\hat{p}$	$\mu_{\hat{p}} = p$ $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
For two populations: $\hat{p}_1 - \hat{p}_2$	$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ When $p_1 = p_2$ is assumed: $s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ where $\hat{p}_c = \frac{X_1 + X_2}{n_1 + n_2}$

Sampling distributions for means:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For one population: $\bar{X}$	$\mu_{\bar{X}} = \mu$ $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	$s_{\bar{X}} = \frac{s}{\sqrt{n}}$
For two populations: $\bar{X}_1 - \bar{X}_2$	$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	$s_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Sampling distributions for simple linear regression:

Random Variable	Parameters of Sampling Distribution	Standard Error* of Sample Statistic
For slope: b	$\mu_b = \beta$ $\sigma_b = \frac{\sigma}{\sigma_x \sqrt{n}}$ where $\sigma_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$	$s_b = \frac{s}{s_x \sqrt{n-1}}$ where $s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n-2}}$ and $s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

*Standard deviation is a measurement of variability from the theoretical population. Standard error is the estimate of the standard deviation. If the standard deviation of the statistic is assumed to be known, then the standard deviation should be used instead of the standard error.

Tables for AP Statistics

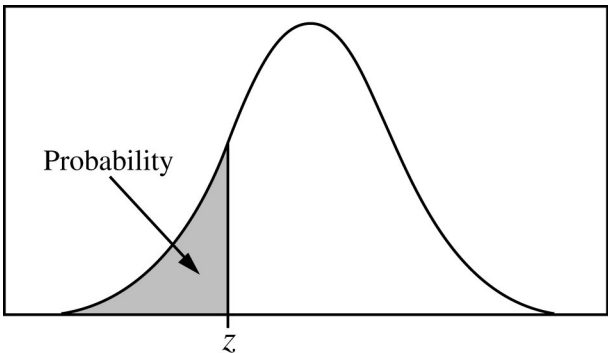


Table entry for z is the probability lying below z .

Table A Standard Normal Probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559

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Table A Standard Normal Probabilities (continued)

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

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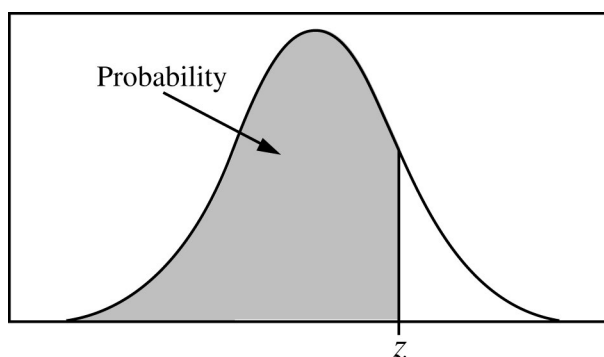


Table entry for z is the probability lying below z .

Table A Standard Normal Probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936

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Table A Standard Normal Probabilities (continued)

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

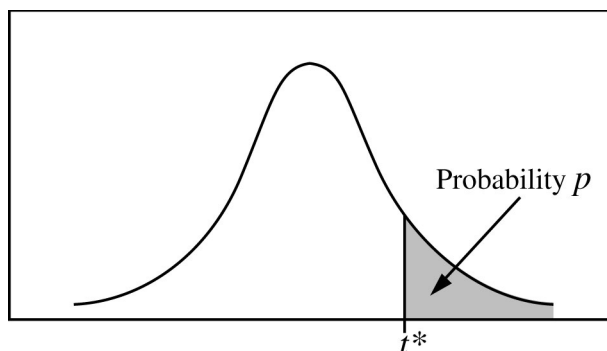


Table entry for p and C is the point t^* with probability p lying above it and probability C lying between $-t^*$ and t^* .

Table B t Distribution Critical Values

df	Tail Probability p											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850

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Table B *t* Distribution Critical Values (continued)

<i>df</i>	Tail Probability <i>p</i>											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
Confidence level C												

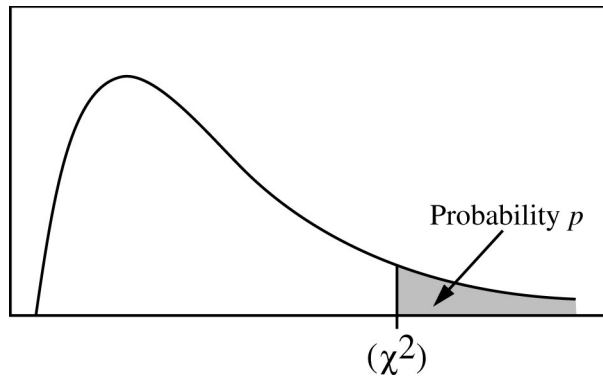


Table entry for p is the point (χ^2) with probability p lying above it.

Table C χ^2 Critical Values

df	Tail probability p											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50

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Table C χ^2 Critical Values (continued)

<i>df</i>	Tail probability <i>p</i>											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.7
80	88.13	90.41	93.11	96.58	101.9	106.6	108.1	112.3	116.3	120.1	124.8	128.3
100	109.1	111.7	114.7	118.5	124.3	129.6	131.1	135.8	140.2	144.3	149.4	153.2

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